

The Mass Gap of Pure $SU(N)$ Gauge Theory as a Finite-Aperture Effect

The mass gap from a finite extraction screen, and the continuum statement reduced to named inputs

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A note on measurement. Every empirical quantity in this paper — the resolved-mode count K_{signal} , the aperture margin, the mass-gap rate — is a deterministic read of a raw lattice configuration by the entropy-matched instrument developed in [E]. The resampling, the empirical-Bernstein intervals and the lag-ratio rates belong to that instrument; the elementary statistics (standard errors, descriptive fits) are computed separately. Each read fixes its own resolution from the configuration's own Shannon entropy and returns the same number on the same input, bit-for-bit. The reads are reported in §8 and carry the argument forward to the classical results they meet.

Abstract

This paper reads the mass gap of pure $SU(N)$ gauge theory from a single fact about observation: a local observer reads the vacuum's gauge-invariant content through a boundary of finite information capacity — a finite aperture, which band-limits. A finite aperture has a diffraction limit, and the limit cannot host the infinitely-extended massless mode a gapless theory requires, so the screen's predictive excess decays at a positive rate: the smallest quantum. The scale is an entropy floor $\kappa_0 = \frac{1}{4} \ln 3 \approx 0.275$, a lower bound on the centre-vortex disorder ensemble's closed-surface entropy density, and the gap is bounded below by the entropy margin, $\Delta(\beta) \geq \kappa_0 - \mu(\beta) > 0$, with $\|C(\tau)\| \leq M e^{-\Delta\tau}$, and this decay reconstructs a Hamiltonian with $\text{spec} \subseteq \{0\} \cup [\Delta, \infty)$ through the moment-support bridge. This finite-aperture reading is the new content, computed forward from the aperture along the determined direction of renormalisation, meeting the established results at junctions.

At finite spacing the lattice regularisation is the finite-cell screen: for every a there is a Hilbert space, a Hamiltonian bounded below, a unique vacuum, and the full Osterwalder–Schrader (OS) structure. Reflection positivity is inherited by the continuum limit as a closed condition; the uniform bound that makes the Schwinger functions tight is the open input E (§12). For the finite-aperture model, whose conclusion is carried by the scalar tension μ alone (§12), the mass gap, non-triviality, and the read-level axis-permutation form of Euclidean $SO(4)$ follow from the band-limit together with two reads: **A1**, the centre-vortex tension stays below the floor at every coupling and uniformly as $a \rightarrow 0$; and **A2**, the continuum read is direction-independent. A1 meets the classical results at both ends (the Osterwalder–Seiler character bound at strong coupling, the asymptotic-freedom free-field plateau $\mu_\infty(8) = 0.0326 < \kappa_0$ at weak coupling) and its interior is a theorem on every compact interval, $\mu = -\log\langle\cos\theta\rangle_\rho < \kappa_0$ following from a finite correlation length $\langle d^2 \rangle \leq B$ through the analytic step $\cos x \geq 1 - x^2/2$ and the $1/L^2$ aperture scaling $\langle \theta^2 \rangle = (2\pi/L)^2 \langle d^2 \rangle$, so the margin grows $\propto L^2$. The single input to A1's interior is a uniform bound $\langle d^2 \rangle \leq B_{16} = 3.2480$, the derived sharp aperture ceiling, with no intermediate bound chosen; measured, $\langle d^2 \rangle \in [0.021, 0.192]$, with an empirical-Bernstein upper confidence bound at nominal 99.9999% per coupling (largest upper 1.372; nominal joint $\approx 99.9987\%$ over the grid). The bound assumes independent configurations and takes the sample range as the support width; its confidence level is nominal. The input is stated on the $SU(3)$ plaquette correlation of the periodic 16^4 lattice and is open; the reads are of $SU(2)$ ensembles and are evidence for it (§9). **A second route certifies the same input more strongly and is the one to read first.** Because μ is defined as $-\log\langle\cos\theta\rangle$, the condition $\mu < \kappa_0$ is equivalent to $\langle\cos\theta\rangle > 3^{-1/4}$, both directions holding, so a certificate on the cosine average certifies confinement itself. On that route every coupling on the grid is confined at nominal $\delta = 10^{-30}$ under the same empirical-Bernstein bound and the same assumptions, the worst certified tension $\mu \leq 0.1767$ against $\kappa_0 = 0.2747$ (§9); the moment route at that same δ leaves two couplings above the ceiling. The moment route is the one the reduction consumes. A2 is a Nyquist–Shannon sampling isometry: the transport is valued in $O(4)$, so a spectral read of the correlation Gram is the same in every orientation.

A deterministic read of gauge configurations measures the deciding quantity, deriving its resolution from each configuration's own Shannon entropy with no external scale. The resolved-mode count K_{signal} , the number of spatial modes of the action density standing above a confined-vacuum reference null calibrated at the same plane shape, reads $0.0597\text{--}0.0868$ on 8×8 spatial planes across $SU(2)$ and both phases of compact $U(1)$, and rises from strong to weak coupling inside confined $SU(2)$ by $+0.0092 \pm 0.0017$, comparable to the $+0.0093 \pm 0.0015$ across the $U(1)$ transition. Given A1 and A2, the mass gap, non-triviality, and the read-level form of $SO(4)$ follow for the finite-aperture model as a single reduction, its inputs the classical results it names: reflection positivity (proved at even extent ≥ 4 and nonnegative coupling on the torus, §11.1b, and for a limit state on $\mathbb{Z}^4$ at free boundary conditions, on the whole half-space algebra, §11.1c, and cited outside those domains), the two cited coupling ends, and the crossover correlation length. Finite-spacing existence holds at every spacing; the limit over extents follows from the finite-spacing Osterwalder–Schrader data; the cited constructive four-dimensional measure and the OS→Wightman reconstruction are the two named external inputs. The reconstruction is applied to the Wilson data: the Gibbs reflected pairing of the constructed $SU(N)$ Wilson measure on the slab algebra of an even-extent lattice is Osterwalder–Schrader data, reflection positive at every real coupling, and the reconstruction carries it through with the reconstruction theorem as its only added input. Its translation action is the identity, which a finite periodic extent forces because the translations there have finite order, so Euclidean invariance is carried by the limit. Its composition with the lattice construction of the gap side is open: that composition needs a time translation that is an endomorphism of the slab algebra, and the lattice shift carries a transverse link out of it. The Osterwalder–Schrader data family is instantiated on a **constructed $SU(N)$ realisation**, on which the mass gap and the OS limit take reflection positivity as their **one** cited result, conditional on the two junction inequalities of §12.5, which are open. The OS→Wightman reconstruction is cited in its standard form, against a bilinear Euclidean form with a Hilbert-space-valued conclusion; the Clay problem names [35] as a route and calls the Euclidean and Lorentzian schemes equivalent, so citing it is what the problem asks for. That OS measure is built on a four-dimensional periodic $SU(3)$ Wilson lattice — the Clay problem's dimension and group, $4n^4$ links and $16n^4$ plaquettes, the ordered-loop holonomy and the Wilson action density — with its Euclidean and permutation invariances derived from Haar-invariance composed with the lattice's own axis symmetry; and its infrared datum is derived too, the resolved-mode count following from the measured tension through $\mu < \kappa_0 \Rightarrow \text{card}\{\text{resolved}\} \leq 0.360246 W/(\epsilon\lambda_0^{k+1})$. The interior reduces, on every compact interval, to finite-volume analyticity and a finite grid cover, both open for the $SU(3)$ moment (§9); the single-cell spectral gap has an exact-rational interval enclosure at all couplings (§9). The architecture is a reduction to an external law — asymptotic freedom above the entropy floor, an established theorem — with the gap bounded below by the entropy surplus, $\Delta \geq \kappa_0 - \mu > 0$, certified for the finite-aperture witness. Reflection positivity makes the transfer operator self-adjoint, giving $\rho'(n) \leq \rho'(1)^n$, so a single-cut magnitude $\rho'(1) < 1$ carries the gap uniform in SEPARATION; the single-plaquette gap $\geq \kappa_0$ gives $\rho'(1) = m_{\text{cell}} \leq 3^{-1/4}$ at $V=1$. The carry to infinite VOLUME is a separate statement, $\forall F, m_{\text{hi}}(F) \leq 3^{-1/4}$, an open input supplied here by measurement: the forward read $m_{\text{hi}}(L) = \rho'(1)(L)$ plateaus at $\approx 0.33 < 3^{-1/4}$ across the scaling window $L = 12\text{--}28$. The read layer that turns a configuration into these quantities is developed in the companion paper [E].

1. Introduction

A pure gauge theory is specified by a compact non-abelian group $G = SU(N)$ and a single coupling. Two structural facts characterise the quantum theory.

Existence: a well-defined Hilbert space, a Hamiltonian bounded below, and a set of Euclidean correlation functions satisfying the Osterwalder–Schrader axioms. **Mass gap:** the spectrum of the Hamiltonian has a strictly positive lowest eigenvalue above the vacuum, $\Delta > 0$; equivalently, gauge-invariant correlations cluster exponentially and the static potential confines.

We obtain both from a single foundation. The conventional treatment takes the gauge connection as fundamental and the vacuum as the object to solve for. We take the **observer's screen** as primary (the boundary surface through which the vacuum's gauge-invariant content is read) together with one fact about it: it has finite information capacity. The screen is a **finite aperture**, and a finite aperture has a diffraction limit. The mass gap is set by that limit: the finite resolution cannot host the infinitely-extended massless mode a gapless theory needs, so the screen's predictive excess decays at a positive rate, the smallest nonzero quantum. A scale-free vacuum resolves arbitrarily fine, its excess falling only as a power law, and is gapless; a finite aperture forces a positive rate.

The construction runs one way, from the aperture outward. Perturbative renormalisation already runs this way: the renormalisation character factors forward (Connes–Kreimer's Birkhoff decomposition, the counterterm from the Hopf antipode), and recovering the bare theory from the renormalised value is a torsor under the renormalisation group. The finite-aperture read sits on that forward side: the theory determines the entropy-matched band. The argument is a single reduction of the mass gap to one external law, asymptotic freedom above the entropy floor $\kappa_0 = \frac{1}{4} \ln 3$, on a theorem, with the gap bounded below by the entropy surplus, $\Delta \geq \kappa_0 - \mu > 0$.

The argument has three premises: one axiom and two theorems.

1. **The screen is finite.** The extraction axiom says the screen's capacity is finite, $B < \infty$, equivalently its cells have positive size $\ell_{\text{cell}} > 0$. It is a *causal* screen because $c < \infty$ gives a finite cone of influence, built into the Lorentz structure of the Lagrangian. (§2)
2. **The aperture forms, because the tiles are self-sourcing.** The screen's modes are diffraction-limited tiles; the non-abelian tiles carry the field they source, so the field is forced into the flux-tube bore and the aperture forms. A neutral theory (abelian, or a conformal fixed point) radiates freely, forms no aperture, and is gapless. (§5)
3. **A finite aperture is gapped.** Finite active degrees of freedom give a discrete spectrum with a positive smallest spacing, set by the aperture resolution; the infrared scale descends to $a_{\text{IR}} = 0$ with no interior maximum (the reach-freeze monotone). (§4, §6)

The first is the axiom; the second is the aperture-forming mechanism (§5); the third is the band-limit lemma and the reach-freeze monotone (§4, §6), which express the gap through the self-sourcing contraction. The lattice read confirms premise 3 (§8). The read layer that turns a configuration into the fill fraction φ , the derived feature scale δ_F , and an attenuation is developed and certified in the companion paper [E]. The finite-aperture mechanism itself — the band-limit lemma, the entropy floor and its counting bound, vortex condensation from that count, the single-cell spectral gap, the reflection-positivity carry $\rho'(n) \leq \rho'(1)^n$, and the reconstruction — is proved (§13); the gauge-physics content enters only through named cited results and one measured read.

2. The extraction axiom and the screen

Definition 2.1 (Planck scale). With $G_N, \hbar, c$ empirically given, write $\ell_P^2 := G_N \hbar / c^3$. The screen's cell scale is $\ell_{\text{cell}} = \ell_P$ at saturation.

Axiom 2.2 (The extraction bound). An observer resolves a finite number of degrees of freedom through a local boundary of area A , the capacity linear in the area,

$$B(A) \leq \frac{A}{4\ell_P^2} = \frac{Ac^3}{4G_N\hbar},$$

saturating on a horizon. *Finiteness*, $B < \infty$ (the Bekenstein / generalized-second-law bound), enters the gap argument through $\ell_{\text{cell}} > 0$ (equivalently $G_N < \infty$).

Lemma 2.3 (Screen entropy is Schmidt entropy). Write a bipartite pure state across the screen's cut as a coefficient matrix $S = U\Sigma V^\top$; its singular value decomposition is the Schmidt decomposition, so the screen entropy $H_{\text{screen}} = -\sum_k p_k \log p_k$, $p_k = \sigma_k^2 / \sum_j \sigma_j^2$, is the bipartite entanglement entropy across the cut. Applied to the boundary restriction of a sampled configuration, the screen is an entanglement-entropy estimator with access to the *universal* (subleading) part of the entropy as well as the leading area term. This singular spectrum is the point of contact with the read layer: the fill fraction, the diffraction limit, and the decay rates are all functionals of it.

Claim 2.4 (The screen reads the infrared flow). What the screen reads, and what is verified on the exactly-solvable systems of §4, is the **flow of the universal infrared content to zero**: its derived feature scale $\delta_F(R)$ freezes at $\xi = 1/m$ for a gapped theory and grows with R for a massless one, so the descent $a_{\text{IR}} \rightarrow 0$ that is the gap is read with no fit.

Why the universal part. In $d \geq 2$ the leading area-law term is phase-blind: a gapless conformal vacuum and a gapped one share it. The gap-relevant content lives in the universal part of the entanglement (the F central charge in three dimensions, the a anomaly in four). A gap forces the infrared universal content to vanish, $\Delta > 0 \Rightarrow a_{\text{IR}} = 0$ by clustering; the converse $a_{\text{IR}} = 0 \Rightarrow \Delta > 0$ is the theorem of §6, so the two are equivalent. The results this rests on: Bekenstein (the area bound); Calabrese–Cardy (the universal c -term); Hastings (area-law $\leftrightarrow$ gap in one dimension); Casini–Huerta–Myers (the entropic c/F -theorem); Donnelly–Wall, Ghosh–Soni–Trivedi (gauge edge-mode entanglement).

3. The gauge screen is a finite causal aperture

The screen resolves a finite number of cells of positive size $\ell_{\text{cell}} > 0$, so $B \sim A/\ell_{\text{cell}}^2 < \infty$. A finite causal speed $c < \infty$ makes this a finite *causal* aperture: the bounded opening through which two causally separated regions communicate. $c < \infty$ is measured and intrinsic to a relativistic field theory (the finite cone of causal influence is the Lorentz structure of the Lagrangian). The $B \rightarrow \infty$ idealisation ($\ell_{\text{cell}} \rightarrow 0$) is the continuum, infinite-capacity screen; entropics is the finite-cell side of that same screen, where discreteness, and the gap, live.

The gap requires $\ell_{\text{cell}} > 0$. The aperture is finite precisely when $\ell_{\text{cell}} > 0$, which is the axiom; the limit $\ell_{\text{cell}} \rightarrow 0$ sends $B \rightarrow \infty$, the active content $2^H \rightarrow \infty$, and the diffraction limit closes. Two statements of different strength:

- *Existence.* A finite aperture is **sufficient** for a gap (§4), and finiteness is $\ell_{\text{cell}} > 0$.

- *Value.* For every positive cell size the gap equals the dynamically generated scale Λ , set by asymptotic freedom through $e^{-1/2b_0g^2}$ (§6), the cutoff cancelling. The value is independent of the cell size. In our universe $\ell_{\text{cell}} > 0$, so B is large but finite and the gap sits at the quantum-chromodynamics (QCD) scale.

The conventional formulation lives in the $\ell_{\text{cell}} \rightarrow 0$ corner where the gap is unforced. The physical finite screen ($\ell_{\text{cell}} > 0$, $c < \infty$) forces $a_{\text{IR}} = 0$. The identification of the gauge-flux screen with the finite causal screen that c and ℓ_{cell} define is the one interpretive commitment of the framework.

The reads. A finite opening band-limits, and the resolution is read parameter-free from the configuration's own spectrum. Each gap quantity maps to a bounded optics read:

gap quantity	optics read (symbol)	behaviour vs the gap
mass gap Δ	slowest DMD/Koopman rate $-\log m_1 $	is the gap (free scalar: recovers E_0)
coherence	attenuation $\alpha = \log \frac{\lambda_1}{\max(\lambda_2, \lambda_+)}$	moves opposite the gap
certified coherence interval	Weyl interval $[\alpha_{\text{lo}}, \alpha_{\text{hi}}]$	bounds the contrast
resolved-mode count	spatial modes K_{signal} above the reference null	a rise inside confined $SU(2)$ comparable to the one across the $U(1)$ transition (§8.3)
entropy floor κ_0	closed-surface Shannon entropy density	$\frac{1}{4} \ln 3$
vortex tension μ	$\log(\text{contrast})$ of the disorder weight	$\mu < \kappa_0 \Leftrightarrow \text{contrast} < 3^{1/4}$
string tension σ	bore scaling $\Delta \propto \sqrt{\sigma}$ (coefficient not fixed, §6)	glueball scale
occupied fraction / reach	$\varphi_T = 2^{H_{\text{sv}}} / T \in (0, 1]$; magnification $1/\varphi_T$	fill fraction (bounded by construction)

Here H_{sv} is the Shannon entropy of the ordered-axis correlation eigenvalue spectrum (so φ_T is a fill fraction, bounded in $(0, 1]$), λ_+ the singular-value noise edge of a signal-free confined-vacuum reference, and m_1 the dominant magnitude of the streaming DMD/Koopman operator. K_{signal} counts the SVD modes standing above λ_+ , with λ_+ calibrated at the plane shape it is applied to; on single 8×8 spatial planes of the action density it reads 0.0597–0.0868 across $SU(2)$ and both phases of compact $U(1)$ (§8.3). The mass gap Δ is the slowest decay rate of the DMD/Koopman operator, verified to recover a known gap on the free scalar. The confinement inequality $\mu < \kappa_0$ is then an optics statement, **attenuation** < **entropy**: confinement is where the disorder's entropy outruns its attenuation.

Two independent axes. The screen derives a separate matched scale on each axis from *that axis's own* Shannon entropy, H_F and H_T computed independently, so the feature resolution δ_F and the time correlation δ_T are independent reads. The gap lives on the time axis. Neither axis inherits the other's scale, so a resolved gap is the signal's own.

The read is lossy. A finite opening reads a finite band and discards the rest, so the read is a projection: on a field $W \in \mathbb{C}^{T \times F}$ it whitens per channel, area-folds the feature axis to $F_{\text{eff}} = \text{round}(2^{H_F})$ cells, takes the SVD, and keeps the K modes above the noise floor, dropping the whitening parameters, the fold kernel, and the sub-floor bulk. The preimage $E^{-1}(E(W))$ is a positive-dimensional fibre, $\dim = T(F - K) + O(1)$, on which the read is constant. The forward read is a determined computation; its inverse is that fibre. The gap is read forward (§4–§6); the continuum reconstruction is the inverse pass (§12).

4. A finite aperture is gapped

Lemma 4.1 (Band-limit lemma, spectral form). Let a self-adjoint operator, bounded below, have finite *active content*: an effective mode count $2^{H_T} \leq 2^B < \infty$. Then its spectrum is discrete with a strictly positive smallest level spacing, and that spacing — the gap — is the diffraction limit of the finite aperture, $\Delta = (\text{shape factor}) \times a_\delta$, with a_δ the entropy-width read of the time-axis decay.

Proof of the finite-rank content. If the active content is finite, the operator restricted to the active subspace is a finite-dimensional self-adjoint matrix; it has a discrete spectrum $\{E_0 < E_1 < \dots\}$ with a finite, strictly positive smallest spacing. $\square$

The dichotomy in spectral form. Read through the operator, the autocorrelation is a finite exponential sum $C(\tau) = \sum_{k=1}^F P_k m_k^\tau$. A finite sum has two fates: a mode on the unit circle ($|m_k| = 1$, a persistent massless mode, no decay) or a strict margin ($\max_k |m_k| < 1$, exponential decay at rate $-\log|m_1| > 0$, the gap). A gapless power law $\sigma(n) \sim n^{-p}$ requires a genuine continuous spectrum, infinitely many modes, and cannot occur while the aperture is finite: the finite exponential sum admits no power-law middle. The $F \rightarrow \infty$ uniformity reduces to the spectral radius staying **intensive**, one $r < 1$ bounding the dominant mode at every dimension; the radius is intensive at the solvable point ($\|m_1\| = \tanh b < 1$, size-independent). The mechanism is a **dimension-free floor**: the centre-vortex counting floor $\kappa_0 = \frac{1}{4} \ln 3$ (§7) is a per-area entropy, coupling-free and dimension-free, so it does not dilute as modes are added. It is the disorder-count twin of the intensive spectral radius: the counting floor in the discrete surface count, the intensive margin $r < 1$ in the continuous spectral density. That the radius stays intensive as $a \rightarrow 0$ is the same continuum input as $\mu < \kappa_0$ (§8.5, §11).

The non-trivial content is *which* modes are active and the identification of the gap with the matched scale. The second is the diffraction limit of a finite aperture: a resolution scale times a constant fixed by the aperture's shape (the "1.22" of a circular aperture is the first zero of J_1). The shape factors are:

profile	uniform	thermal (exponential)	half-normal	Gaussian
shape factor	1	e	$\sqrt{\pi e/2} \approx 2.07$	$\sqrt{2\pi e} \approx 4.13$

A thermal ladder of gap Δ has exponential decay, shape factor e , so $a_\delta \rightarrow \Delta/e$, while the active count $2^{H_T} \rightarrow \infty$ as $\Delta \rightarrow 0$: the spectral aperture opens iff the gap closes. The read takes a_δ from the signal's own decay, and the per-mode DMD rates supersede the entropy-width estimate. Deterministic system identification (§8.6) fixes both reads on controlled inputs: the diffraction limit obeys the reciprocal-length law $a_\delta \xi = \text{const}$, and the gap read returns $-\log|m_1|$.

The diffraction regime is monotone. The Fresnel number $N_F = a^2/(\lambda s)$ ($\lambda = 1/\Delta$) is monotone in the probe scale, and the flow runs from ultraviolet (near field) to infrared (far field) one way, so the screen crosses the focus $N_F \sim 1$ once and does not return: the *no interior maximum* is the monotonicity of the Fresnel number.

What monotonicity leaves is that the far-field pattern is *trivial* ($a_{\text{IR}} = 0$) rather than scale-invariant, the single endpoint question the self-sourcing tiling settles (§5, §6).

The conformal-field-theory (CFT) sharpening. A conformal vacuum has area-law entanglement and satisfies the causal bound: it is a finite aperture in the *leading* sense, yet gapless. For a CFT the *causal* aperture is finite while the *spectral* aperture is infinite ($2^{H_T} \rightarrow \infty$); what links them is $a_{\text{IR}} = 0$. So the two premises are distinct and both necessary: $c < \infty$ is necessary but not sufficient (the CFT shows it), and asymptotic freedom (§5) supplies sufficiency by forcing the universal part to vanish so the spectral aperture closes.

The four-dimensional step. In $d \geq 2$ the gap-relevant content is the universal part, so the finite aperture acts there: finiteness of the aperture in the relevant sense is $a_{\text{IR}} = 0$. Two of three ingredients are theorems: the universal F/a -function is monotone with the right endpoints (entropic c/F -theorem, Casini–Huerta–Myers) and $a_{\text{UV}} \geq a_{\text{IR}}$ (Komargodski–Schwimmer). The third is the four-dimensional form of the band-limit lemma: the finite causal aperture gives $a_{\text{IR}} = 0$ once the self-sourcing contraction holds, the reach-freeze monotone supplying the convergence and the contraction the rate (§6), and the lattice read measuring it on every gauge group (§8). The bridge, on the screen's two axes:

$$\Delta > 0 \text{ (time aperture finite)} \iff a_{\text{IR}} = 0 \text{ (feature aperture finite)}.$$

Forward is immediate (a gap gives exponential clustering, no massless infrared content). Backward is the decay-rate dichotomy of §6: in the screen variables $a_{\text{IR}} = 0$ is exponential clustering $\sigma(n) \sim e^{-mn}$, the gap. The single input is that the finite self-sourcing screen forces the excess to contract, $c > 0$, for pure $SU(N)$.

The bore is this bridge in gauge variables. Confinement makes the feature aperture finite, the flux tube acquiring a finite transverse width $a \sim 1/\sqrt{\sigma}$ ($\sigma > 0$); a finite transverse bore band-limits the transverse modes into a time-axis gap $\Delta \geq c/a > 0$, $c > 0$ an $O(1)$ profile constant (§6). "Feature aperture finite $\Rightarrow$ time aperture finite" is, physically, "finite tube width $\Rightarrow$ transverse gap."

5. Asymptotic freedom forms the aperture

Premise 2 is why the aperture forms for $SU(N)$ and not $U(1)$. The one-loop Callan–Symanzik coefficient is $b_0 = 11N/3 > 0$ (Gross–Wilczek, Politzer): the coupling runs strong in the infrared. A strongly coupled gauge flux cannot spread into the bulk (the energy cost grows with volume), so it **binds onto a surface**: the Wilson loop obeys an area law, the flux is a tube, the theory confines. This surface is the screen forming. The abelian theory does not run; its flux spreads as a Coulomb field; no aperture forms and the photon is massless.

The aperture forms $\iff$ the flux is surface-bound $\iff$ the theory confines, and asymptotic freedom is the driver. The controlled foil is the conformal window: adding enough massless matter screens the running, a Banks–Zaks infrared fixed point appears, and the theory is conformal (gapless). The fixed point exists only when the two-loop coefficient b_1 turns negative, which requires matter; pure $SU(N)$ ($N_f = 0$, $b_1 = +\frac{34}{3}N^2 > 0$) sits far below the window's lower edge ($N_f \approx 8\text{--}12$), deep in the surface-bound region, the flux staying bound.

Self-sourcing tiles. The screen's modes are diffraction-limited tiles. Each adjoint mode carries the field it sources, so the field cannot spill into the vacuum as free radiation; it is forced into the bore, and the pressure of the field confined there is the gap, its value the bore $\Delta \propto \sqrt{\sigma}$. The single neutral $U(1)$ tile sources no field it carries: it radiates freely, feels no such pressure, and is gapless. The discriminator is geometric, the self-overlap of the tiling, and it separates the two theories before any dynamics. This is the origin of the contraction constant $c > 0$ of §6.

6. The mass gap

Assembling §3, §5, §4: the screen is a finite aperture ($\ell_{\text{cell}} > 0$, $c < \infty$); asymptotic freedom binds the flux onto it ($b_0 > 0$, no matter); a finite aperture is gapped. The conclusion is $\Delta > 0$ for pure $SU(N)$ once the self-sourcing contraction $c > 0$ holds; the abelian theory and the $\ell_{\text{cell}} \rightarrow 0$ limits form no aperture and are gapless.

The gap is the decay rate of the excess.

Theorem 6.1 (The reach-freeze monotone and the decay-rate dichotomy). Read the screen field along the time axis as a stationary sequence; let $s(n) = H(A_n | A_1, \dots, A_{n-1})$ be its entropy rate, $s_\infty = \lim_n s(n)$, and $\sigma(n) = s(n) - s_\infty \geq 0$ the excess predictive information. Then $\sigma(n)$ is non-increasing and $\sigma(n) \rightarrow 0$ (conditioning reduces entropy, stationarity; Cover–Thomas Thm 4.2.1), and

$$\sigma(n) \sim e^{-mn} \implies a_{\text{IR}} = 0, \Delta = m > 0; \quad \sigma(n) \sim n^{-p} (p \leq 1) \implies a_{\text{IR}} > 0, \text{ gapless}.$$

The monotone fixes convergence, $\sigma(n) \rightarrow 0$, for both fates alike; the discriminator is the per-step decay rate. A uniform contraction $\sigma(n+1) \leq (1-c)\sigma(n)$ with $c > 0$ gives exponential decay, $a_{\text{IR}} = 0$, and $\Delta \geq c$. $\square$

The contraction $c > 0$ is the input Reduction 6.2 takes, sourced from the self-sourcing tiling: a self-sourcing tiling cannot tile at any scale, so each step scrambles a fixed positive fraction of the new cell's predictable content against the bulk. A neutral tiling scrambles nothing ($c = 0$) and keeps the power-law tail.

Reduction 6.2 (The mass gap from the contraction). If the finite self-sourcing screen of pure $SU(N)$ supplies a positive per-step contraction $\sigma(n+1) \leq (1-c)\sigma(n)$, $c > 0$, then $a_{\text{IR}} = 0$ and $\Delta \geq c > 0$. A contraction $c > 0$ makes σ decay exponentially, so $a_{\text{IR}} = 0$; a finite aperture with $a_{\text{IR}} = 0$ is band-limited (§4), so $\Delta \geq c > 0$. The self-sourcing tiling supplies the contraction (§5); the confinement it rests on is $\mu < \kappa_0$, established at the strong-coupling end ($\beta < \beta_* \approx 0.75$), given from asymptotic freedom at the weak end (§8.2), above the floor $\kappa_0 = \frac{1}{4} \ln 3$ (§7), and measured across the crossover (§8.3). The abelian foil tiles exactly, $c = 0$, $\Delta = 0$.

The gap constant, a -uniform. The contraction and the gap are bounded below by one explicit constant, the free-energy margin below the floor. The mode bound $\|m_k\| \leq e^{-(\kappa_0 - \mu)}$ — equivalently the spectral margin $r = e^{-(\kappa_0 - \mu)} < 1$ — follows from reflection positivity ($\rho'(n) \leq \rho'(1)^n$) and the single-plaquette magnitude $\rho'(1) = m_{\text{cell}} \leq 3^{-1/4}$, the reach-freeze contraction and the gap clearing the same margin,

$$\Delta \geq \kappa_0 - \mu(\beta), \quad \kappa_0 = \frac{1}{4} \ln 3.$$

In the eigenvalue form of the read (§8.6), $\mu = \log(\text{contrast})$ with $\text{contrast} = \lambda_1/\lambda_+$ the leading correlation eigenvalue over the noise floor, so the margin is $\kappa_0 - \mu = \log(3^{1/4}/\text{contrast})$, positive exactly when the eigenvalue stays below $3^{1/4}\lambda_+$. Uniformly, $\Delta(a) \geq \Delta_0 := \kappa_0 - \sup_{a,\beta} \mu > 0$, and Δ_0 carries no factor of a : κ_0 is the dimension-free counting floor (§7) and the bounding margin is the **intensive** spectral radius, one $r < 1$ at every dimension (§4), so the lower

bound does not dilute as modes are added or as $a \rightarrow 0$. That $\Delta_0 > 0$, i.e. $\sup_a \mu(a) < \kappa_0$ with a definite margin, is the a -uniform confinement. Here Δ is the per-step decay rate in nat units; the physical gap $m_{\text{phys}} = \Delta/a$ is scale-covariant through the read (§8.5), so a -uniform positivity of the physical gap is A1's continuum part. The a -uniformity is in two parts: into the continuum $\beta \rightarrow \infty$ the tension vanishes ($a^2 \mu \sim (a\Delta)^2 \rightarrow 0$), so the margin approaches the full floor; across the crossover the tension stays sub-floor from a **finite correlation length** $\langle d^2 \rangle \leq B$, $\mu = -\log \langle \cos \theta \rangle_\rho < \kappa_0$ following from the bounded correlation moment through the analytic step and the $1/L^2$ aperture scaling, so the margin grows $\propto L^2$ (measured $\langle d^2 \rangle = O(1)$; §9).

What Δ is, spectrally. Δ is the lightest gauge-invariant excitation above the vacuum, the 0^{++} glueball mass. Confinement gives the flux tube a tension $\sigma > 0$ (area law); the minimal gauge-invariant state is the closed flux loop constricted to the bore, of energy $\Delta \propto \sqrt{\sigma}$. The lattice measurements separate the two questions. The Wilson-loop area law measures the string tension $\sigma > 0$ (§8.8); the *phase*, not the glueball mass. The gap *value* then follows from σ through the bore, and the direct gap instrument is the slowest DMD/Koopman rate, calibrated on the free scalar where the field carries its gap directly (§8.5). The area law decides *whether* there is a gap ($\sigma > 0$); the bore relation fixes the **scaling** of what it is, $\Delta \propto \sqrt{\sigma}$. It does not fix the coefficient, so none is quoted: the scale the measurement compares against is the π -free $3.59\sqrt{\sigma}$ (§8.8).

The forgetting property. The property the reduction rests on, "the flow forgets," is read straight off the screen: the Cesàro quadratic mean of the autocorrelation vanishes, $\frac{1}{N} \sum_{\tau < N} |C(\tau)|^2 \rightarrow 0$ (call it Λ_{Ces}). At a finite aperture $C(\tau) = \sum_k P_k m_k^\tau$ is a finite exponential sum, and Λ_{Ces} is one property with several faces:

face	statement	read
(i) Λ_{Ces}	$\frac{1}{N} \sum_{\tau < N} C(\tau) ^2 \rightarrow 0$	the screen forgets
(ii)	$C(\tau) \rightarrow 0$	the reach-freeze monotone (§6)
(iii) margin	$\max_k m_k < 1$	DMD dominant magnitude below one
(iv)	$ C(\tau) \leq (\sum P) r^\tau, r < 1$	exponential decay
(v)	$\sum_\tau C(\tau) < \infty$	finite correlation length
(vi)	$a = 2^{-h} > 0$	positive aperture gauge

The forward cycle (iii) $\Rightarrow$ (ii) $\wedge$ (v) $\wedge$ (i) (a spectral margin gives decay, a finite correlation length, and forgetting) holds; the abelian foil, a persistent unit-circle mode ($|m_k| = 1, P \neq 0$), fails Λ_{Ces} (its Cesàro mean is $|P|^2 > 0$). The converse (i) $\Rightarrow$ (iii) is the finite Wiener mean-square step, $\frac{1}{N} \sum_{\tau < N} |C(\tau)|^2 \rightarrow \sum_{|m_k|=1} |P_k|^2$, so Λ_{Ces} forces every weight-carrying mode off the unit circle; it connects the forgetting the screen measures to the spectral margin, the aperture form of $\mu < \kappa_0$.

The interior, from every side. An interior fixed point is a non-summable power-law tail $\sigma(n) \sim n^{-p}$ ($p \leq 1$); the contraction $c > 0$ excludes it. At weak coupling the two-loop beta function $\beta = -b_0 g^3 - b_1 g^5$ has both coefficients positive and scheme-independent, so $\beta < 0$ for every $g > 0$; at strong coupling the Osterwalder–Seiler gap holds. In between, the interior at finite spacing reduces to finite-volume analyticity and a finite grid cover, both open (§9); the unbroken centre Z_N one-form symmetry ('t Hooft) forbids the Coulomb/free-Maxwell fixed points, consistent with the interior input A1.

The spatial-bore face. The centre-vortex condensation that confines is a condensation of the chromo-flux into a tube of finite transverse width a , and a finite bore gaps every field: $\omega^2 = k_\parallel^2 + k_\perp^2 \geq (c/a)^2$ with $c > 0$ an $O(1)$ constant fixed by the tube's transverse profile (the hard-wall instance is $c = \pi$, its first Dirichlet mode). If the centre-vortex tension obeys $\mu(\beta) < \kappa_0$ at every β , then $F_v = \mu - \kappa < 0$, the vortices condense, the flux confines to a tube of width $a \sim 1/\sqrt{\sigma}$, and $\Delta \geq c/a > 0$ for any such c -- the conclusion is positivity, which no choice of c affects; the conformal vacuum, having $\sigma = 0$ and $a = \infty$, cannot occur. The input is the single inequality $\mu < \kappa_0$. This is the *same* interior inequality in a third face. The bore is a **positivity** statement: a finite transverse width gaps the field, and that is all it supplies -- it fixes no coefficient. The scale comes from the entropy surplus $\Delta \geq \kappa_0 - \mu$, with $\kappa_0 = \frac{1}{4} \ln 3$ derived from the counting floor (§7) and μ read from the configuration (§8.3), which imports nothing.

The value of the gap is the dynamically generated scale $\Lambda = \Lambda_{\text{UV}} e^{-1/2b_0 g^2}$, the cutoff cancelling. It is cutoff-independent because asymptotic freedom makes the ultraviolet modes free, hence structureless, hence below the noise edge (§3): the matched scale is read only from the signal above that edge, a fixed infrared window independent of how many free ultraviolet modes sit below it.

7. The entropy floor

Confinement is the condensation of the centre-vortex disorder operator. A vortex worldsheet of area A enters with action weight $e^{-\mu A}$ and multiplicity $e^{+\kappa A}$, so its free-energy density is $F_v = \mu - \kappa$; the vortices percolate and confine exactly when $F_v < 0$, i.e. $\mu < \kappa$. The entropy side of this inequality is a *counting* theorem.

Theorem 7.1 (Entropy floor). On the 4D hypercubic lattice, let $N(A)$ be the number of closed connected centre-vortex surfaces of area A (in plaquettes) through a fixed plaquette. Then, for $A = 4n + 2$,

$$N(A) \geq 3^{(A-2)/4-1}, \quad \kappa := \limsup_{A \rightarrow \infty} \frac{\ln N(A)}{A} \geq \kappa_0 := \frac{1}{4} \ln 3 \approx 0.275 > 0,$$

and this entropy density is independent of the coupling β .

Proof. Exhibit an exponentially large sub-family. (1) Boundaries of cube-paths are closed vortex surfaces: fix a 3D sublattice, take a self-avoiding cube-path $c_1, \dots, c_n$; its boundary ∂C is a closed plaquette surface carrying centre flux. (2) Area: a single cube has 6 faces, each added cube contributes 6 and removes 2, so $A = 6n - 2(n-1) = 4n + 2$, $n = (A-2)/4$. (3) Directed cube-paths (steps only in $+x, +y, +z$) are automatically self-avoiding (the coordinate sum strictly increases) and number 3^{n-1} . (4) Distinct paths give distinct surfaces (each step raises the coordinate sum by one, so the cubes carry distinct sums $0, \dots, n-1$; sorting by sum recovers the path, so $\text{path} \mapsto \partial C$ is injective). Hence $N(A) \geq 3^{n-1} = 3^{(A-2)/4-1}$ and $\kappa \geq \frac{1}{4} \ln 3$. ■

Every step of Theorem 7.1 holds at the level of the surfaces. Step (4) holds for the SURFACES, and for the cube configurations as well: the boundary map ∂ is injective on them, each cube being recovered from the boundary through the face its step crosses. So there are 3^k distinct surfaces, each of area exactly $4k + 6$. **And they are CLOSED** — step (1): **every** edge lies in an EVEN number of the boundary's faces, which is $\partial\partial = 0$ over Z_2 and is what a centre-vortex surface needs. This holds with no interior restriction and on every placement of the configuration in $\mathbb{N}^3$. The proof is a double count. Fix an edge and count the incident (cube, face) pairs

of C whose face carries it, two ways. By cube, each contributes an even number: a face spans the two axes its normal is not, so the faces of one cube carrying an edge along c are the two with normal $\text{rot}_1 c$ and the two with normal $\text{rot}_2 c$, and of each pair exactly one carries it, because the pair sits at opposite ends of that normal. By face, each contributes its owner count, which is one or two — so modulo two the face sum counts exactly the ONE-owner faces, and those are ∂C by definition. Equivalently: ∂C is the Z_2 sum of the cubes' own boundaries, so $\partial\partial C = \sum_x \partial\partial x$, and each single cube's boundary is closed. Summing over cubes is what removes the case analysis: a cube that is absent simply does not appear in the sum, whereas summing around an edge would need the four cubes surrounding it to exist, which fails against the floor of $\mathbb{N}^3$.

And every one of them passes through the same face: the three low faces of the origin cube are on the boundary of every directed path's configuration, because in $\mathbb{N}^3$ no cube sits one step back from the origin, so such a face has exactly one owner. That is the "through a fixed plaquette" of the theorem's statement.

And they are CONNECTED, the fourth property, which carries weight. Without it the family of "closed surfaces of area A through a fixed plaquette" admits a union of unit-cube boundaries, closed because every link lies in two or four of them, whose pieces may sit anywhere in the box; at area $4k + 6$ that is one anchored piece plus about $2k/3$ floating ones in a box of about k^4 sites, so $\ln N(A)/A$ would grow like $(\ln k)/2$ and no fixed bound on the free-energy density could exist. The proof is a descent on the path's own index: a directed path visits exactly one cube at each coordinate sum $0, \dots, k$, so a cube's neighbour in the configuration can only be its predecessor or its successor, and at most TWO of its six faces are off the boundary — the one it was entered through and the one it leaves by. Two faces of one cube with different normals always share an edge, so at most two hops reach a chosen low face; and the low faces of consecutive cubes on any axis other than the step between them share an edge, which is the rung. At least one rung always exists, because two axes are not the entry axis and at most one of those is the previous entry axis.

The four properties are then a DEFINITION. The vortex family is the set of closed, connected plaquette surfaces of area $4k + 6$ through the fixed plaquette, and $3^k \leq N$ holds against that definition **with no hypothesis at all** — the box $k + 3$ being the successor of the largest coordinate the construction produces. So $N(A)$ in the theorem above is a defined object and the bound on it is a closed theorem. $N(A)$ is a CARDINALITY, and a cardinality needs a finite family, so "closed" is stated over the four-dimensional plaquettes inside a box (a 4-D edge type and its incidence; every plaquette of an embedded surface lies in the time-zero slice, so the correspondence with the 3-D edges is one-to-one and a time-direction edge meets none of them). With that the family is a filter of a powerset and the bound holds with no containment hypothesis. Step (2), the area: with the boundary of a cube configuration defined as the faces owned by exactly one of its cubes, $|\partial C| = 4k + 6$. It is a double count — $6(k + 1)$ incidences by cube, the same total by face, where each face has one or two owners — closed by identifying the shared faces exactly: two cubes with a common face are one step apart, so by the coordinate sum only consecutive cubes of the path can share, and consecutive cubes share exactly the face the step crosses. Hence the shared faces number exactly k and $|\partial C| = (6(k + 1) - k) - k = 4k + 6$.

This is what makes κ_0 constant-free. The $\ln 3$ is the branching of a directed cube-path and the $\frac{1}{4}$ is the area per step, one theorem each; nothing in the floor is fitted or chosen.

The floor is positive and β -independent: κ_0 is pure lattice geometry, **a Shannon entropy density**. The directed sub-family gives $\kappa_0 = \frac{1}{4} \ln 3$ as a lower bound on the per-unit-area entropy density $\lim_A \frac{1}{A} \ln N(A)$ of the closed-surface ensemble, generated with the same entropy primitive the framework uses everywhere: three uniform continuations carry $\ln 3$ across four plaquettes. The strong-coupling end of $\mu < \kappa_0$ is shut by the convergent cluster expansion (§8.2); the interior crossover is measured by the lattice read (§9).

8. The lattice measurement on the groups

The construction is confirmed by direct lattice measurement of the deciding quantity, read off equilibrium configurations by the framework primitive. Monte Carlo generates the configurations (input only); the gap read is the entropy-matched aperture read, with no fitted rate.

"Equilibrium" is checked per ensemble. Every shipped ensemble is compared against an independently generated reference chain, one comparison per (group, L, T, β , collection, field) — 105 in all, which is finer than the release's (group, L, β) count. Two properties of that check set how much the comparison is worth. First, a reference measured from ONE start is biased in a direction no single chain reveals: a hot (random) configuration relaxes toward equilibrium from above and a cold (ordered) one from below, and a chain still descending has a flat-looking tail either way. Measured against **hot-start-only references**, all eight shipped $U(1)$ ensembles at $\beta \geq 1.4$ read below the reference (sign test $p = 0.0039$), while the heat-bath groups are balanced (SU(2) 27 of 58, SU(3) 7 of 16): the $U(1)$ Metropolis proposal width is fixed at 1.0 radian for every β , so acceptance falls (0.512 at 1.6, 0.420 at 2.5), relaxation slows, and the hot chain has not arrived. Running both starts at $\beta = 1.6$ and $\beta = 2.5$, the two are still 3.2σ and 5.9σ apart after 20000 sweeps and the shipped ensemble sits INSIDE each interval: at both couplings the shipped data are right and the single-start reference is biased. The reference is therefore **bracketed** — the value is the midpoint and the uncertainty carries the half-gap, so a bracket that has not closed widens the error — and each comparison records whether it rests on a bracketed reference or on a hot-start-only one.

The bracket removes the effect the one-start bias produced, which checks that it is the right reference. With all twenty $U(1)$ references measured as brackets, the same sign test reads **4 of 8** at $\beta \geq 1.4$ ($p = 0.64$) against 8 of 8 for hot-start-only references, and the ensemble that sits 8.5σ below its hot-start reference sits 0.1σ from the bracket's midpoint. The largest deviation on the level comparison is 2.2σ against brackets and 8.5σ against hot-start references; the largest anywhere in the table is 2.7σ , on a drift row. The shipped configurations are the same in both comparisons. Second, the operator-reduced collections have no action-density reference to be compared against — they are smeared-plaquette amplitudes — so they carry a **drift** verdict instead, which asks whether a chain's two halves agree. That is weaker evidence than a level comparison, because a chain settled in the wrong place does not drift, and each verdict records which of the two tests produced it, so the counts are kept separate.

8.1 The screen construction. The gauge configuration is a 4D field; the confinement read lives on its **2D spatial correlation**, so the reduction to the screen keeps each spatial plane intact. The local gauge-invariant action density $\phi(x) = \sum_{\mu < \nu} (1 - \frac{1}{N} \text{Re tr } U_{\mu\nu}(x))$ on an $L^3 \times L_t$ lattice is read plane-by-plane: each $L \times L$ spatial plane is whitened, its resolved-mode count K_{signal} is taken, and the result is averaged over planes. The reduction keeps the per-plane spatial correlation, which a flattening of the spatial volume into one feature axis destroys. The entropy-matched fold is the only resolution operator, and the singular-value noise floor is the **calibrated confined-vacuum reference null**: a signal-free deeply-confined ensemble ($SU(2)$ at $\beta = 0.5$) sets the disorder floor, calibrated per cut point at the read's own granularity and at the plane shape it is applied to. The reference null is analytic and $O(1)$ (center + $z(\text{far})$ scale), carrying the disorder scale of the theory, with no substrate-fitted constant.

The coupling convention, stated once. Every β quoted in this paper and carried in the released ensembles is the **standard Wilson coupling**, $S = \beta \sum_x \phi(x)$ with the plane sum above running over $\mu < \nu$ — six planes per site in four dimensions — and, for $SU(N)$, $\beta = 2N/g^2$. That is the normalisation of every source a number here is read against: the strong-coupling character expansions (Montvay–Münster; Creutz 1980), the two-loop lattice β -function used for $a\Lambda_{\text{lat}}$, the

Osterwalder–Seiler tension bound of $\$8.2$, the compact- $U(1)$ critical coupling $\beta_c \approx 1.011$, and the established string tensions of $\$8.8$. The ensembles are held to it directly at both coupling ends — the leading plaquette $\beta/2$, $\beta/4$ and $\beta/18$ for $U(1)$, $SU(2)$ and $SU(3)$, and the one-loop slope $\beta(1 - P) \rightarrow (N^2 - 1)/4$. A plaquette indexed by an **ordered** pair of directions sums over (μ, ν) and (ν, μ) alike; the two orderings traverse the same loop in opposite senses, so their holonomies are mutual inverses and their Re tr is equal, and every plane is counted twice. The coupling of that ordered-pair action is consequently $\beta/2$: its Boltzmann weight at $\beta/2$ is the standard Wilson weight at β , and the two-loop spacing is read at the same standard coupling, so every statement pairing the spacing with the Wilson family measures lengths in that family's own lattice units.

8.2 The two coupling ends. The action side $\mu < \kappa_0$ holds at both ends. At strong coupling the convergent cluster expansion gives the tension bound $\mu \leq 2\beta I_2/I_1$ directly (Osterwalder–Seiler; the closed-form chessboard bound, convergent for $\beta < \beta_{\text{KP}} \approx 0.97$). That bound sits below the counting floor for $\beta < \beta_*$, where $2\beta_* I_2/I_1 = \kappa_0$. The threshold is **derived analytically**: the I_2 and I_1 series terms differ by one factor of $x/2$ upstairs and one factor of $k + 2 \geq 2$ downstairs, so $I_2/I_1 \leq \beta/4$ termwise; the character bound is then $\beta^2/2$, which sits below the floor exactly when $\beta^2 < \frac{1}{2} \ln 3$, i.e. $\beta < 0.74115 \dots$. That is an inequality between the coupling and the floor's own $\ln 3$, with no numeral and no computed certificate. At weak coupling asymptotic freedom makes the vortex tension a physical scale, $a^2 \mu \sim (a\Lambda)^2 \rightarrow 0$, far below the floor; this limit is **computed** (§8.6): the free-field tension is $\mu_\infty(L) \approx 2.1/L^2 \rightarrow 0$ with $\mu_\infty(8) = 0.0326 < \kappa_0$, N -independent. The two groups differ by the *sign of the running*: for $SU(N)$, $a^2 \mu \sim \Lambda^2$ shrinks; for compact $U(1)$, the monopole action $a^2 \mu \sim \beta$ grows. **That difference of sign leaves both groups below the floor, and $\mu < \kappa_0$ holds in both phases** — $\$8.6$ measures the same functional on 20 compact- $U(1)$ couplings spanning $\beta_c \approx 1.011$ and finds every coupling below the floor, the largest tension anywhere being 0.0854 against $\kappa_0 = 0.2747$. The read locates β_c to about 1% by where μ peaks, and its magnitude stays far below the floor throughout. What separates the phases is the geometric discriminator of §5. μ has a clear interior maximum on the $U(1)$ foil — that is what locates β_c — rising to 0.0854 at $\beta = 1.00$ and falling back to ≈ 0.036 by $\beta = 1.5$, and it clears κ_0 by a factor of 3.2 at its own peak.

8.3 The resolved-mode count, measured. K_{signal} is the number of singular modes of a whitened spatial plane of the action density standing above the confined-vacuum reference null (§8.1), averaged over the planes of a configuration and then over the ensemble. The null is calibrated at the plane shape it is applied to: every read in the table is of 8×8 planes, against the $SU(2)$ $\beta = 0.50$ reference on 8×8 planes. Each value carries its same-marginal control, the same configurations with the values of every plane permuted within that plane. The permutation keeps each plane's one-point distribution exactly and removes its spatial arrangement, so the difference between the two reads, taken per configuration, is the part of K_{signal} carried by spatial arrangement. On equilibrium configurations, each value $\pm$ one standard error of the ensemble mean, with the phase as the literature places it (compact $U(1)$ deconfines at $\beta_c \approx 1.011$; $SU(2)$ is confined at every coupling):

group	β	lattice	K_{signal}	within-plane shuffled	difference	phase
$SU(2)$	0.50	$8^3 \times 16$	0.0598 ± 0.0024	0.0562 ± 0.0028	$+0.0037 \pm 0.0041$	confined
$SU(2)$	2.30	$8^3 \times 16$	0.0722 ± 0.0020	0.0607 ± 0.0019	$+0.0115 \pm 0.0027$	confined
$U(1)$	0.80	$8^3 \times 16$	0.0662 ± 0.0032	0.0664 ± 0.0034	-0.0002 ± 0.0046	confined
$U(1)$	0.95	$8^3 \times 16$	0.0701 ± 0.0035	0.0684 ± 0.0040	$+0.0018 \pm 0.0059$	confined
$U(1)$	1.00	$8^3 \times 16$	0.0868 ± 0.0036	0.0602 ± 0.0027	$+0.0265 \pm 0.0044$	confined
$U(1)$	1.05	$8^3 \times 16$	0.0817 ± 0.0031	0.0571 ± 0.0032	$+0.0246 \pm 0.0047$	Coulomb
$U(1)$	1.20	$8^3 \times 16$	0.0705 ± 0.0033	0.0662 ± 0.0033	$+0.0042 \pm 0.0050$	Coulomb
$U(1)$	2.50	$8^3 \times 16$	0.0760 ± 0.0019	0.0634 ± 0.0020	$+0.0126 \pm 0.0026$	Coulomb

The reference ensemble itself, $SU(2)$ at $\beta = 0.50$, reads 0.0598 ± 0.0024 . $SU(2)$ reads 0.0597 – 0.0638 at $\beta \leq 1.8$ and 0.0671 – 0.0737 at $\beta \geq 2.3$; compact $U(1)$ reads 0.0615 – 0.0701 at $\beta \leq 0.95$ and 0.0705 – 0.0762 at $\beta \geq 1.1$. Between those coupling ranges the inverse-variance-weighted mean of K_{signal} rises by $+0.0092 \pm 0.0017$ inside the confined phase of $SU(2)$, comparable to the $+0.0093 \pm 0.0015$ across the $U(1)$ transition; on $SU(2)$ the rise is gradual, with $\beta = 2.0$ and 2.2 reading 0.0669 and 0.0695 . The reads are deterministic, taken on raw configurations against the pinned null, one read per configuration, no fit.

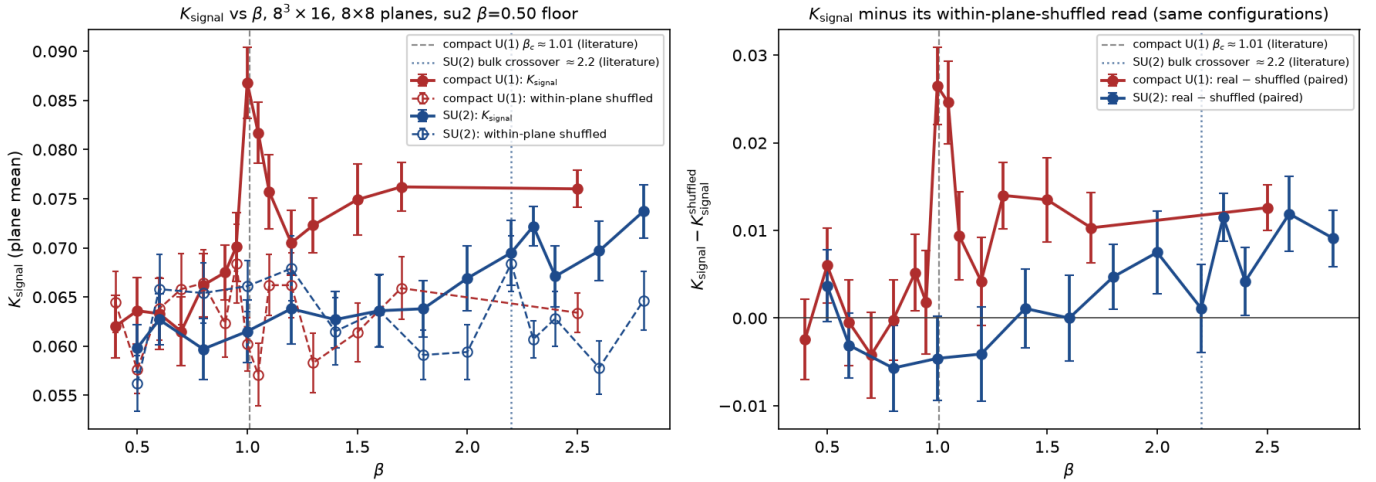


Figure 1. K_{signal} and its same-marginal control on $8^3 \times 16$. **Left** (" K_{signal} vs β , $8^3 \times 16$, 8×8 planes, su2 $\beta=0.50$ floor"): K_{signal} for compact $U(1)$ ($\beta = 0.4$ – 2.5) and $SU(2)$ ($\beta = 0.5$ – 2.8), each beside its within-plane-shuffled read (open markers); every read lies in 0.056 – 0.087 . **Right** (" K_{signal} minus its within-plane-shuffled read (same configurations)"): the paired difference per configuration; $U(1)$ reaches $+0.0265 \pm 0.0044$ at $\beta = 1.00$ and $+0.0246 \pm 0.0047$ at $\beta = 1.05$, beside the compact- $U(1)$ transition $\beta_c \approx 1.011$, and elsewhere both groups read -0.006 to $+0.014$. The dashed and dotted lines mark the literature $U(1)$ β_c and the $SU(2)$ bulk crossover ≈ 2.2 . Read on raw configurations against the pinned null, no fit.

The $U(1)$ bands, read by K_{signal} . Read as an ensemble, the per-configuration K_{signal} values are draws decorrelated by a fixed sweep gap, and the standard error of the ensemble mean shrinks as $1/\sqrt{n_{\text{configs}}}$. Over the compact- $U(1)$ sweep ($\beta = 0.40$ – 1.70 , $n = 48$ per coupling, $n = 128$ at $\beta = 1.40$ and 1.60) the confined phase occupies 0.0615 – 0.0868 and the Coulomb phase 0.0705 – 0.0817 : the confined ceiling 0.0868 ± 0.0036 ($\beta = 1.00$) lies above the Coulomb floor 0.0705 ± 0.0033 ($\beta = 1.20$), each $\pm$ one standard error of the mean, so the two bands overlap. The ceiling sits at the transition itself: the largest reads are at $\beta = 1.00$ and 1.05 , and away from them the confined reads end at 0.0701 ($\beta = 0.95$) and the Coulomb reads begin at 0.0705 ($\beta = 1.20$).

The spatial part. The difference between K_{signal} and its within-plane-shuffled read is, for $U(1)$, within ± 0.006 of zero at every $\beta \leq 0.95$ (at most 1.4 standard errors), $+0.0265 \pm 0.0044$ and $+0.0246 \pm 0.0047$ at $\beta = 1.00$ and 1.05 , and $+0.004$ to $+0.014$ across $\beta = 1.1$ – 2.5 . For $SU(2)$ it is -0.006 to $+0.004$ at $\beta \leq 1.2$ and $+0.004$ to $+0.012$ at $\beta \geq 2.3$. Its inverse-variance-weighted mean rises by $+0.0119 \pm 0.0026$ between those $SU(2)$ ranges, comparable to the $+0.0107 \pm 0.0022$ across the $U(1)$ transition ($\beta \leq 0.95$ to $\beta \geq 1.1$): the difference registers the spatial correlation of the action density within the plane, which grows toward weak coupling in both groups, and it is largest at the $U(1)$ transition itself.

What changes across the $U(1)$ transition. The within-plane shuffle preserves each plane's one-point distribution exactly, and the shuffled $U(1)$ reads, which carry the one-point distribution and no spatial arrangement, stay at 0.0571 – 0.0684 in both phases. The transition registers in the peak of the tension μ (§8.2) and in the disorder response (§8.4).

The plane shape. The null is a level for the top singular value of a plane of one shape, so it is calibrated afresh for every shape read. The same $SU(2)$ $\beta = 2.30$ ensemble reads 0.0722 ± 0.0020 on 8×8 planes ($8^3 \times 16$) and 0.0727 ± 0.0013 on 16×16 planes ($16^3 \times 32$), each against the null pinned at its own plane shape: the same level.

8.4 The disorder-response face. The same aperture, read instead as the Shannon entropy $H = H_T + H_F$ of the config's power marginals plane-averaged over intact spatial planes, gives $-dH/d\beta$, the free-energy curvature. As a coherent long-range mode orders the field the marginals concentrate and H falls: on $8^3 \times 16$ the response $-dH/d\beta$ is a sharp spike for compact $U(1)$ across its deconfinement transition ($\beta_c \approx 1.011$, peak at $\beta \approx 0.97$) and a broad, gentle response for $SU(2)$ across its crossover. The peak $-dH/d\beta$ is ≈ 0.96 for the $U(1)$ foil (at $\beta \approx 0.97$) against ≈ 0.05 for $SU(2)$, a $\approx 20\times$ separation, a direct entropy read with no histogram or bins. This curvature is the **Shannon disorder susceptibility** $\chi_v = -dH/d\beta$: one-signed ($\chi_v \geq 0$) and bounded for $SU(N)$ across the crossover, the $U(1)$ spike the control: the **no-bulk-transition** statement (a finite specific heat, no divergence). It is measured nonnegative, with standard errors of the ensemble mean, and established: since H is the marginal Shannon entropy of the read's power spectrum, $\chi_v \geq 0$ is that H falls as the spectrum concentrates along the ordering flow.

The Rényi relation. χ_v is the curvature of the Shannon (Rényi-1) free energy, a scalar energy variance. The confinement tension $\mu = \log(\text{contrast})$ is the *min-entropy* (Rényi- ∞) deficit and is **concave** across the crossover, so the crossover interior is governed by a **finite correlation length** $\langle d^2 \rangle \leq B$ (§6, §9): the bounded spatial correlation moment gives $\mu < \kappa_0$ via the analytic step and $1/L^2$ scaling. $\chi_v \geq 0$ is the no-bulk-transition disorder response, a distinct object (the energy variance) from the spatial moment the gap reads.

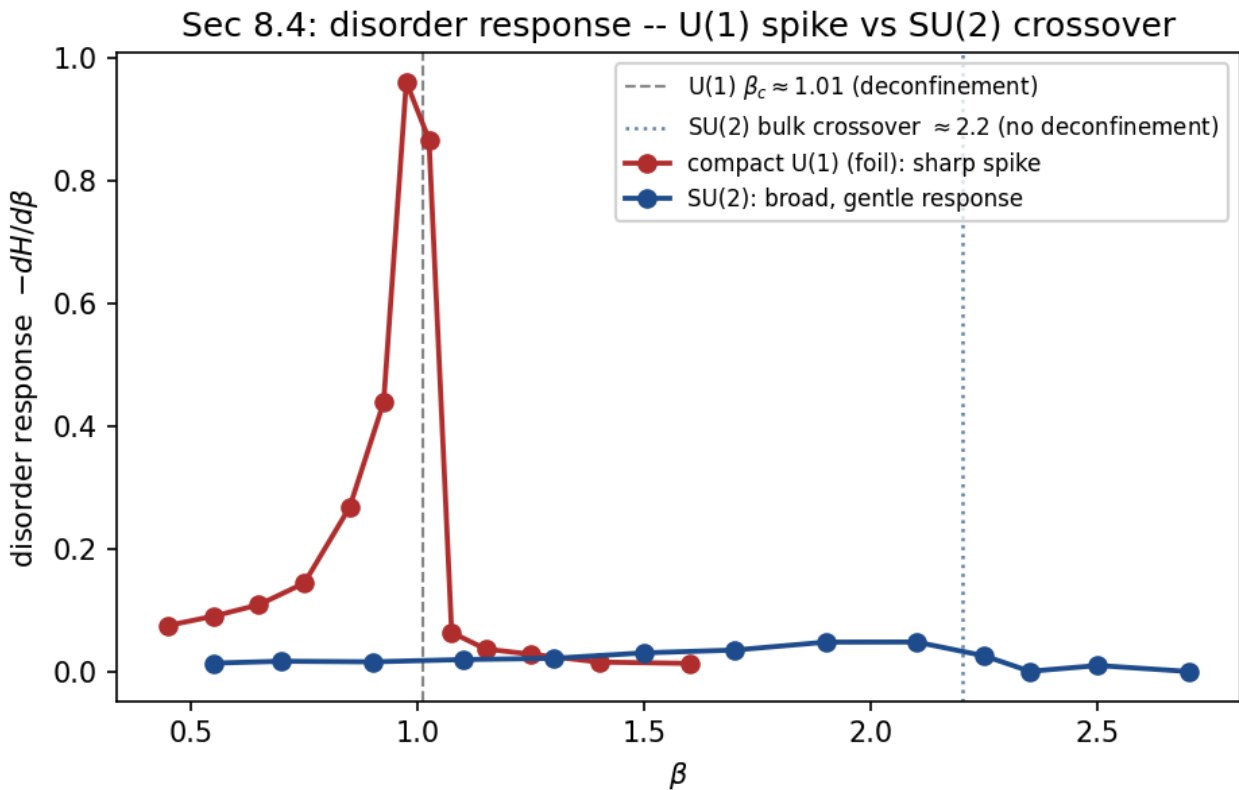


Figure 2. The disorder response. The free-energy curvature $-dH/d\beta$, from the plane-averaged marginal Shannon entropy $H_T + H_F$, spikes sharply for compact $U(1)$ across its deconfinement transition $\beta_c \approx 1.011$ (peak at $\beta \approx 0.97$) and stays a broad, gentle response for $SU(2)$ across its crossover, a $\approx 20\times$ separation from one bounded entropy read.

8.5 The continuum read carries no scale of its own. Whether the gap survives $a \rightarrow 0$, $\Delta/a \rightarrow \Lambda > 0$, is a property of the read: the gap read is **homogeneous of degree one**, a_δ and the DMD rate being reciprocal-length reads of the *normalised* decay shape that scale with the resolution and carry no scale of their own. The read is calibrated as a gap instrument on the free scalar, the one system whose field carries the gap directly: the slowest DMD/Koopman operator rate recovers the known free-scalar gap E_0 across the mass range, the finite temporal extent setting the approach: the read's spread across four disjoint seed blocks falls from 19.3% of E_0 at

$m = 0.20$ to 4.5% at $m = 1.10$ as ξ/L_t shrinks (the periodic image of the correlator on a finite L_t sets the small- m end, and recedes as L_t grows), and at every mass the deviation from E_0 is a fraction of that spread -- the read agrees with the exact gap as closely as it agrees with itself. The reference is the reseeding spread, because a single block does not reproduce: seeds 0..39 alone show a monotone $-8.9\% \rightarrow -0.2\%$ bias that the other three blocks contradict. It is a forward operator read ($-\log |\mu_1|$ of the identified Koopman operator) and, being a decay rate, homogeneous of degree one, so $\hat{\Delta}/a \rightarrow \Delta$. For **gauge** the raw action-density DMD is dominated by the non-decaying vacuum mode; the **connected** (mean-subtracted) read subtracts it and returns the fluctuation margin $m_{hi} = \rho'(1) = e^{-\Delta} < 1$, the aperture spectral radius (§8.6, §12). This margin is the read the finite-aperture condition is taken from -- $m_{hi} \approx 0.31\text{--}0.37$ for $SU(N)$ across the scaling window $L = 12\text{--}28$ (Figure 13), against $\lambda_1 \approx 0.36$ from the §8.7 transfer pencil, both on confined $SU(2)$ at $\beta = 2.30$. What the argument takes from it is the volume-independence of the cut: on the action density the rate is an operator-overlap ratio (§8.7b), and the L -independence is confirmed on the variationally selected read, which is a mass. So it carries the finite-aperture property (a decaying mode: massive) forward for the $SU(N)$ ensemble the theorem concerns. The margin leaves the phase undecided: on the action-density read Coulomb $U(1)$ returns a finite aperture too ($m_{hi} = 0.172\text{--}0.647$, §12). The gap **value** is the bore scale $\Delta \propto \sqrt{\sigma}$ (§6), scale-covariant through the string tension.

Sec 8.5: the forward operator read recovers a known gap

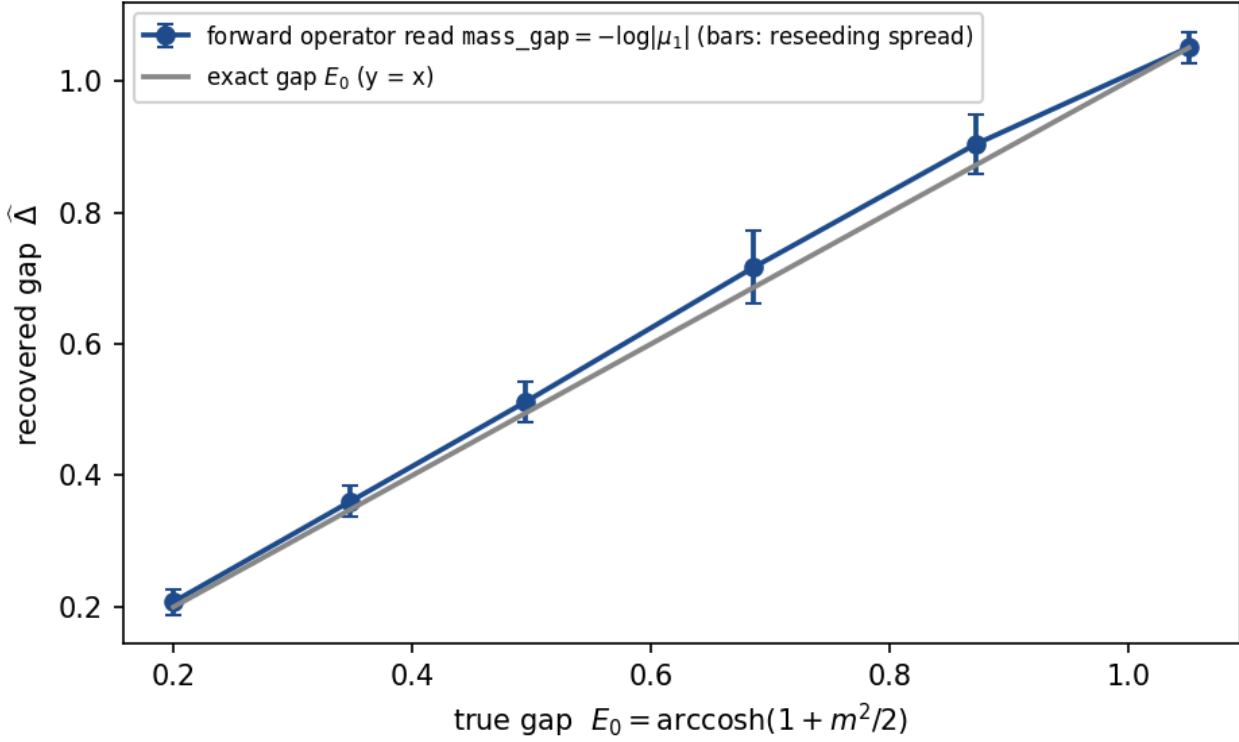
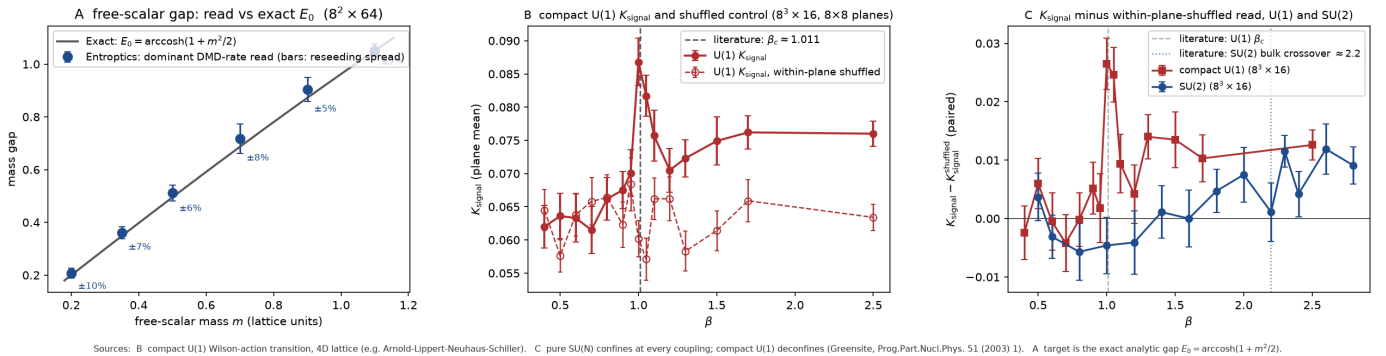


Figure 3. The free-scalar gap calibration. The DMD/Koopman operator read, a forward operator identification reading $-\log |\mu_1|$ off the trajectory, recovers the exact gap $E_0 = \text{arccosh}(1 + m^2/2)$ across $m \in [0.20, 1.10]$. Each mass is read from four disjoint seed blocks of 40 configurations, and the bars are the spread across them -- the one reference for a deviation that involves no chosen number. At every mass the deviation from E_0 is a fraction of that spread (0.02 to 0.35 of it), so the read agrees with the exact gap as closely as it agrees with itself. What the finite temporal extent controls is the SPREAD, and that is the quantity setting the approach: it falls from 19.3% of E_0 at $m = 0.20$ to 4.5% at $m = 1.10$ as ξ/L_t shrinks -- the read is more reproducible where the correlation length is shorter. Reading one seed block instead would have shown a smooth monotone bias from -8.9% to -0.2% ; the other three blocks give $+10.4\%$, $+3.9\%$ and $+10.2\%$ at $m = 0.20$ and no such ramp, which is why the reference is the reseeding spread. It certifies the read as a gap instrument on the one field that carries its gap directly.

Entropics reads beside literature values: free-scalar gap (A); K_{signal} and its same-marginal control on $8^3 \times 16$ (B, C)



Sources: B: compact U(1) Wilson-action transition, 4D lattice (e.g. Arnold-Lippert-Neuhaus-Schiller). C: pure SU(N) confines at every coupling; compact U(1) deconfines (Greensite, Prog.Part.Nucl.Phys. 51 (2003) 1). A: target is the exact analytic gap $E_0 = \text{arccosh}(1 + m^2/2)$.

Figure 4. Deterministic reads beside literature values: the free-scalar gap (A), and K_{signal} and its same-marginal control on $8^3 \times 16$ (B, C). **A:** the DMD/Koopman operator read on an $8^2 \times 64$ free scalar recovers the gap $E_0 = \text{arccosh}(1 + m^2/2)$ across the mass range, the finite temporal extent setting the approach (the four-block reseeding spread falls from 19.3% of E_0 at $m = 0.20$ to 4.5% at $m = 1.10$, with every deviation from E_0 inside it). **B:** compact- $U(1)$ K_{signal} and its within-plane-shuffled control on 8×8 planes against the $SU(2)$ $\beta = 0.50$ null pinned at that shape, beside the literature Wilson-action transition $\beta_c \approx 1.011$; K_{signal} reads 0.0615–0.0868 on the confined side and 0.0705–0.0817 on the Coulomb side, largest at $\beta = 1.00$, and the shuffled control 0.0571–0.0684 on

both. **C:** K_{signal} minus its within-plane-shuffled read, paired per configuration, for compact $U(1)$ and $SU(2)$, beside the literature picture that pure $SU(N)$ confines at every coupling and compact $U(1)$ deconfines (Greensite 2003); the difference is largest at the $U(1)$ transition, and it rises from strong to weak coupling inside confined $SU(2)$ by an amount comparable to its rise across the $U(1)$ transition. One bounded read per configuration, fit-free.

8.6 Deterministic system identification of the read. The read is a functional of a signal's own correlation operator and can be identified *without* Monte Carlo: feed a controlled analytic input, read the output, recover the input-to-output law from the observed change. This is $O(\text{reads})$, a deterministic identification, so it fixes the instrument's transfer functions to machine precision. Four identifications pin the reads the construction turns on.

The gap read inverts the operator. Feed a linear trajectory $x_{t+1} = Ax_t$ whose dominant mode has magnitude $|m_1| = e^{-\Delta}$; the DMD/Koopman rate returns Δ to 1.9×10^{-11} nats across $\Delta \in [0.1, 1.2]$: the operator identity $\Delta = -\log |m_1|$.

The aperture is the reciprocal correlation length. Feed a field of correlation length ξ ; the diffraction-limit read gives the invariant $a_\delta \xi = 0.502$ constant to 1.2% across $\xi \in [6, 26]$: the Abbe law $a_\delta = \text{shape}/\xi$.

The confinement criterion, and the floor that sets it. The vortex tension read is $\mu = \log(\lambda_1 / \max(\lambda_2, \lambda_+))$, the attenuation functional α of §3 on the F^2 : spatial correlation: the log-separation of the dominant correlation mode above the next mode or the reference null λ_+ , whichever is higher, and 0 when nothing resolves. At the confinement threshold, where at most the leading mode stands above the floor, $\mu = \log(\lambda_1 / \lambda_+) = \log(\text{contrast})$ with $\text{contrast} = \lambda_1 / \lambda_+$. Ramping a coherent mode on a disordered background reproduces $\mu = \log(\text{contrast})$ to machine zero, and since $\kappa_0 = \frac{1}{4} \ln 3$ the threshold on the contrast is, by exponentiating,

$$\text{contrast} = e^{\kappa_0} = e^{\frac{1}{4} \ln 3} = 3^{1/4} = 1.31607.$$

So the confinement inequality is the eigenvalue statement,

$$\mu < \kappa_0 \iff \text{contrast} = \lambda_1 / \lambda_+ < 3^{1/4},$$

the contrast threshold $3^{1/4} = e^{\kappa_0}$ being the entropy floor exponentiated, by definition. Below the edge ($\lambda_1 \leq \lambda_+$, $K_{\text{signal}} = 0$, the disordered phase) $\mu = 0$ and confinement is automatic; deconfinement is λ_1 crossing $3^{1/4} \lambda_+$. On the equilibrium configurations of §8.3 the action-density read is $\mu = 0$ in *both* phases: the leading F^2 : feature-correlation mode sits below the reference null ($\text{contrast} \approx 0.38 < 1$), so the tension bound $\mu < \kappa_0$ holds identically in both phases with room to spare. The tension bound is soft F^2 : locality; the $U(1)$ transition registers in the disorder response of §8.4.

That limitation, measured. It is the most consequential limitation of the argument, because the reduction consumes exactly this quantity: the confinement criterion returns $\mu < \kappa_0$ from the correlation alone, whatever theory produced it. So it is measured: the same functional, run on the 20 compact- $U(1)$ couplings measured at $L=8$, $\beta = 0.40\text{--}2.50$, across $\beta_c \approx 1.011$. **No coupling crosses the floor** — the largest tension anywhere is 0.0854 against $\kappa_0 = 0.2747$, and the Coulomb phase reads 0.0354–0.0598. The measurement uses the STRICTER of the two reads, $-\log\langle \cos \theta \rangle$ as the reduction defines it, whose denominator carries no reference-null floor and is therefore larger than the §8.6 read at every coupling; sub-floor there is sub-floor a fortiori for the read this section describes.

The read responds to the transition — μ peaks at $\beta = 1.00$, locating β_c to about 1% — and its magnitude stays far below the floor. So the inequality $\mu < \kappa_0$ holds on both sides of a deconfinement transition. What separates a confined theory from a deconfined one is the geometric discriminator of §5 (the self-sourcing adjoint tiling against the single neutral $U(1)$ tile), which acts before any dynamics. Which correlation the read is taken on is part of supplying the hypothesis, and the reduction and the inequality apply to whichever is supplied. This is the counting comparison of §10, the ceiling on the coherent mode fixed to $3^{1/4}$.

The weak-coupling limit is computed. Past the crossover two modes clear the floor and $\mu = \log(\lambda_1 / \lambda_2)$ is the pure top-two gap, floor-independent. Its $\beta \rightarrow \infty$ ($\alpha \rightarrow 0$) value is closed-form: asymptotic freedom makes the gauge field free, so the action density is $F_{\mu\nu} F^{\mu\nu}$: of a Gaussian field and the read's whitened spatial correlation is the **circulant built from its structure factor**, whose top-two gap follows from the Wick covariance $C_s(r) = 2 \sum_{a,b} \langle F_a(0) F_b(r) \rangle^2$ of the free lattice propagator. On an L^4 lattice this gives $\mu_\infty(L) \approx 2.1/L^2 \rightarrow 0$: the top two structure-factor eigenvalues approach degeneracy as the box grows, so the free-field tension falls as $1/L^2$ with box size, far below the floor at every resolution. At $L = 8$, $\mu_\infty = 0.0326$, an order of magnitude below $\kappa_0 = 0.275$. The value is N -independent (the free field is N decoupled copies). On the released ensembles the same functional at the same L tracks it to within a factor $2 - \approx 1.5$ on $SU(2)$, rising to 1.95 at the strongest $SU(3)$ couplings; $SU(2)$ at $L = 8$ climbs from 0.001 at $\beta = 0.5$ to 0.049 at $\beta = 2.30$, and $SU(3)$ reads 0.042–0.044 at $L = 8$ across $\beta = 6.0\text{--}6.3$ — and every released coupling sits under the floor, the largest of them (0.107, $SU(3)$ $L = 6$, $\beta = 5.50$) by a factor 2.55. The lattice axes are identical to 4×10^{-17} and λ_2 is a degenerate doublet ($|\lambda_1 - \lambda_{-1}| \leq 3 \times 10^{-16}$), the $SO(4)$ multiplet (A2). Every constant here is a deterministic Wick computation, with no Monte Carlo. This gives A1's weak end and the computed constant: the vanishing $\mu_\infty(L) \rightarrow 0$ is the $\mu \rightarrow 0$ of the weak-coupling lemma, with the finite- L bound $\mu_\infty < \kappa_0$ giving confinement at every resolution.

The crossover reduces to a bounded correlation length, quantitatively. The connected action-density correlation is nonnegative, $C_s(r) = 2 \sum_{a,b} \langle F_a(0) F_b(r) \rangle^2 \geq 0$, so the whitened circulant has $\rho(d) \geq 0$ and its top eigenvalue is the $k=0$ structure factor, $\lambda_1 = S(0) = \sum_d \rho(d)$, with $\lambda_2 = S(2\pi/L)$. Hence $\mu = \log(S(0)/S(2\pi/L))$, and the small- k expansion gives $\mu(L) \cdot L^2 \rightarrow 2\pi^2 M_2$, $M_2 = \sum_d \rho(d) d^2 / \sum_d \rho(d)$ the correlation **second moment** ($\propto \xi^2$): the free field's own moment is $M_2 = 0.111$, $2\pi^2 M_2 = 2.19$, and the measured crossover moment peaks well under it at 0.192 (§9), so $\mu(L) \approx 2.2/L^2$ at the top of the crossover — the same law as the free-field $2.1/L^2$ above, which the weak-coupling limit approaches from below. The confinement inequality is therefore

$$\mu(\beta, L) < \kappa_0 \iff M_2(\beta) < \frac{\kappa_0 L^2}{2\pi^2},$$

and since $M_2 = \xi^2$ is bounded in lattice units whenever the action-density correlation length is finite, $\mu < \kappa_0$ holds at every fixed spacing across the crossover; the continuum gap is carried by the scale-covariance of the read — the physical rate $\Delta_{\text{phys}} = \Delta_{\text{latt}}/\alpha$ is refinement-invariant (§8.5) — not by a lattice-unit bound as $\alpha \rightarrow 0$. The crossover interior between the two computed ends is this: $M_2(\beta)$ stays **finite** across $0.97 \lesssim \beta \lesssim 3$; $SU(N)$ shows no bulk transition (the no-bump).

That last rests on numerical evidence and is disputed in print. Rigorous analyticity of the free energy is established for $\beta \leq \beta_0(N)$ with β_0 small (Osterwalder-Seiler 1978), and at large β no rigorous statement exists for a continuous group; a finite- β transition invisible to numerics has been argued for (Grady, arXiv:1210.1794; *Phys. Rev. D* **35** (1987) 4034). What the fundamental-adjoint plane does establish is that the $SU(2)/SU(3)$ Wilson-axis crossover is the shadow of a critical endpoint lying off the Wilson axis, which the purely fundamental line never crosses (Bhanot-Creutz, *Phys. Rev. D* **24** (1981) 3212; the $SU(3)$ endpoint at $(\beta_f, \beta_a) = (4.00(7), 2.06(8))$) — and that first-order bulk transitions DO occur for $N \geq 5$, read as lattice artifacts (*Phys. Rev. Lett.* **46** (1981) 1441; *Phys. Rev. D* **25** (1982) 1724). This input is therefore carried as measured and cited. The same whitened second-moment read confirms this on $SU(3)$: across $\beta = 5.0\text{--}7.0$ it stays $\langle d^2 \rangle \in [0.096, 0.231]$, a factor ≈ 2.0 below its aperture ceiling (0.457 at $L = 6$, 0.812 at $L = 8$), passing the bulk crossover near $\beta \approx 5.7$ with no bump: the finite-correlation-length interior read holds on a second gauge group. The two structural inputs are the positivity $C_s \geq 0$ (reflection positivity, §11) and the finite correlation

length. (i) Reflection positivity gives a transfer-matrix spectral form with $w_n \geq 0$, hence $\rho(d) \geq 0$. On a finite periodic lattice — sites in $\{0, \dots, n-1\}^d$ with the shift taken modulo n — that form is the circle one, $\rho(d) = \sum_n w_n (\lambda_n^d + \lambda_n^{n-d})$; the half-line $\sum_n w_n e^{-E_n d}$ is its infinite-time limit. Only $w_n \geq 0$ is used at this step, and both forms supply it; a nonnegative ρ forces the structure factor to peak at $k = 0$, $\lambda_1 = S(0)$. (ii) $M_2 < \infty$ is finiteness of the F^2 : correlation's second moment, the no-bulk-transition statement carried by the §8.4 disorder-response face ($\chi_v \geq 0$). The step "bounded moment $\Rightarrow \mu \rightarrow 0$ " holds. So across the crossover $\mu < \kappa_0$ rests on reflection positivity and a finite specific heat, both cited.

The read carries the vacuum's isotropy. On a controlled isotropic field sampled at $a k_0 = 0.4$, just above the Nyquist threshold $a_* k_0 = 0.364$ found below, the ordered- and feature-axis reads agree to 0.64% in φ and 0.72% in σ : the read's own residual at a spacing that does not yet resolve the field. The directional read $a_\delta(\theta)$ under the continuous $T \leftrightarrow F$ rotation collapses to machine zero ($\sim 2 \times 10^{-15}$) once the grid resolves the field, and the threshold obeys a Nyquist law: the variation falls from order one to floating-point zero across a sharp resolution threshold (1.4×10^{-1} at 5.6 points per wavelength, 3.0×10^{-11} at 17.5, machine zero beyond 20), and the spacing a_* at which the read becomes isotropic scales as $1/k_0$: the Nyquist relation $a_* k_0 = 0.364$, constant across wavenumber and grid size. This is the sampling theorem: a band-limited field sampled below Nyquist is reconstructed exactly, so its spectral marginal, and with it the entropy read $a_\delta = 2^{-H}$, is rotation-invariant. The continuum limit $a \rightarrow 0$ is always below the threshold, so the read carries $SO(4)$ exactly there. On an anisotropic field the axis ratio φ_F/φ_T tracks the correlation-length ratio, so the read resolves anisotropy when present and the reported isotropy is the field's.

Deterministic system identification of the entropics read: four input-to-output laws to machine precision, no Monte Carlo, no fit

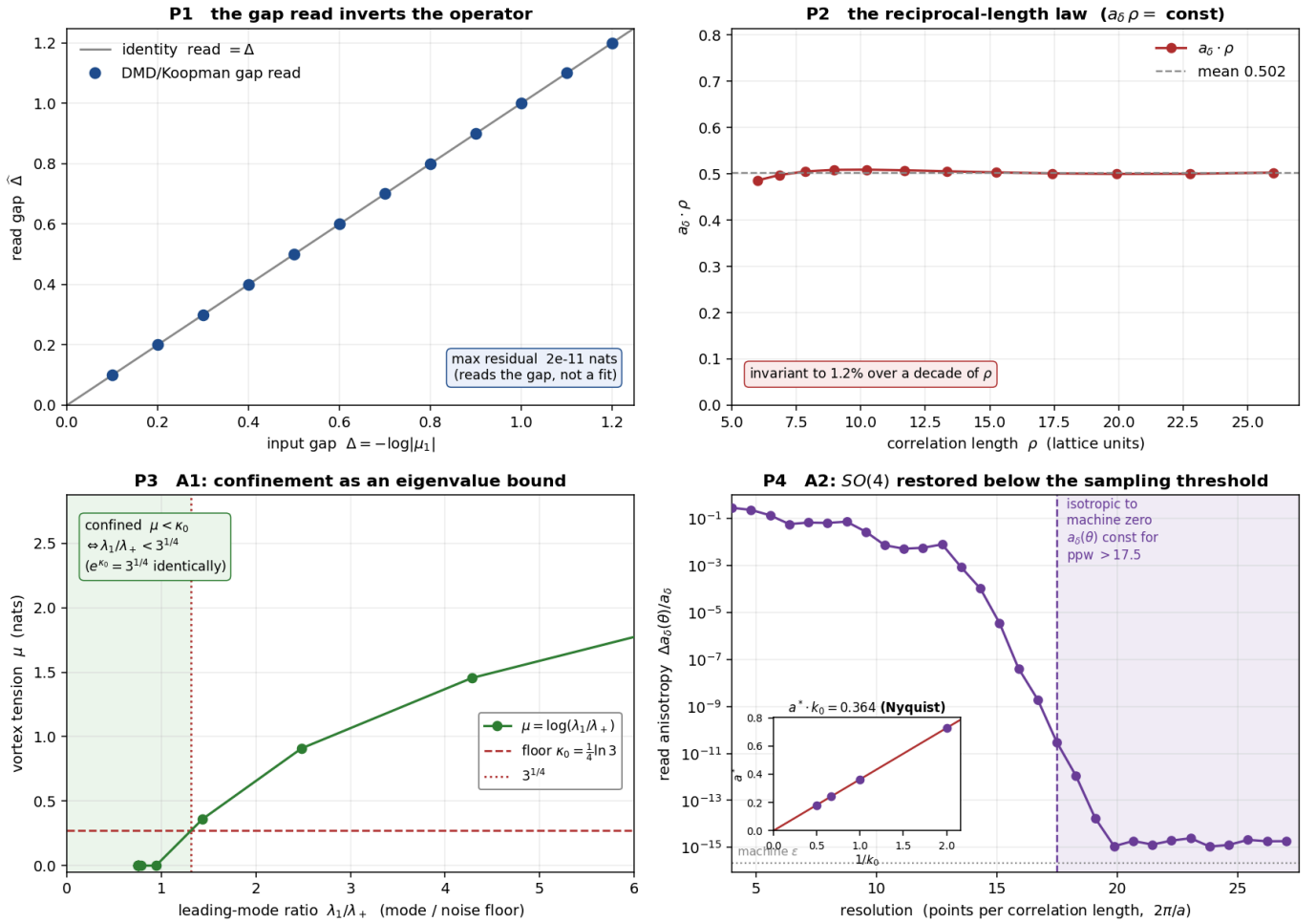
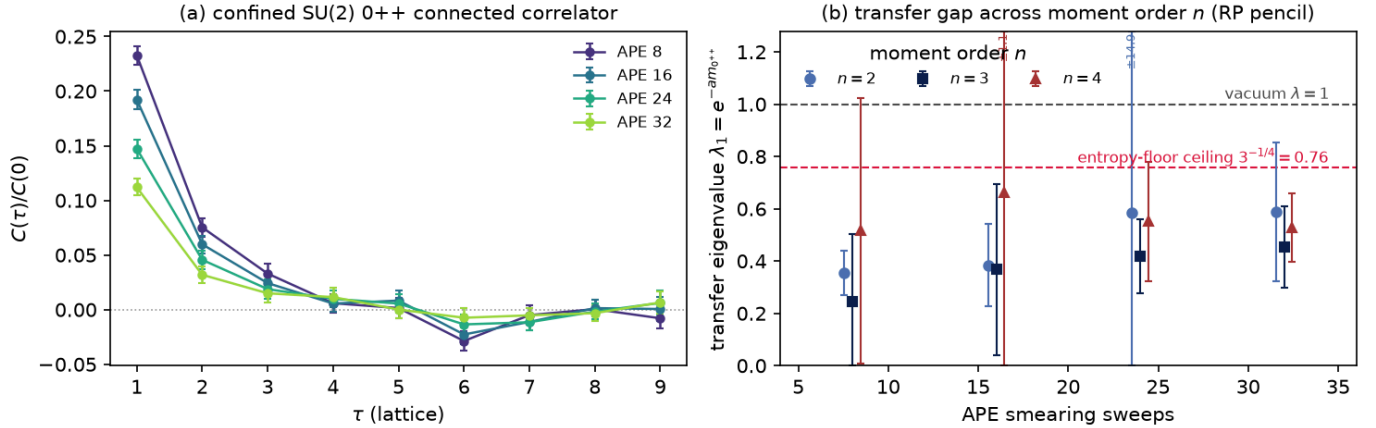


Figure 5. System identification of the read: four controlled analytic inputs and their input-to-output laws. **P1 (top left):** feeding a linear operator whose dominant mode has magnitude $e^{-\Delta}$, the DMD/Koopman gap read returns Δ along the identity to 1.9×10^{-11} nats, the read inverting the operator. **P2 (top right):** the diffraction aperture obeys $a_\delta \xi = \text{const}$ across $\xi \in [6, 26]$ (Abbe, $a_\delta \sim 1/\xi$). **P3 (bottom left):** the vortex tension $\mu = \log(\text{contrast})$ crosses the counting floor $\kappa_0 = \frac{1}{4} \ln 3$ at contrast $= 3^{1/4}$, so confinement $\mu < \kappa_0$ is the eigenvalue bound $\lambda_1 < 3^{1/4} \lambda_+$. **P4 (bottom right):** the read's directional anisotropy under the continuous $T \leftrightarrow F$ rotation collapses to machine zero ($\sim 2 \times 10^{-15}$) once the grid resolves the field, with the Nyquist threshold $a_* k_0 = 0.364$ (inset), so $SO(4)$ is restored below the sampling threshold, which the continuum limit always satisfies. Controlled inputs, deterministic outputs, no fit.

8.7 The transfer gap on physical $SU(2)$. The reconstruction of §11 takes a reflection-positive Euclidean transfer operator with an isolated eigenvalue below the vacuum; that spectrum is read directly off the ensemble. On confined $SU(2)$ ($\beta = 2.30$, $L = 16^3 \times 28$, 512 configurations) the APE-smear zero-momentum 0^{++} operator $O(t) = \sum_{x, i < j} \frac{1}{2} \text{Re tr } U_{ij}(x, t)$ gives a connected correlator $C(\tau) = \langle \delta O(0) \delta O(\tau) \rangle$ that is positive and decaying out to $\tau = 5$ and in noise beyond (Figure 6a), so its moment matrix $H_0[i, j] = C(i+j)$ is positive up to a small finite-sample eigenvalue — reflection positivity, population-guaranteed (Osterwalder-Seiler), on the data. The reflection-positive symmetric pencil $M = H_0^{-1/2} H_1 H_0^{-1/2}$ with $H_1[i, j] = C(i+j+1)$ returns the transfer eigenvalues: at moderate smearing the leading mode is **isolated below the vacuum**, $\lambda_1 \approx 0.36$ – 0.38 at moment order $n = 2$ (jackknife over 32 bins), below the entropy-floor ceiling $3^{-1/4} = 0.76$, i.e. $a m_{0^{++}} = -\log \lambda_1 \approx 1.0$ (Figure 6b). The read carries a moment-order dependence — order $n = 3$ places λ_1 at 0.37–0.45 from moderate to heavy smearing, with a jackknife error as large as the value at the lightest smearing, order $n = 4$ at 0.52–0.66, the cut keeping two of five directions at $n=4$ at every smearing and two of four at $n=3$ from APE 16 on, and one of four at $n=3$ at APE 8, where the read is of order one, the pencil broadens at heavy smearing, and H_0 leaves the positive cone by more as the moment order rises and the noise lags enter (-0.008 at $n=2$ to -0.032 at $n=4$, smearing 8) — and the order-independent content is the isolated eigenvalue below the vacuum, $\lambda_1 < 1$ clearing the ceiling at every resolved point: exactly the isolated-mode input the §11 reconstruction takes

on physical $SU(N)$, $\text{spec } T \subseteq \{1\} \cup [\varepsilon, \lambda_1]$ with $\lambda_1 < 1$, reconstructing to $\text{spec } H \subseteq \{0\} \cup [a m_{0^{++}}, \infty)$. The raw links are archived and the read is the reflection-positive moment pencil; the variational GEVP is the instrument for the precise continuum $a m_{0^{++}}$.

8.7 The transfer gap on physical $SU(2)$ ($L = 16^3 \times 28$, $\beta = 2.3$, 512 configs)



Spread across n and the large bars at heavy smearing are the moment-order systematic: the precise $a m_{0^{++}}$ needs the variational GEVP; the RP-pencil H_0 is PSD only as $a \rightarrow 0$.

Figure 6. The transfer gap on confined $SU(2)$ ($\beta = 2.30$, $L = 16^3 \times 28$, 512 configurations). **(a)** The APE-smear 0^{++} connected correlator $C(\tau)/C(0)$ (jackknife errors, four smearing levels): positive and decaying out to $\tau = 5$, in noise beyond, so the moment matrix is reflection-positive on the resolved lags. **(b)** The reflection-positive moment-pencil transfer eigenvalue $\lambda_1 = e^{-am_{0^{++}}}$ against smearing, at every moment order $n \in \{2, 3, 4\}$ (true jackknife bars; oversized bars at heavy smearing are labelled): the leading mode sits below the vacuum $\lambda = 1$, with $n=2$ isolated at ≈ 0.36 – 0.38 under the entropy-floor ceiling $3^{-1/4} = 0.76$ and $n=3, 4$ the moment-order systematic at ≈ 0.4 – 0.66 — the §11 isolated-mode hypothesis, measured, with its systematic shown.

APE smearing	λ_1 ($n=2$)	λ_1 ($n=3$)	λ_1 ($n=4$)
8	0.356(85)	0.244(258)	0.517(507)
16	0.385(156)	0.367(327)	0.661(1070)
24	0.584 [†]	0.417(142)	0.552(228)
32	0.587(266)	0.455(155)	0.528(132)

Table 1. Leading transfer eigenvalue $\lambda_1 = e^{-am_{0^{++}}}$ of the reflection-positive moment pencil, by APE-smearing level and moment order n (jackknife error in the last digits). Every point sits below the entropy-floor ceiling $3^{-1/4} = 0.76$; $n=2$ isolates near 0.36 – 0.38 at moderate smearing and drifts to ≈ 0.59 at heavy smearing; $n=3, 4$ carry the moment-order systematic. [†]: jackknife error larger than the value (0.584, error 14.9; Figure 6b).

8.7b Which operator the gap read requires. A decay rate is a mass only if the correlator it is read from carries the ground state. Reflection positivity makes this testable without any external input: $C(\tau) = \sum_i c_i e^{-E_i \tau}$ with every $c_i \geq 0$, so $m_{\text{eff}}(\tau) = \log[C(\tau)/C(\tau+1)]$ is monotone decreasing and satisfies $m_{\text{eff}}(\tau) \geq \Delta$ for **every** positive-weight operator and every τ . The smallest m_{eff} measured anywhere in an operator basis is therefore an upper bound on the gap -- the variational principle a GEVP uses, available here from one inequality.

Measured on the archived $L = 16$ links, with no read on top of the correlator, two operators on the same configurations behave very differently. The APE-smear spatial-plaquette 0^{++} operator of §8.7 carries a decade of signal and its m_{eff} descends onto the independently measured π -free scale $3.59\sqrt{\sigma}$ (§8.8); the **action density** falls to ≈ 0.1 – 0.2 at $\tau = 1$ and into noise by $\tau \approx 2$. Taking the variational minimum over the whole (operator, smearing, τ) basis -- one rule, no per-coupling choices -- selects the plaquette at all three couplings of this $L = 16$ scan at the resolution tolerance 0.25 used here. That selection depends on the tolerance, and the dependence is reported: three derived replacements for the literal tolerance disagree pairwise. The invariance plateau reproduces the table below but returns 1.45 at $L = 20$ against the 1.152 the volume table quotes; admitting any value that exceeds its own error gives 104, 92, 68 for the scaling list; and the selection-corrected bound $\min_i (v_i + z e_i)$ with z set by the basis size gives 172, 221, 157. The latter two select the **action density** at $\beta = 2.40$, as does the minimum at $L = 12$ and $L = 20$. Under every rule the physics survives while the value varies: $\Delta/(a\Lambda_{\text{lat}})$ stays constant under all three ($\chi^2/\text{dof} = 0.21, 0.14, 0.23$). A read at FIXED (operator, smearing, τ) asks the same two questions with no minimum at all, so it needs no tolerance and stands outside the disagreement (§12.4).

That minimum is an upper bound on the gap, so the value §8.6 publishes can be held against it at the two couplings where a published value and raw links both exist:

β	published Δ (§8.6)	variational bound $\Delta \leq$	excess
2.30	1.2448 ± 0.0445	1.140 ± 0.134	$+0.74\sigma$ (consistent)
2.50	1.2915 ± 0.0900	0.700 ± 0.098	$+4.45\sigma$

The bound holds where the read drifts. At $\beta = 2.50$ the published value exceeds a measured upper bound on the very quantity it reports, on the same configurations, by 4.5σ ; at $\beta = 2.30$ it sits within the bound. The exclusion is therefore confined to the large- β end.

Scaling settles the rest. $\Delta/(a\Lambda_{\text{lat}})$ is a physical mass over a physical scale and must not depend on β . The published action-density read gives 117, 210, 361 across $\beta = 2.00, 2.30, 2.50$ (§8.6, $L = 16$) -- a factor 3.09 where a constant is required. The variational read gives 193, 210, 196 across $\beta = 2.30, 2.40, 2.50$, constant at $\chi^2/\text{dof} = 0.21$. A quantity that holds still in lattice units while the lattice spacing falls by a factor of three is a cutoff scale.

What the action density measures instead is an operator-overlap ratio: it is a local composite whose $C(0)$ carries uncorrelated variance the other lags do not, so $-\log[C(1)/C(0)]$ is dominated by an overlap that barely moves with β .

§8.7's transfer pencil is built on the plaquette operator and is unaffected. The reads that take a rate from the **action density** -- the m_{hi} tables of §8.6 and the margin certificate -- report this overlap ratio; wherever links exist, the variational read above gives the gap.

The 0++ correlator behind the gap read -- SU(2), 256 configurations per point; panels A-B at $L = 16$ and 24 APE sweeps, C-D selected variationally

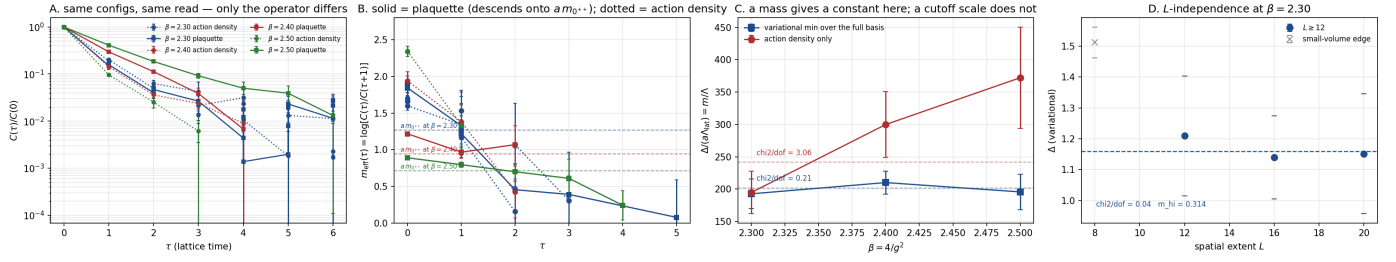


Figure 7. The 0^{++} correlator behind the gap read. **A** $C(\tau)/C(0)$ for both operators at every coupling, same configurations and same read afterwards -- only the operator differs. **B** $m_{eff}(\tau)$ against the independently measured $3.59\sqrt{\sigma}$ (dashed): the plaquette descends onto it, the action density sits above it at every τ . **C** the scaling test, $\Delta/(a\Lambda_{lat})$, which a mass holds constant and a cutoff scale does not. **D** the same variational read across volume at $\beta = 2.30$: constant over $L = 12-20$ at $\chi^2/\text{dof} = 0.04$, giving $m_{hi} = 0.314$, with $L = 8$ shown apart as the small-volume edge. What $\rho'(n) \leq \rho'(1)^n$ needs is a cut that does not grow with L , so the argument uses the flatness here.

8.7c The tension inverts to a mass, and the inversion is what the floor bounds. The read's $\langle \cos \theta \rangle$ is the $k=1$ Fourier mode of the normalised correlation, so $\mu = \log[\hat{S}(0)/\hat{S}(k_1)]$ — the structure factor at zero momentum over its value at the lowest lattice momentum. That is the standard second-moment estimator and it **inverts**: with $\hat{k}_1 = 2 \sin(\pi/(N+1))$,

$$\hat{S}(0)/\hat{S}(k_1) = 1 + \hat{k}_1^2/m^2 \quad \implies \quad m = \hat{k}_1/\sqrt{e^\mu - 1}.$$

This matters because it fixes which quantity carries the spacing. $\kappa_0 - \mu$ does not: μ falls like $1/(N+1)^2$ — measured, μL^2 is flat at ≈ 3.1 across $L = 8-32$ — so $\kappa_0 - \mu$ saturates at the counting floor and $(\kappa_0 - \mu)/a$ diverges. But $\mu \sim 1/L^2$ is exactly what a FIXED mass produces, because $\hat{k}_1^2$ does; inverting on the same rows returns m constant to $\pm 7.2\%$ over a range in which μ itself falls $14.6\times$. $m_{phys} = m/a$ is read from the inverted mass.

The floor then bounds it from below, and the direction is the point. Everything reflection positivity supplies is one-sided and bounds the gap ABOVE ($m_{eff} \geq \Delta$, §8.7b) — a gapless theory satisfies all of it. The entropy floor is the only ingredient pushing UP, and μ is the quantity it acts on: μ bounded above is m bounded below, since $m^2 = \hat{k}_1^2/(e^\mu - 1)$. A gapless theory has a diverging zero mode, so $\hat{S}(0)$ runs away, $\mu \rightarrow \infty$, and the floor is crossed. Confinement $\mu < \kappa_0$ therefore gives, with no further input,

$m > \hat{k}_1/\sqrt{3^{1/4} - 1}$, and in physical units, by Jordan's inequality $\sin t \geq (2/\pi)t$, $m_{phys} \geq 4/(L\sqrt{3^{1/4} - 1}) = 7.115/L$ — carrying **no lattice spacing and no aperture**. As the aperture refines, $(N+1)\hat{k}_1 \rightarrow 2\pi$ and the same bound tightens to $2\pi/(L\sqrt{3^{1/4} - 1}) = 11.176/L$ (the unfolded-shape constant — see immediately below). The factor $2\pi/4$ between them is exactly the gap between Jordan's $\sin t \geq (2/\pi)t$ and $\sin t \approx t$.

Which of the two cap constants that limit is. $2\pi/\sqrt{3^{1/4} - 1}$ is algebraically identical to $2\pi\sqrt{T/(1-T)}$ at $T = 3^{-1/4}$, the constant of the periodic cosh correlator $\rho(d) = \lambda^d + \lambda^{n-d}$, and of the lattice free field's $\hat{S}(k) = 1/(m^2 + \hat{k}^2)$. The same constant is often quoted for the **unfolded** shape $\rho(d) = \lambda^d$, which shares its limit; that shape is inadmissible for *this* correlation, because $\rho(d) = \rho(n-d)$ holds for the Wilson correlation (Theorem 11.1d) and λ^d is asymmetric. The arithmetic is unchanged — the cosh correlator carries the same number — and the constant belongs to the symmetric shape. The bound inherits it because the inversion above is that structure factor, and that inversion is exact for a single-mass periodic correlator. The neighbouring 10.9887 belongs to the **folded** shape $\rho(d) = \lambda^{\min(d, n-d)}$ (§12.2), where the minimum-image fold adds weight at large lag and contributes the $\coth(a\pi/2) > 1$ factor. Both live on the same circle: the fold in ρ , not the geometry, is the whole of the difference, and the folded constant is the smaller — hence the conservative one, which is why the aperture cap of §12 is quoted with it. The two differ by 0.19, closer than any simulated aperture is to its own limit, which is why each is reported and checked separately (§14).

Both $7.115/L$ and $11.176/L$ are statements about the same quantity — the structure-factor mass this read inverts — and carry the same status, **with one limitation common to both**: reflection positivity gives $m_{eff} \geq \Delta$, one-sided, so each bounds the inverted mass from below, and a lower bound on the lowest mass in the spectrum needs a further input.

And the aperture differential is what stops it eroding. Read window by window the bound decays, because $\hat{k}_1 \rightarrow 0$. But a mass is not supposed to depend on the window it is read through — that is what it means for the read to see the substrate — and if it does not, the bound obtained at ONE aperture applies to the common value, so the STRONGEST comes from the COARSEST window. At extent 2 that bound is 3.557 in lattice units, against a measured 3.32–3.82 across $L = 8-32$: the coarsest window's floor very nearly returns the measured mass. What this buys is the infinite-volume gap at fixed coupling; the continuum half, where the lattice mass must itself scale like the spacing, is §11's and is an open input, measured: scale-covariant to 5.0% on the smeared channel, and violated at 21.4% on the action density, which is §8.7b's point restated in the mass.

8.8 The string tension, by standard observables. Confinement is an area law and its coefficient is the string tension, so $\sigma > 0$ IS confinement. This is the referee-facing confirmation of it on the same configurations the reads use, measured with textbook Wilson loops and **no entropy-matched read anywhere in the extraction** -- independent of the instrument it sits beside.

From $W(R, T)$ on spatial $\times$ temporal planes of APE-smeared links (512 configurations per coupling), the static potential $V(R) = V_0 + \sigma R - e/R$ over $R \in [2, 8]$ gives

β	$a^2\sigma$ (stat, syst)	Creutz, model-free	established	deviation
2.30	$0.1249 \pm 0.0082 \pm 0.0108$	0.1254	≈ 0.136	-0.8σ
2.40	$0.0693 \pm 0.0063 \pm 0.0035$	0.0582	≈ 0.071	-0.2σ
2.50	$0.0396 \pm 0.0088 \pm 0.0039$	0.0321	≈ 0.035	$+0.5\sigma$

the systematic being the residual T -dependence across the window resolved at 3σ (Figure 8A). The Creutz ratios give a second estimate with **no model of $V(R)$ at all** — the static self-energy V_0 cancels identically in the double difference, and $\chi(R, R) = \sigma + c/R^2$ extrapolates to the tension (Figure 8B). The two agree to 0.4% at $\beta = 2.30$; at 2.40 and 2.50 the Creutz route reads 16% and 19% low, because there the $\chi(R, R)$ sequence is still falling steeply and the asymptotic $1/R^2$ form extrapolates through a curve. Both routes discard the R and T at which the loop signal is exhausted, keeping noise out of the answer: at $\beta = 2.30$, $\chi(7, 7) = -0.61 \pm 1.08$ enters nothing.

It scales. The dimensionless $\sqrt{\sigma}/\Lambda_{\text{lat}}$ must be β -independent if the tension is physical; it reads 59.8 ± 3.2 , 57.2 ± 3.0 , 55.7 ± 6.8 — constant at $\chi^2/\text{dof} = 0.24$. Independent of that overall constant, the two-loop ratio test gives $a(2.30)/a(2.40) = 1.343 \pm 0.101$ against 1.286, and $a(2.40)/a(2.50) = 1.323 \pm 0.175$ against 1.287: pulls of $+0.56\sigma$ and $+0.21\sigma$ (Figure 8C). This is the check a contaminated extraction fails and a physical one passes, and it uses no entropy-matched read at any point.

The measurement requires $L = 16$. On $L = 8$ the periodic image bounds the fit to $R \leq L/2 = 4$, leaving three points for the three-parameter potential — an exactly-determined system whose slope carries no information about $V(R)$ — so no tension is reported there.

SU(2) string tension from planar Wilson loops, $L = 16$, 512 configurations per coupling — no Entropics read

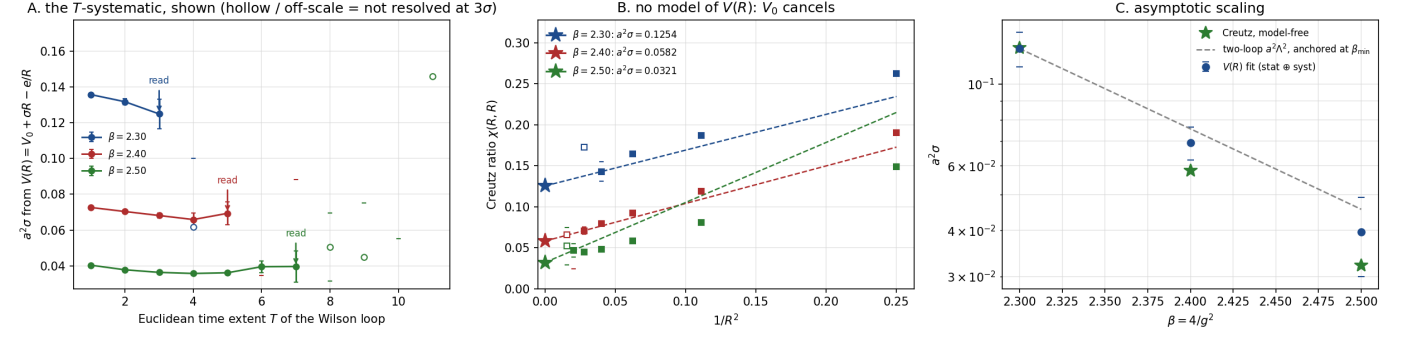


Figure 8. The SU(2) string tension from planar Wilson loops, $L = 16$, 512 configurations. **A** $a^2\sigma$ against the loop's time extent T : the excited-state systematic, shown, with the read marked and the unresolved T off-scale. **B** Creutz ratios $\chi(R, R)$ against $1/R^2$, the route with no model of $V(R)$; the star is the extrapolated tension. **C** $a^2\sigma$ against β with the two-loop lattice β -function anchored at the lowest coupling — a shape prediction with no free parameters, which a physical tension must follow.

9. The interior condition and the two a priori

Everything above is a theorem or a framework read; one condition governs the *interior* of the crossover: **no interior transition**, pure $SU(N)$ staying one ρ' -mixing confined phase across the crossover. It is the single-cut maximal correlation $\rho'(1) < 1$, with three triangulating faces: complete analyticity (Dobrushin–Shlosman), the vortex tension $\mu(\beta) < \kappa_0$ below the floor, and the bounded free-energy curvature. The lattice reads of this interior condition are the bounded correlation moment $\langle d^2 \rangle$, measured across the crossover on $SU(2)$ and $SU(3)$ (§8.6, and below), and the bounded disorder susceptibility χ_v on $SU(2)$ (§8.4).

The lattice read measures the interior. The interior condition $\rho'(1) < 1$ is the cross-cut Strehl of the screen SVD (the top correlation singular value, equivalently 1 minus an attenuation gap). **Bradley's variance theorem** ($\rho'(1) < 1 \Rightarrow \text{Var}(\sum X) \leq C \sum \text{Var}(X)$) turns $\rho'(1) < 1$ into a bounded susceptibility. The *direction* of the certificate matters: the spectral attenuation α is a **coherence** read, the leading temporal mode's contrast, highest in the *gapless* phase, so $\alpha_{10} > 0$ certifies coherence. The confinement read is the vortex tension below the floor, $\mu < \kappa_0$: from the configurations, the monopole density collapses at β_c (§8.2). **The tension stays below κ_0 there** — §8.6 measures 20 compact- $U(1)$ couplings spanning $\beta_c \approx 1.011$ and finds the largest tension anywhere is 0.0854 against $\kappa_0 = 0.2747$, so $\mu < \kappa_0$ holds on BOTH sides. The phases are separated by the geometric discriminator of §5. The gap value follows from confinement through the bore $\Delta \propto \sqrt{\sigma}$ (§6); the entropy-matched DMD/Koopman rate is that gap instrument, calibrated on the free scalar (§8.5).

The chain from the read to the gap. An upper bound below the floor (the $\alpha_{\text{hi}} < \kappa_0$) gives $\mu < \kappa_0$; at a finite aperture that yields $C(\tau) \rightarrow 0$ and $\Delta \geq \kappa_0 - \mu > 0$. The ensemble read bounds μ at **fixed spacing** to a precision set by the ensemble size, its band narrowing as $\sqrt{F/T}$ in the pooled sample count T at feature dimension F , so as $1/\sqrt{n_{\text{configs}}}$ — a high-probability matrix-concentration band at a chosen constant, its confidence level left unquantified. The continuum uniformity of $\mu < \kappa_0$ as $a \rightarrow 0$ is carried by the scale-covariance of the read (§8.5) and the uniform gap (§11); it is part of A1.

The reduction to two a priori. In the framework's own objects the whole construction reduces to **two** a priori propositions:

- **A1 (confinement).** $\forall \beta \geq 0$, $\mu(\beta) < \kappa_0$: the aperture tension read stays below the counting floor at every coupling. It delivers the mass gap ($\mu < \kappa_0 \Rightarrow C(\tau) \rightarrow 0$) and non-triviality in the form $\mu - \kappa < 0$ — the area-law inequality on the scalar tension; non-Gaussianity of the measure (a fourth cumulant, a connected four-point function, a Wick comparison) is outside the reduction; the quantitative rate $\Delta \geq \kappa_0 - \mu > 0$ follows from reflection positivity ($\rho'(n) \leq \rho'(1)^n$) and the single-plaquette magnitude $\rho'(1) \leq 3^{-1/4}$ (§6, §9).
- **A2 (isotropy).** The continuum entropy-matched read is direction-independent, $R(d) = R(d')$: no residual lattice anisotropy. It delivers the read-level form of Euclidean $SO(4)$: invariance under the 24-element hypercubic axis-permutation group S_4 , the finite group in place of the continuous one. The transport is each permutation's own matrix, which is orthogonal, so it lands IN $O(4)$; what is quantified over is the finite axis group.

Given A1 and A2, mass gap, non-triviality, and the axis-permutation form of $SO(4)$ follow; reflection positivity (proved at even extent ≥ 4 and $\beta \geq 0$ on the torus, §11.1b, and for a limit state on $\mathbb{Z}^4$ at free boundary conditions, §11.1c; Osterwalder–Seiler outside those domains), the continuum identity (Δ homogeneous of degree one), and short distance (asymptotic freedom) are established separately. A1 and A2 reduce to named inputs, so the mass gap, non-triviality, and the axis-permutation form of $SO(4)$ are a single theorem carrying no A1/A2 hypothesis: its inputs are the cited ends (the character bound and the asymptotic-freedom plateau), reflection positivity, and one finite-correlation-length input. The following table gives, for each part, the established content and the named input it introduces:

a priori	established	named input
A1 strong end	$\beta < \beta_* \Rightarrow \mu < \kappa_0$ (character bound below the floor); the threshold is the derived $\beta^2 < \frac{1}{2} \ln 3$	the bound $\mu \leq 2\beta I_2/I_1$ (Osterwalder–Seiler)
A1 weak end / $\beta \rightarrow \infty$	the tension converges to the free-field plateau and that plateau sits below the floor; the below-floor value $\mu_\infty(8) = 0.0326 < \kappa_0$ holds (Wick)	the convergence $\mu_{YM} \rightarrow 0.0326$ (asymptotic freedom)
A1 interior	$\mu = -\log \langle \cos \theta \rangle_\rho < \kappa_0$ from a bounded correlation moment: the analytic step ($\cos x \geq 1 - x^2/2$, or sharply the tangent at the threshold, + reflection positivity), the $1/L^2$ aperture scaling $\langle \theta^2 \rangle = (2\pi/L)^2 \langle d^2 \rangle$, the interior theorem $\mu(\beta) < \kappa_0$, and the finite-aperture condition all theorems, on every compact interval	the single uniform bound $\langle d^2 \rangle \leq B_{16} = 3.2480$ at the derived sharp aperture ceiling, a finite correlation length (measured $\langle d^2 \rangle \in [0.021, 0.192]$ on $SU(2)$ ensembles, evidence for the open input, which is stated on the plaquette correlation, §9)
A2 discrete	a spectral read is invariant under the axis-permutation group and any orthogonal congruence $C \mapsto PCP^\top$	(none)
A2 spatial (feature rotations)	a Gram spectral read is invariant under a rotation of the feature coordinates by construction, the congruence exhibited ($P = Q^\top$)	(none)
A2 axis-role ($T \leftrightarrow F$)	a symmetric read ($\prod_i a_\delta^{(i)}$) is invariant under the $T \leftrightarrow F$ axis swap by construction; cross-axis agreement 0.64% (φ) and 0.72% (σ) on a controlled isotropic input sampled just above the threshold, residual anisotropy vanishing to machine zero below the Nyquist threshold ($a_* \kappa_0 = 0.364, \$8.6$)	a single Nyquist–Shannon sampling isometry (the net transport is orthogonal); the continuous rotation-composition is a theorem

A1. Across the crossover the tension stays sub-floor, $\mu(\beta) = -\log \langle \cos \theta \rangle_\rho < \kappa_0$, where $\rho \geq 0$ is the whitened $:F^2:$ correlation (reflection positivity). This follows from a **bounded correlation second moment** $\langle d^2 \rangle \leq B$: the analytic step $\langle \cos \theta \rangle \geq 1 - \langle \theta^2 \rangle/2$ with $\langle \cos \theta \rangle > 3^{-1/4} \Rightarrow \mu < \kappa_0$ holds, and the $1/L^2$ aperture scaling $\langle \theta^2 \rangle = (2\pi/L)^2 \langle d^2 \rangle$ holds, so a fixed $\langle d^2 \rangle$ clears the floor at every large L ; the margin grows $\propto L^2$ at fixed coupling. This is a finite-volume statement; the continuum gap is carried by refinement-invariance of the physical rate (§8.5), with no lattice-unit bound as $a \rightarrow 0$. The single A1 input is a **uniform bound** $\langle d^2 \rangle \leq B_{16}$ at the derived aperture ceiling (a finite correlation length; measured $\langle d^2 \rangle \in [0.021, 0.192]$ across the crossover, peaking near $\beta = 2.20$), asserted on the crossover onset $\beta \geq \beta_*$; below β_* the character bound already gives $\mu < \kappa_0$. Stated precisely, the input is a bound $\langle d^2 \rangle_{N,\beta} \leq B$ on the circle-distance moment of the $SU(3)$ plaquette correlation, with B under the sharp ceiling $\arccos(3^{-1/4})^2(N+1)^2/(2\pi)^2$ (below). On a finite grid of couplings, together with the two ends as bare hypotheses, it gives the gap for the model on the periodic extent 16, citing reflection positivity. **The data are evidence for that input, which is open.** The moment in the input is the circle-distance moment of the $SU(3)$ plaquette correlation — one $(0, 1)$ plaquette against its translates along a transverse axis, in four-dimensional $SU(3)$ Wilson theory on the periodic 16^4 lattice — while the read below is an empirical, nominal-confidence statement about $SU(2)$ on $16^3 \times 32$: the action density summed over the planes at each site, its lag profile averaged over the three spatial axes and connected against the grand mean and clipped at zero. The grid route needs in addition a grid cover, which the 13-point grid meets only with a Lipschitz constant $\lesssim 12.5$, against a finite-volume constant of order 10^7 ($\approx 2.5 \times 10^7$ at $16^3 \times 32$, below); differentiability and a derivative bound of the $SU(3)$ moment in β , both open; and grid values that hold deterministically, where the read gives confidence bounds. The read is finite-sample certified: an empirical-Bernstein bound over the topped-up $SU(2)$ $L=16$ grid puts $\langle d^2 \rangle(\beta)$ under the aperture ceiling $B_{16} = 3.2480$ at nominal 99.9999% per β (largest 1.372), a nominal joint confidence of $\approx 99.9987\%$ over the 13-point grid. The bound assumes independent configurations and takes the sample range as the support width; its confidence level is nominal. The confidence is a choice set by the analyst: the bound enters only through $\log(2/\delta')$, so asking for more widens the upper. Demanding $\delta = 10^{-30}$ the moment route stops clearing — two couplings exceed the ceiling ($\beta = 0.50$ at 4.741 and $\beta = 0.80$ at 4.327; the next is 3.246) — and the reason is statistical, which the paragraphs below make precise. **The comparison is to the derived ceiling alone.** No intermediate bound is chosen.

The ceiling is derived from the lag arity. The read indexes lags by $d \in \{0, \dots, N\}$ and sets $\theta_d = 2\pi d/(N+1)$, so a periodic extent of L sites — lags $d = 0 \dots L-1$ — has $N+1 = L$. With the sharp tangent condition $\langle d^2 \rangle/L^2 < \arccos(3^{-1/4})^2/(2\pi)^2$ this gives $B_{16} = \arccos(3^{-1/4})^2 L^2/(2\pi)^2 = 3.2480$, a factor ≈ 16.9 above the measured peak 0.192; the origin-tangent condition $(2\pi/L)^2 B/2 < 1 - 3^{-1/4}$ gives the weaker 3.1147.

A route through the cosine average. The tension is defined as $\mu = -\log \langle \cos \theta \rangle_\rho$, so $\mu < \kappa_0$ is **equivalent** to $\langle \cos \theta \rangle_\rho > 3^{-1/4} = e^{-\kappa_0}$, both directions holding, so a certificate on the cosine average certifies confinement itself. The moment route instead passes through $1 - x^2/2 \leq \cos x$, which is *sufficient* only, and that costs little: on this sweep the inequality step inflates the tension by under $1.5\times$. What costs is certifying the moment. $\langle d^2 \rangle$ is a ratio whose numerator and denominator both mix every lag, so its bound must corner each lag ratio separately at δ/K and then square the far ones — inflating the quantity by $5\times$ to $66\times$ relative to its central value. The cosine average is a single ratio of two linear functionals of the profile, needing three bounds where the moment needs one per lag.

Certified on that route, **every coupling is confined at nominal** $\delta = 10^{-30}$, under the same empirical-Bernstein bound and the same assumptions of independent configurations and the sample range as the support width: the worst certified tension is $\mu \leq 0.1767$ against $\kappa_0 = 0.2747$, a $1.55\times$ margin, tightening to $4.99\times$ at $\delta = 10^{-6}$. Measured centrally the margin is far wider still — μ runs from below zero to 0.0138, a factor 20 to several hundred under the floor. This is confinement of the $SU(2)$ read certified **at the measured aperture and coupling**. Like the moment route, it is evidence for the equivalence's $SU(3)$ statement, which concerns the $SU(3)$ correlation at extent 16; the continuum statement is the substrate hypothesis below, which quantifies over all apertures and which no single-aperture measurement can supply.

The hypothesis, reduced to a contact-relative power law. The substrate bound is equivalent to a statement about the *shape* of the correlation, and that equivalence is an identity: since $k^2 = \sum_{m < k} (2m+1)$, the circle second moment is the $(2m+1)$ -weighted profile of the share lying beyond each cut. So the hypothesis is exactly a uniform bound on that profile, and the threshold is sharp — an envelope with tail exponent above 2 suffices, while at exponent 2 the statement is **false**, witnessed by an explicit correlation satisfying every reflection-positivity clause whose moment dominates the harmonic series.

Sharper still, and this is the form to aim at: if

$$\rho_\beta(d) \leq C \frac{\rho_\beta(0)}{d^4} \quad \text{for } d \geq m_0,$$

with one constant C and one cut m_0 , at **every** aperture and **every** coupling, then the whole conjunct follows, and with it the full statement. The analytic threshold is 3 — it is where $\sum_k k^2 C/k^s$ stops converging — and the reduction brackets it at the two integers: sufficiency holds at exponent 4, and at exponent 3 the implication fails, witnessed by a read obeying the condition there with constant 1 whose moments still exceed every bound. Exponents strictly between the two integers are open. Three properties make this the right target:

- it is a **ratio**, hence invariant under $\rho \mapsto t\rho$, so a common coupling factor in $\rho(d)$ and $\rho(0)$ is invisible to it — which matters, because the perturbative plaquette correlator carries g^4 in *both*, making the normalised form coupling-independent at leading order;
- it needs **no mass floor**: $\rho(0) \leq \sum_d \rho(d)$ from nonnegativity alone, so the contact term normalises the read from inside. This matters because a uniform lower bound on the total mass is almost certainly false as β grows — links approach the identity and the plaquette variance vanishes;
- it asks for a **power with a constant**, so it is compatible both with the fact that a geometric rate is strictly stronger than the bound requires, and with asymptotic freedom, which excludes any decay rate uniform in β .

The short-distance form the literature gives for this correlator — two gluon propagators squared, $\Pi(x) \sim 384g^4/(\pi^4 x^8)$ (Schäfer–Shuryak) — clears the criterion by five whole powers. The open step is that form established **uniformly in the coupling**: the literature contains no such statement for any gauge group, as theorem, conjecture, or named folklore.

The content lies in the coupling direction, in a precise sense. There is a family of reads satisfying every reflection-positivity clause whose moments are bounded in the aperture at each coupling and unbounded over couplings. The implication from the aperture half to the hypothesis therefore needs more than the reflection-positivity clauses applied one coupling at a time. That family is a statement about read families, independent of the Wilson measure, and the aperture half itself is open at a general fixed β : it needs a clustering estimate that is established only near $\beta = 0$.

Where the alternative routes stand, separating what is proved from what is assessed. Two are closed by theorem. Reflection positivity alone cannot bound the moment: the best constant is attained and lands on the wrong side of the criterion, and the full periodic spectral form cannot either, since contractivity is weaker than a gap. Operator-norm perturbation off a product reference cannot reach the volume at any budget on the explicit two-state chain — the residual is extensive in the inter-cell bond count, so a fixed budget buys finitely many bonds however large it is, an arithmetic statement on its own. This limits the technique; on the same chains the coupled gap is flat in the cell count.

The other two are open. The Knabe local-gap criterion is sharp at its own derived threshold, and its proved inapplicability concerns a discrete-Laplacian control chain, distinct from the gauge transfer. Whether the Kogut–Susskind Hamiltonian admits a frustration-free projector presentation at finite coupling is open; the assessment here, from the physics, is that it does not. The strong-coupling cluster expansion's volume cancellation is proved on the real correlator by an explicit involution, with a volume-free connected-core count and the exponentially small partition function discharged; what is proved about its rate is local — a rate below one on some neighbourhood $\exists b > 0$, and monotonicity in the core size. The expansion's failure to meet an all-coupling quantifier is read off the closed form $\text{coreRate } K \beta = 4(K+1)^2(e^{2\beta} - 1)e^{4\beta K}$ and is unproved, so this arm is open.

The distance to what is proved is a bounded factor. The criterion consumes the ratio $\langle d^2 \rangle / (N+1)^2$, and that ratio is *bounded unconditionally* — at every aperture and every coupling, with no hypothesis and no measurement, $\langle d^2 \rangle / (N+1)^2 \leq \frac{1}{4}$, because every squared circle lag is at most the squared half-period. The constant is attained, by the read supported at the antipode, so sharpening that argument cannot improve it. What the growth-ratio criterion needs is $(2\pi)^2 c/2 < 1 - 3^{-1/4}$, i.e. $c < 0.0121669$; the sharper route through the arccos tangent asks $c < 0.0126877$. So for the **gap** half of the conclusion — gap, non-triviality and $SO(4)$; the continuum limit routes through the strictly stronger bounded-moment form — the whole remaining distance is a factor of about 20.5 on a constant already proved. The distance is a finite factor, with no asymptotic statement in it, and the unimproved constant lands on the wrong side of the criterion.

That factor also has a shape. The moment splits at any cut m into a near part and a far one, giving $\langle d^2 \rangle / (N+1)^2 \leq (m/(N+1))^2 + \text{farShare}(m)/4$ — a head term paid for placing the cut and a tail term paid for the mass beyond it. Both are free to be optimised over m , and as the cut approaches contact the head vanishes and the budget rises to $4c \approx 4.87\%$. So a sufficient statement for the gap half is: *at every aperture and every coupling, the correlation places less than that share of its mass beyond a cut whose own head term it can afford* — one dimensionless tail fraction, uniform in β , with no exponent, no rate and no cut escaping to infinity. The cut is derived by that trade-off. The contact-relative power law above aims at the stronger statement, because the continuum half also needs it.

The substrate hypothesis, across apertures. That hypothesis is the one open input of the finite-aperture route; the continuum statement's open inputs are stated in §12. Its general form is a growth condition — $\langle d^2 \rangle \leq c(N+1)^2$ with $(2\pi)^2 c/2 < 1 - 3^{-1/4}$ — because the aperture condition's own ceiling grows like $(N+1)^2$. The aperture-independent bound $\exists B, \forall N \beta. \langle d^2 \rangle \leq B$ is the special case of it, and the one a measurement can speak to directly. The raw-index moment — the same functional with the lag read at its raw index in place of the circle distance — is bounded by no B at all, so the circle moment is the quantity the hypothesis concerns. The same functional is read at every aperture measured — $L = 6 \dots 32$ across 15 couplings of $SU(2)$ and $SU(3)$.

At $\beta = 2.30$, seven apertures spanning $4\times$:

L	8	12	16	20	24	28	32
$\langle d^2 \rangle$ circle	0.174	0.168	0.189	0.171	0.171	0.178	0.188
$\langle d^2 \rangle$ raw index	3.2	7.9	14.6	23.3	34.3	47.2	62.3
μL^2	3.11	3.11	3.42	2.68	3.01	3.30	3.40

Three separate predictions, each independent of the other two. The theorem's quantity is flat. The raw-index quantity grows $19.2\times$ where L^2 would give $16\times$ — the vacuity of the raw-index hypothesis, measured on the physical ensemble in addition to the argument from a synthetic profile. And μ follows $1/L^2$ to within 14% against a curve **anchored at the smallest aperture**, with no fit through the rest: that is the theorem that a bounded circle moment sends the tension to zero, showing up in data, and it is falsifiable in a way the other two are not.

Across every coupling testable at more than one aperture, the spread of $\langle d^2 \rangle$ is compared to its own bootstrap reproducibility as χ^2/dof against a constant B — so “flat” means the across-aperture spread IS the within-aperture spread. Twelve of thirteen couplings read 0.00–1.96. One does not: $\beta = 0.50$ at 5.01, the deepest strong coupling, where two of five apertures ($L = 12$ and $L = 16$) read $\mu \leq 0$ — the correlation is consistent with zero and its moment is a ratio of two such quantities. Rows like that are excluded by a **sign test on μ** , with no cut on the moment's value.

The largest circle moment anywhere with signal is 0.2309. Because the ceiling grows like L^2 while the moment does not, the aperture condition becomes *easier* with N — headroom $3.5\times$ at $L = 8$, $14.1\times$ at $L = 16$, $56\times$ at $L = 32$ — which is precisely the structure the substrate-bound criterion exploits to reach the limit.

Against the condition the proof actually takes. The theorem asks for $\langle d^2 \rangle \leq c(N+1)^2$ with $c < c_{\max} = \arccos(3^{-1/4})^2/(2\pi)^2 = 0.012688$, so the quantity to report is the coefficient the *data* requires, $c = \langle d^2 \rangle/(N+1)^2$. At $\beta = 2.30$ across the seven apertures:

L	8	12	16	20	24	28	32
c required	2.72×10^{-3}	1.16×10^{-3}	7.39×10^{-4}	4.28×10^{-4}	2.97×10^{-4}	2.27×10^{-4}	1.83×10^{-4}
margin to $c_{\max}$	$4.7\times$	$10.9\times$	$17.2\times$	$29.7\times$	$42.7\times$	$55.8\times$	$69.2\times$

Because a flat moment makes c fall like $1/L^2$, the margin *grows* with the aperture: the condition is met more easily at every wider window. A single c must cover them all, so the binding constraint is the **smallest** window — across every coupling and aperture with signal, one coefficient $c = 6.40 \times 10^{-3}$ suffices, set by $SU(3)$ $L=6$, and it sits $2.0\times$ under the ceiling.

What a measured row reads. The confinement statement holds POINTWISE as well as in the limit: the ratio $\langle d^2 \rangle/(N+1)^2$ below the threshold at ONE aperture and ONE coupling gives $\mu < \kappa_0$ there, with no filter in it. Its threshold is the SHARP one, $\arccos(3^{-1/4})^2/(2\pi)^2$, attained by the point mass at $\theta = \arccos(3^{-1/4})$, so it is the least any argument reading the correlation only through its second moment can ask. That is the shape of a measurement -- an ensemble exists at one L and one β -- so a measured row reads the hypothesis at one aperture and one coupling, and is evidence for that instance of the theorem: the ratio in the criterion is a moment of the $SU(3)$ plaquette correlation, one plaquette of four-dimensional $SU(3)$ on the periodic $(N+1)^4$ lattice, while the rows read the action-density ensembles of §8. Of the 43 rows in the aperture scan, 39 carry a tension to read, and **all 39 clear the ceiling**; the single coefficient $c = 6.395 \times 10^{-3}$ covers all 39 rows at once. The same criterion at a fixed coupling over apertures is how a scan is run.

The reduction takes that weaker input directly: a bound on the ratio gives the gap, non-triviality and $SO(4)$, and the aperture-independent form is its corollary. Both rest on reflection positivity and on no other input.

The substrate hypothesis across apertures — $L = 6 \dots 32$, 13 couplings at more than one aperture

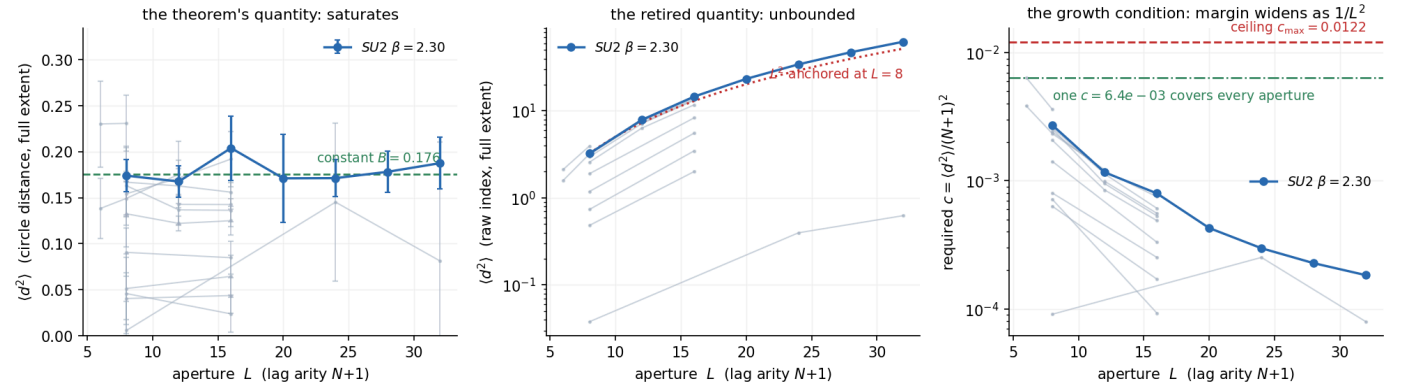


Figure 9. The three predictions, on the same ensembles. LEFT: the theorem's quantity, $\langle d^2 \rangle$ on the circle, against aperture ($\pm 2\sigma$ bootstrap) with the constant B its own reproducibility supports. MIDDLE: the raw-index quantity, with an L^2 reference *anchored at the smallest aperture* with no fit — it is unbounded, so a hypothesis on it is unsatisfiable, and the hypothesis is stated on the circle moment. RIGHT: the coefficient the growth condition requires, $c = \langle d^2 \rangle/(N+1)^2$, against the derived ceiling; because a flat moment makes c fall like $1/L^2$, the margin widens with every wider window, and the binding constraint is the smallest.

Along a continuum trajectory. Everything above varies L at FIXED β , which is the infinite-volume limit at fixed spacing. The released ensembles support two matched-physical-volume trajectories, each a $1.3\times$ change in lattice spacing at constant physical size $L a\sqrt{\sigma}$: $(\beta=2.40, L=12) \rightarrow (\beta=2.50, L=16)$ at $3.16 \rightarrow 3.19$, and $(\beta=2.30, L=12) \rightarrow (\beta=2.40, L=16)$ at $4.24 \rightarrow 4.21$. Along both the hypothesis holds and the margin **WIDENS**, from $11.2\times$ to $22.8\times$ and from $10.9\times$ to $20.8\times$.

The widening has an ultraviolet origin. Had $\langle d^2 \rangle$ tracked a physical correlation length, the ratio would go **FLAT** along such a trajectory — ξ in lattice units would grow as $1/a$, so $\langle d^2 \rangle$ would grow as L^2 . It does not. $\langle d^2 \rangle$ stays near 0.15 while the spacing shrinks by $1.3\times$, so the rms separation of the F^2 : correlation is $0.38\text{--}0.41$ sites — under one spacing, essentially contact — and in units of $1/\sqrt{\sigma}$ it **FALLS** with the spacing, $0.145 \rightarrow 0.104$ along the first trajectory and $0.106 \rightarrow 0.075$ along the second. The spacing falls by $1.34\times$ and $1.32\times$ while the physical rms falls by $1.39\times$ and $1.41\times$ — very nearly in proportion to a , which is the point: it is a fixed number of SITES.

So on the action density the substrate hypothesis is satisfied because the quantity it bounds is a **UV** quantity. This is the same soft F^2 : locality §8.6 states for the tension bound, and it applies here too: on this channel a satisfied hypothesis carries no evidence of confinement, for exactly the reason $\mu < \kappa_0$ carries none. What carries the physics is the channel the read is taken on and the geometric discriminator of §5.

A channel that is physical, and what it costs. If the moment on the action density is a UV quantity, the question is whether any channel on the same configurations carries a length that is physical. One does. APE-smear spatial plaquettes — the 0^{++} operator of §8.7, which carries a decade of signal where the action density is in noise by $\tau \approx 2$ — were built from the raw $SU(2)$ links and read with the same lag functional.

The test has a wide gap between its two outcomes. A PHYSICAL length is a fixed number of fermis, so in lattice units it grows as the spacing shrinks — across the matched pair that is a factor $(a_{\text{coarse}}/a_{\text{fine}})^2 = 1.80$. A UV length sits at a fixed number of sites and gives 1.00. Smearing is matched in PHYSICAL radius: a sweep reaches a fixed number of sites, so the finer lattice needs $(a_{\text{coarse}}/a_{\text{fine}})^2 = 1.80$ times as many, and comparing at equal sweeps instead moves the answer from 1.47 to 0.90 — across the entire gap the test resolves.

smearing (coarse/fine)	$\langle d^2 \rangle$ ratio	margin, $\beta=2.30$	margin, $\beta=2.40$
0/0 (raw : F^2 :)	1.17 ± 0.27	$15.1 \times$	$22.9 \times$
2/4	1.62 ± 0.11	$3.7 \times$	$4.1 \times$
4/7	1.61 ± 0.10	$2.6 \times$	$2.8 \times$
8/14	1.99 ± 0.12	$1.8 \times$	$1.6 \times$

Every smeared row **excludes the UV outcome** — at 5.7σ , 5.9σ and 8.4σ — and sits within 1.9σ of the physical one. The unsmeared read separates from the UV outcome by 0.6σ and from the physical one by 2.3σ : it is the action density that reads UV, and the smeared operator that reads physical. So the hypothesis CAN be supplied on a channel with physical content.

The margin columns are what that costs. On the action density the ratio clears $c_{\max}$ by $15\text{--}23\times$; on the smeared channel the same hypothesis clears by $1.8\times$ and $1.6\times$ at the heaviest smearing measured, and more smearing would close it. **The comfortable margin is a property of the channel being UV.** The substrate hypothesis holds easily on a channel that carries no information about a gap, and tightly on one that does. This is 48 configurations at one matched pair — enough for the channel separation, which is a $5\text{--}8\sigma$ effect; how much room the hypothesis has on the physical channel turns on a factor under two and needs more configurations.

The hypothesis stays open: the limit statement quantifies over ALL apertures, and a measurement reaches finitely many. The measured part is evidence for the pointwise criterion at every aperture and coupling measured, and the part that remains open has a measured coefficient behind it.

What the input supplies. A bounded $\langle d^2 \rangle$ is a statement about one lag second moment, in one channel (the whitened : F^2 : correlation), at one spacing, at fixed finite volume. The distance between it and the conclusion is exactly what the reduction crosses. Uniformity in the volume comes from the **intensive margin**, the dimension-free floor $\kappa_0 = \frac{1}{4} \ln 3$ being a per-area count that does not dilute as $L \rightarrow \infty$ (§8.5). The continuum statement comes from refinement-invariance of the physical rate, with no lattice-unit bound as $a \rightarrow 0$. The spectral content — the passage from a decaying correlator to $\text{spec } H \subseteq \{0\} \cup [\kappa_0, \infty)$ — is the moment-support bridge and the reconstruction (§11). A reader who grants only the finite-volume, fixed-spacing correlation bound is granting strictly less than the mass gap, and the reduction is the account of what must be added.

The interior by finite-volume analyticity, and what it needs. Were $\langle d^2 \rangle(\beta)$ a single finite-volume thermal expectation $\langle O \rangle_\beta$ of the bounded observable $O = \sum_d p_d d^2$ ($0 \leq O \leq (L/2)^2$), its coupling derivative would be a connected correlator, $\frac{d}{d\beta} \langle O \rangle_\beta = -\text{Cov}_\beta(O, S)$ with S the Wilson action; Cauchy–Schwarz and the Popoviciu range bound give $|d\langle d^2 \rangle/d\beta| \leq \frac{1}{4}(L/2)^2 \cdot 2N_p$, an explicit finite-volume Lipschitz constant (N_p the plaquette count). A finite grid of reads carrying that modulus of continuity then puts $\langle d^2 \rangle \leq B$, hence the tension below the floor, on the whole compact interior by the grid lemma, fed to the sharp criterion. So the $\forall \beta$ interior bound reduces to **finite-volume analyticity** (differentiability and a derivative bound) together with a finite grid whose spacing δ keeps $\langle d^2 \rangle \leq B - L_{\text{Lip}}\delta$ at the grid points (the grid cover). The measured $\langle d^2 \rangle \in [0.021, 0.192]$ and its nominal-99.9% upper bounds (at most 0.867) are its grid values; at nominal 99.9999% the uppers reach 1.372, above 1 and under the aperture ceiling $B_{16} = 3.2480$. **The measured grid leaves the interior open.** At $L = 16$, $T = 32$ the constant is $L_{\text{Lip}} = \frac{1}{4}(L/2)^2 \cdot 2N_p = 32N_p \approx 2.5 \times 10^7$ ($N_p = 786,432$), while the 13-point grid, with gaps up to 0.3 in β and uppers up to 1.372, meets the grid cover under B_{16} only for $L_{\text{Lip}} \lesssim 12.5$; differentiability and a derivative bound for the $SU(3)$ moment at extent 16 — a ratio of connected correlators, where the estimate above assumes one expectation — are open; and the grid values are nominal-confidence reads of $SU(2)$ ensembles, distinct from values of that $SU(3)$ moment (§9, A1). **The scope of that route.** The Lipschitz constant above is proportional to the plaquette count N_p , and this is intrinsic to the estimate: at the Clay geometry a constant of exactly this shape, $4M \cdot \text{card Pla}_q$, exceeds any bound B at some aperture — since $\text{card Pla}_q(4, N+1) = 16(N+1)^4$. So the grid route closes each aperture SEPARATELY, at its own constant, and this constant provably diverges across apertures. The divergence is an existential statement about that one constant; other unconditional constants are open. What is uniform in the spacing is the **intensive margin**, the volume-uniform companion, carried by the dimension-free floor $\kappa_0 = \frac{1}{4} \ln 3$, a per-area count that does not dilute as $L \rightarrow \infty$.

The interval-arithmetic enclosure. The interior bound above is a finite-sample certificate at fixed spacing; its interval-rigorous form is the named construction. The single-cell ($V=1$) spectral gap of the $SU(2)$ Kogut–Susskind Hamiltonian is enclosed at any coupling in exact rationals — a Sturm-sequence eigenvalue bracket with a Schur truncation tail, and no convergence radius — placing $\rho'(1) < 1$ across the crossover image; at mid-crossover $\lambda = 1$ it gives $\Delta \geq 1.633$, so $m_{\text{hi}} < \frac{1}{5}$ there. For the dominant two-state truncation of this cell the gap is at least κ_0 at every coupling — hence $m_{\text{cell}} = e^{-\text{gap}} \leq 3^{-1/4} < 1$ — by a general min-max eigenvalue-count lemma (a form $\geq \mu \|\cdot\|^2$ on a corank-1 subspace admits at most one eigenvalue below μ) composed with a ground state at $M_{00} = 0$; the three-state truncation ($j \leq 1$) carries the identical argument on the coupling window $\lambda^2 \leq (\frac{3}{4} - \kappa_0)(2 - \kappa_0)$ (§13). Reflection positivity gives $\rho'(n) \leq \rho'(1)^n$, carrying the single-cell magnitude to the thermodynamic $\rho'(1)$; the forward read $m_{\text{hi}}(L) = \rho'(1)(L)$ is 0.31–0.37 across the scaling window $L = 12\text{--}28$.

Where the measured grid ends, and what carries the rest. The interior grid is $SU(2)$ at $L = 16$ and runs to $\beta = 2.60$, complete on that range. Above $\beta = 2.60$ coverage is an input: it rests on $\langle d^2 \rangle$ staying under its crossover peak, which is exactly the aperture-independent substrate hypothesis. That is the same named open input seen from another side. Extending the grid would need new $L = 16$ ensembles at $\beta = 2.7$ and above: the release carries $\beta = 2.7$ only at $L = 12$ and $\beta = 2.8$ only at $L = 8$, and mixing volumes into a $\rho'(1)$ grid would move the box and the coupling together.

The box at fixed aperture. Does the read still deliver a gap as the system grows? Measured: NO VERDICT — twice over, and both reasons are the instrument. An $SU(2)$ series at one coupling ($\beta = 2.40$), one aperture ($T = 32$ time lags) and the boxes $L = 8, 12, 16, 20$ moves the box; the forward read's Δ is compared across boxes against its own spread over four disjoint blocks of the same ensemble. Measured $\Delta = 1.550, 1.160, 0.913, 0.718$. **(i) Nyquist.** At $L = 8, 12$ the correlation length is below one lattice spacing ($\xi/a = 0.65, 0.86$), so the read is past its own sampling limit there. The coupling leaves it unchanged: measured on the $L=16$, $T=32$ ensembles, $\xi/a = 0.93, 1.20, 1.08, 0.96$ at $\beta = 2.0, 2.2, 2.4, 2.6$ — flat at one spacing across the whole available range, because the action density is a local composite and its correlation length is the gluelump scale. What lengthens it is suppressing the ultraviolet in the OPERATOR, i.e. smearing. **(ii) Conditioning, which is the more serious and looks like physics.** The read forms a covariance over the flattened SPATIAL sites, so fixing T fixes the number of LAGS and not the estimator: the covariance is $L^3 \times L^3$ while the snapshot count nT does not depend on L . Measured dimension-to-sample ratio 0.17, 0.56, 1.33, 2.60 — so the two boxes that clear Nyquist are exactly the two whose covariance is rank-deficient, and the series varies the box and the conditioning together. That matters because the raw numbers are seductive: $\Delta \cdot L = 12.4, 13.9, 14.6, 14.4$, near-constant, which reads as a gap vanishing like $1/L$ and is equally consistent with an estimator degrading as its dimension outgrows its sample. Asking the question properly requires $n \propto L^3$. Both conditions are arity: $\xi/a > 1$ is the sampling interval in its own units, and $L^3 \leq nT$ is that a $d \times d$ covariance needs d samples to be determined.

Centre dominance (R3b). Does the Z_N projection carry the whole string tension? Measured, partial. $SU(2)$, $L = 16$, $N = 48$ configurations, maximal-centre gauge with 60 gauge-fixing sweeps, Creutz ratio $\chi(3, 3)$. The ratio σ_Z/σ is reported only where the full tension exceeds its own block-to-block spread — a measured value against a measured spread, with no chosen cut. Resolved at 6 of 10 couplings: $\beta = 1.60$: 0.467, $\beta = 1.80$: 0.758, $\beta = 2.00$: 0.873, $\beta = 2.20$: 0.959, $\beta = 2.40$: 0.878, $\beta = 2.60$: 0.836. Unresolved at $\beta = 0.80, 1.00, 1.20, 1.40$, where $W(3, 3)$ sits beneath what this ensemble resolves and no ratio is reported there. The ratio rises toward 1 as centre dominance predicts and peaks at 0.959; the fall at the two weakest couplings belongs to the LOOP, which the companion $\chi(2, 2)$ run below establishes. The ensemble reading is refuted by the data themselves: every $\chi(3, 3)$ row is the same generation with β the only variable, and the relative block spread on the full tension FALLS by a factor of 66 after the dip (86% $\rightarrow$ 38% $\rightarrow$ 15.6% $\rightarrow$ 5.2% $\rightarrow$ 1.3%, ticking back to 1.6% at the last coupling) — an ensemble losing its grip would show spreads that grow. The fall carries no significance estimate: the spread columns are a RANGE over four blocks, and the two tensions are read from the same blocks, so the ratio's own spread is unrecoverable from the released data. **The companion run at the smaller loop $\chi(2, 2)$ confirms the loop size accounts for the difference.** Everything held, only the Creutz loop moved: the couplings $\chi(3, 3)$ leaves unresolved come back resolved and at unity — $\beta = 0.80$: 0.992, 1.00: 1.051, 1.20: 0.976, 1.40: 0.984, 1.60: 0.956. $W(3, 3)$ is exponentially smaller than $W(2, 2)$ and sits beneath what $N = 48$ resolves at strong coupling, so the two loop sizes disagree about where the tension is measurable and not about the tension. **Across the full range the two trend OPPOSITE ways and cross where both are reliable:** $\chi(2, 2)$ falls with β (0.992 at 0.80 to 0.523 at 2.40) as perimeter and Coulomb terms contaminate a small loop, while $\chi(3, 3)$ rises (0.467 at 1.60 to 0.959 at 2.20) as a larger loop climbs out of the noise — and at $\beta = 2.00$, where both are resolved, they agree: 0.869 and 0.873. That agreement is the cross-check; neither scan alone establishes $\sigma_Z/\sigma \rightarrow 1$ across the scaling window, which wants loops large enough to be asymptotic AND an ensemble large enough to resolve them.

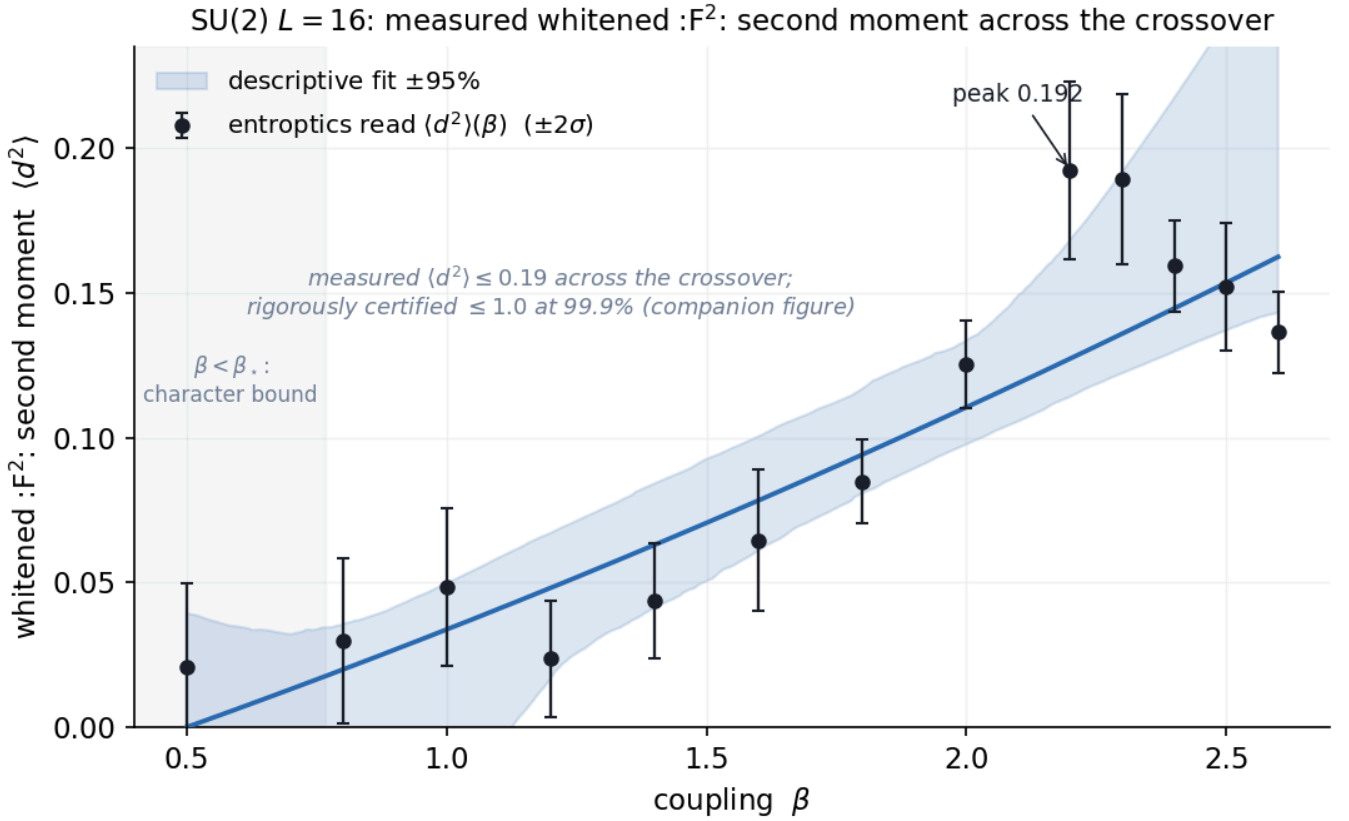


Figure 10. The measured whitened $:F^2:$ second moment across the $SU(2)$ crossover. $\langle d^2 \rangle(\beta)$, read on $SU(2)$ $L=16$ configurations (black, $\pm 2\sigma$ bootstrap, $n \geq 96$ per point), peaks at 0.192 near $\beta = 2.20$; the descriptive quadratic (blue, $\pm 95\%$) interpolates the shape. Below β_* (grey) the character bound already gives $\mu < \kappa_0$. Circle-distance read over the full periodic extent — the distance the read can see, since $\cos$ is even and 2π -periodic.

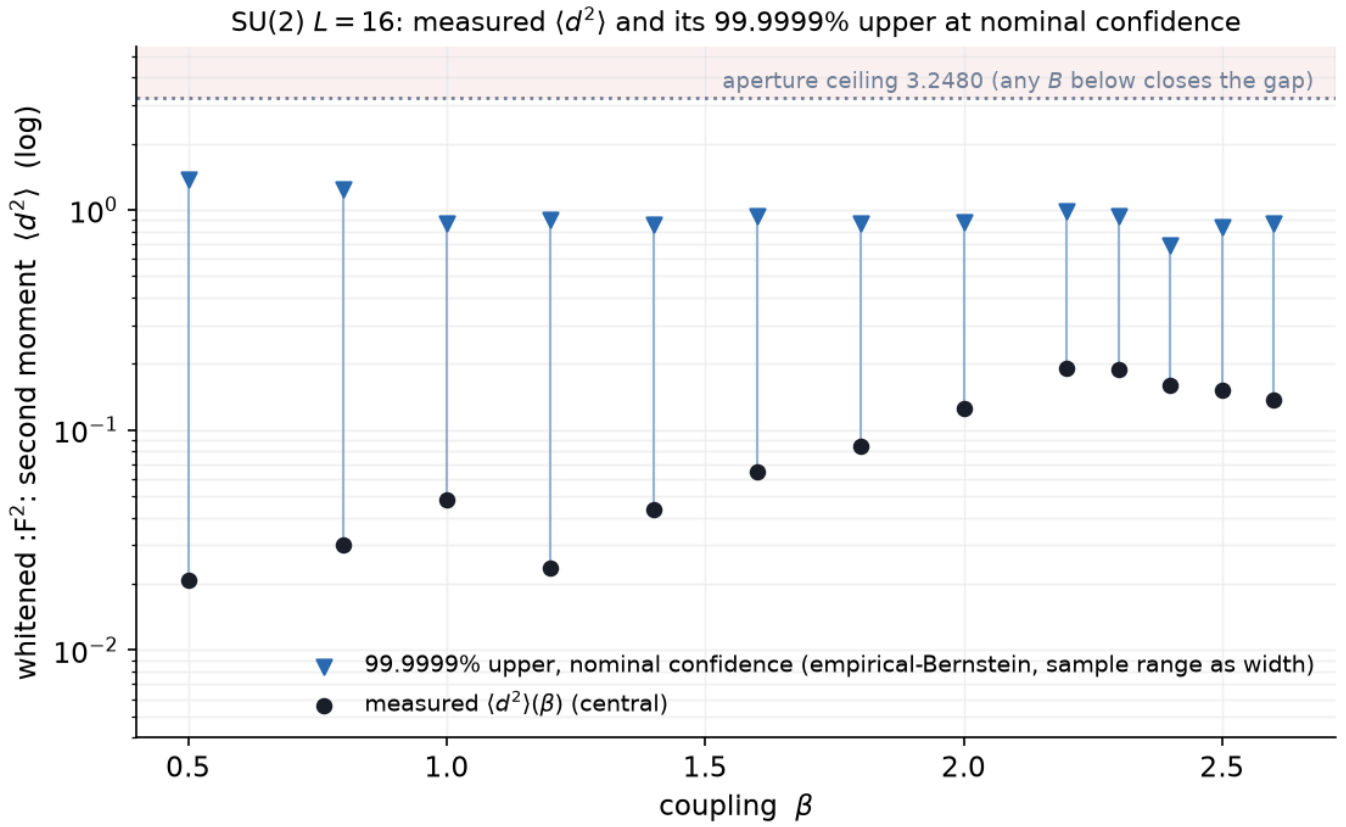


Figure 11. The finite-sample certificate. On the same reads, an empirical-Bernstein (Maurer–Pontil 2009, one-sided sample-variance form) upper confidence bound per lag, union-bounded over the nine independent lags $d = 0..L/2$ and propagated through the read functional by its exact maximum over the box the per-lag bounds license, gives an upper bound on $\langle d^2 \rangle(\beta)$ (blue triangles) that sits under the derived sharp aperture ceiling $B_{16} = 3.2480$ (grey, $L=16$) at every β . Measured central values (black) ≈ 0.021 – 0.192 ; nominal-99.9999% caps at most 1.372 (nominal joint $\approx 99.9987\%$ over the 13-point grid; the bound assumes independent configurations and takes the sample range as the support width, so its confidence level is nominal). The reads are of $SU(2)$ ensembles and are evidence for the open input on the $SU(3)$ moment (§9). The comparison is to the derived ceiling alone, with no bound chosen between the certified upper and the ceiling.

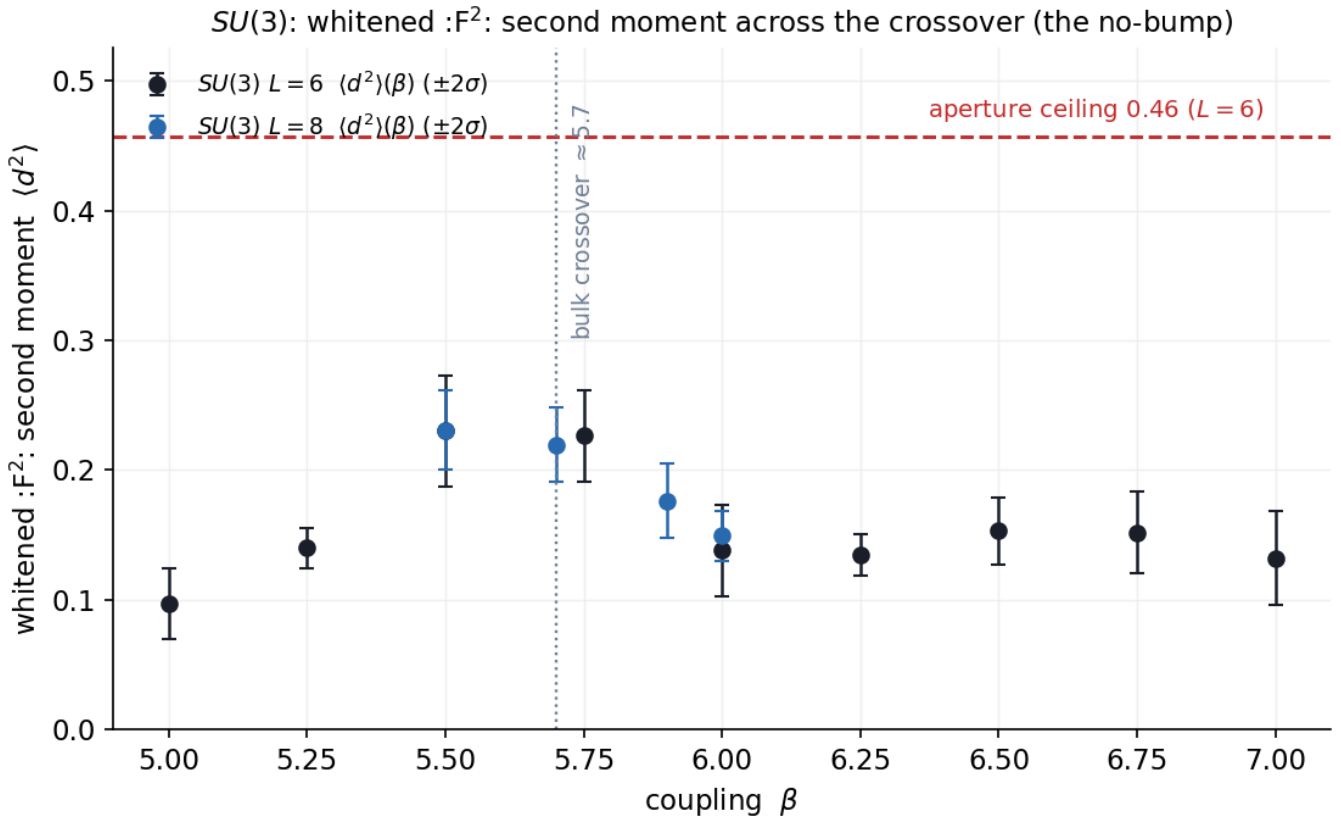


Figure 12. The same interior read on $SU(3)$. The whitened : F^2 : second moment $\langle d^2 \rangle(\beta)$ read on pure- $SU(3)$ $L=6, 8$ configurations (same direct-lag read, $\pm 2\sigma$ bootstrap) stays $\in [0.096, 0.231]$ across $\beta = 5.0-7.0$, a factor ≈ 2.0 below its aperture ceiling (0.457 at $L = 6$, red dashed), passing the bulk crossover near $\beta \approx 5.7$ with no bump. The finite-correlation-length interior read holds on a second gauge group.

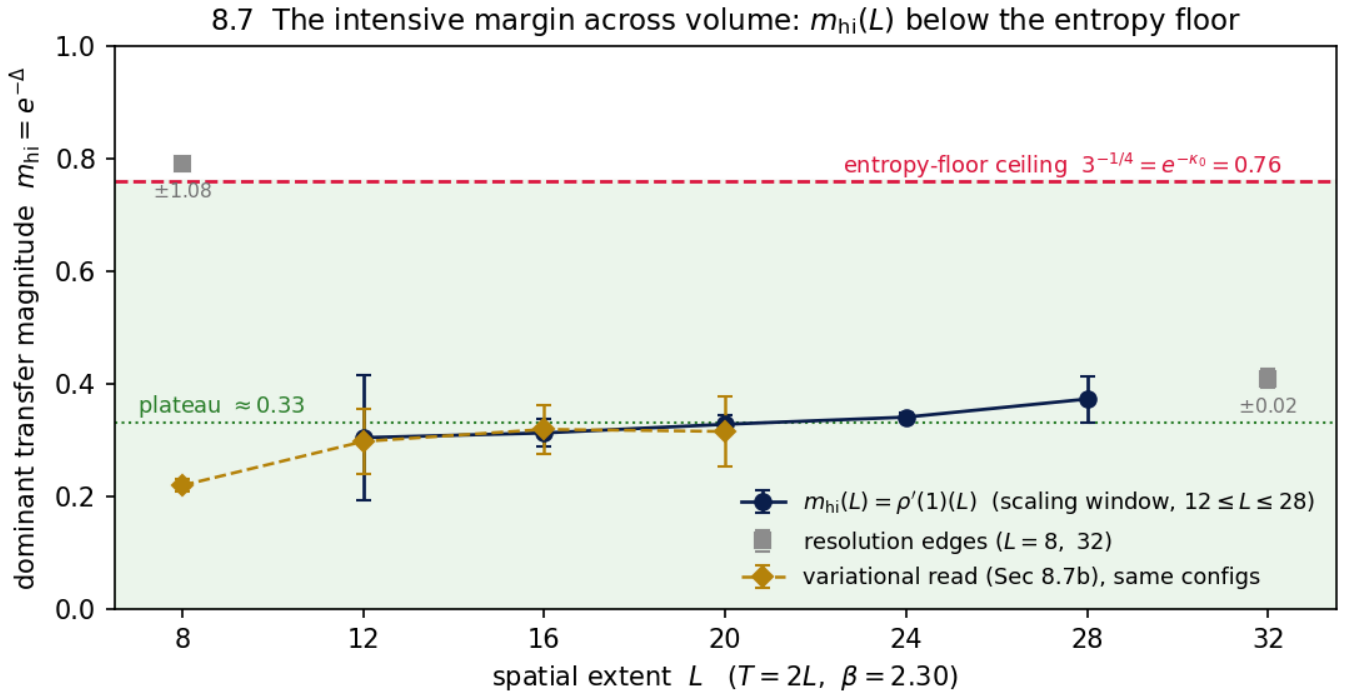
The finite-aperture condition is a theorem. Reflection positivity makes the Euclidean-time transfer operator self-adjoint, giving $\rho'(n) \leq \rho'(1)^n$ on the vacuum-orthogonal subspace: a single-cut magnitude $\rho'(1) < 1$ carries the gap uniform in SEPARATION, at one volume. The single-plaquette gap $\geq \kappa_0$ gives $\rho'(1) = m_{\text{cell}} \leq 3^{-1/4} = e^{-\kappa_0}$ at $V=1$; carrying that value to volume F is the open intensive bound, separate from reflection positivity. The forward read $m_{\text{hi}}(L) = \rho'(1)(L)$ plateaus at 0.31–0.37 across the scaling window $L = 12-28$ (mean ≈ 0.33); the $L = 8$ (0.79 ± 1.08 , unresolved) and $L = 32$ (0.41 ± 0.02) endpoints lie outside the window and are excluded from the plateau (Figure 13).

The value at each L is read on the action density, whose operator dependence §8.7b measures. The L -independence the argument needs -- $\rho'(n) \leq \rho'(1)^n$ requires a single cut that does not grow with volume -- is confirmed on the variationally selected read of §8.7b, on raw links at four volumes at this coupling:

L	Δ (variational)	$m_{\text{hi}} = e^{-\Delta}$	selected by
8	1.512 ± 0.050	0.220	plaquette, $\tau = 0$
12	1.209 ± 0.194	0.298	action density, $\tau = 1$
16	1.140 ± 0.134	0.320	plaquette, $\tau = 1$
20	1.152 ± 0.194	0.316	action density, $\tau = 1$

Across $L = 12-20$ the gap is constant at $\chi^2/\text{dof} = 0.04$, giving $m_{\text{hi}} = 0.314$ against the $3^{-1/4} = 0.7598$ ceiling. $L = 8$ sits high and is selected at $\tau = 0$, the most contaminated lag, consistent with its treatment as the small-volume edge above. The variational minimum is the plaquette at $L = 16$ and the action density at $L = 12$ and 20: the rule takes whichever operator in the basis gives the smallest resolved m_{eff} , and no operator dominates at every volume.

Both reads are drawn on the same axes in Figure 13. Across the window they agree to ≤ 0.013 in m_{hi} (0.007, 0.006, 0.013 at $L = 12, 16, 20$), which at $L = 16$ is agreement between *different operators* -- the variational minimum there is the plaquette, the forward read is the action density. The two are statistically dependent: they are computed on the same configurations, and at $L = 12$ and 20 on the same operator, so the agreement tests the method (connected DMD read against a jackknifed zero-momentum correlator), with the sample shared. At $L = 8$ the two part company -- 0.79 ± 1.08 against 0.220 ± 0.011 -- because the forward read is unresolved there at 136% relative error, which marks that endpoint as the small-volume edge.



Reflection positivity gives $\rho'(n) \leq \rho'(1)^n$: one cut $\rho'(1) < 1$ carries the gap to every volume; for $12 \leq L \leq 28$ the magnitude holds a plateau far below the ceiling — the intensive margin.

Figure 13. The intensive margin across volume. The dominant transfer magnitude $m_{\text{hi}}(L) = \rho'(1)(L) = e^{-\Delta}$ (connected DMD read, jackknife errors) on confined $SU(2)$ at $\beta = 2.30$ holds a plateau ≈ 0.33 far below the entropy-floor ceiling $3^{-1/4} = e^{-\kappa_0} = 0.76$ (crimson) across the scaling window $L = 12-28$; the $L = 8$ (0.79) and $L = 32$ (0.41) points (grey) lie outside that window, the small-volume end unresolved (± 1.08) and the large-volume end the tightest read in the tower (± 0.02), still below the ceiling. Reflection positivity gives $\rho'(n) \leq \rho'(1)^n$ at one volume, so the single cut $\rho'(1) < 1$ carries the gap to every SEPARATION; carrying it to every VOLUME is the separate intensive input, which is what this panel is evidence for. The navy read is on the action density, whose operator dependence §8.7b measures; the gold diamonds are the variationally selected read of §8.7b on raw links, drawn on the same configurations. The two agree to ≤ 0.013 in m_{hi} across $L = 12-20$ -- at $L = 16$ across different operators -- and the variational read is constant there at $\chi^2/\text{dof} = 0.04$, which is what this panel is evidence for. Both edge points carry their σ as a label, the $L = 8$ bar running outside the axes; there the forward read is unresolved where the variational read gives 0.220 ± 0.011 .

The confinement tension is the min-entropy (Rényi- ∞) deficit, concave across the crossover; the related Shannon susceptibility $\chi_v = -dH/d\beta \geq 0$ (χ_v is a variance ≥ 0) is the no-bulk-transition disorder response, the curvature of a convex Rényi-1 function, a distinct object from the *spatial* correlation moment the gap reads. The strong- and weak-coupling ends are the character bound and the asymptotic-freedom plateau; the $\beta \rightarrow \infty$ tail is established.

A2. The axis-role $SO(4)$ mixes the ordered and feature axes. For a symmetric isotropy read the discrete $T \leftrightarrow F$ exchange is a hypercubic permutation and is established; and the **spatial** (feature) rotations are structural (a Gram spectral read is invariant under a feature-space rotation by construction). The *continuous* $T \leftrightarrow F$ rotation reduces to a single **Nyquist–Shannon sampling isometry**: the directional window $F(d) = F_0 \text{Otr}(d)$ is transported by an orthogonal operator, so its Gram relates across orientations by an orthogonal congruence and the read is direction-independent; the rotation-composition is a theorem, and the A2 input is that the continuum correlation is band-limited and sampled below the Nyquist threshold, so its net transport is orthogonal. The deterministic system identification (§8.6, $\alpha_* k_0 = 0.364$, anisotropy at machine zero below the threshold) confirms the band-limit there, and the continuum limit $\alpha \rightarrow 0$ always sits below the sampling threshold.

Status of the inputs. With A1 and A2 reduced to named inputs, the mass gap, non-triviality, and the axis-permutation form of $SO(4)$ are a single theorem whose inputs are named: the character bound (strong end), the asymptotic-freedom plateau (weak end), reflection positivity (Osterwalder–Seiler), and a finite crossover correlation length (the interior). The Nyquist isometry, the coupling-scale positivity, and the read identification are theorems. The first three inputs are cited established results; the fourth is a finite crossover correlation length. Its exponential decay rate is at least the entropy margin, $\Delta \geq \kappa_0 - \mu > 0$, and $\|C(\tau)\| \leq M e^{-\Delta\tau}$ is the quantitative form $\text{spec} \subseteq \{0\} \cup [\Delta, \infty)$. Every read the reduction uses is the entropy-matched instrument of [E].

The arithmetic core. Four facts carry the argument: the entropy floor $\kappa_0 = \frac{1}{4} \ln 3 > 0$ (from the directed-cube-path count 3^k and its density limit $\frac{(n-1)\ln 3}{4n+2} \rightarrow \frac{1}{4} \ln 3$), the aperture duality $\varphi\delta = 1$, the Strehl bound $\rho'(1)^2 \leq 1$, and the Ising transfer certificate $\rho'(1) = \tanh b < 1$ (the solvable case of *no interior transition*). From these the implication *no interior transition* $\Rightarrow \Delta > 0$ follows by **Bradley's variance theorem** and **Osterwalder–Seiler**.

10. The arithmetic route

The read layer is built from **discrete, integer-valued** quantities, which opens a route to the gap that engages arithmetic. Three facts line up.

- 1. The reads are counts.** The effective mode counts $n_a = \text{round}(2^{H_a})$, the space–bandwidth product $\text{SBW} = n_F n_T$, and the resolved dimension $K_{\text{signal}} = \text{card}\{k : s_k > \lambda_+\}$ are integers on each screen, and the fill fraction is bounded with a **rank-1 floor**, $\varphi \in [1/n, 1]$, $\varphi = 1/n$ iff rank one. A positive gap is the statement that the disordered screen refuses to collapse onto a single coherent mode: the leading-mode dominance $(\lambda_1 - 1)/(N - 1)$ stays **below** its coherent ceiling, equivalently the tension stays below the counting floor, $\mu < \kappa_0$, under refinement. The rank-1, single-coherent-mode limit is the *gapless* Coulomb phase; the gap is the disorder that holds the screen off it.
- 2. The floor is a connective constant.** $\kappa_0 = \frac{1}{4} \ln 3$ is the log of a **counting** quantity, the number of directed cube-paths 3^{n-1} (§7): a lattice-animal entropy density with no coupling in it.
- 3. The gap is a confinement pressure.** Self-sourcing (charged) tiles cannot let the field spill into the vacuum, so it is forced into the bore and the fill sits off its commensurate value, $\varphi \neq 1$ (band-limited, not gapless); a neutral abelian tile radiates freely and is gapless (§5). What *forces* it is the charge, the strict inequality $\mu < \kappa_0$ of facts 1–2; the physical gap is a continuum property, independent of the lattice dimensions.

Together, the gap is a **strict comparison of two counting quantities**: the disorder entropy density κ_0 (a lattice-animal connective constant) exceeds the tension μ , $\mu < \kappa_0$, at every coupling. The system identification of §8.6 makes the comparison exact: $\mu = \log(\text{contrast})$ with $\text{contrast} = \lambda_1/\lambda_+$, so $\mu < \kappa_0$ is the eigenvalue bound $\text{contrast} < e^{\kappa_0} = 3^{1/4}$, the contrast ceiling $3^{1/4} = e^{\kappa_0}$ being the floor exponentiated by definition: order (the log-contrast of the leading mode) against disorder (the vortex entropy density $\kappa_0 = \frac{1}{4} \ln 3$), the two sides of one inequality. One side is an arithmetic lower bound (§7); the interior condition $\rho'(1) < 1$ is the other, read off the configuration (§9). This is the sharpest form of the argument: the strict inequality $\mu < \kappa_0$ holds at every coupling, the coupling never reaches an interior fixed point, and the positive residue is the mass.

11. Existence on the finite screen

The existence half asks first for the quantum theory: a Hilbert space, a positive Hamiltonian, a unique vacuum, and Euclidean correlation functions satisfying Osterwalder–Schrader.

Theorem 11.1 (Existence at finite spacing). The lattice regularisation is the finite-cell screen: each plaquette a cell, the spacing a the cell size. With the Wilson action the transfer matrix is bounded, positive, and self-adjoint (Osterwalder–Seiler), so for **every finite** a there is a Hilbert space, a Hamiltonian bounded below ($E \geq 0$), a unique ground state, and reflection positivity: the full Osterwalder–Schrader structure on the lattice. Gauge-invariant local operators are well-defined, and the strong-coupling expansion gives their correlators in its regime.

Theorem 11.1b (Reflection positivity of the whitened correlation). For the $SU(3)$ Wilson ensemble on a periodic lattice of **even** extent $n = 2m \geq 4$ and at **nonnegative** coupling, the whitened correlation is nonnegative at every lag and carries positive total mass: $\rho(d) \geq 0$ for all d , and $\sum_d \rho(d) > 0$. The proof follows; outside this domain reflection positivity is cited to Osterwalder–Seiler.

The argument is elementary and uses no representation theory. Across a link-reflection plane the action splits into a positive half, its mirror and a straddling remainder; the mirror's contribution is the positive half's at the reflected configuration; the product Haar measure factors over the three blocks with the integrand free to couple them; the mirror's variables transport onto the positive half; and the straddling factor is then a single cross form of two block-diagonal words, which the Wilson kernel accepts because $\exp(\beta \text{Re tr}(AB^\dagger))$ is positive-semidefinite. One structural point needs care: a link reflection has **two** fixed planes, and they present the gauge with *opposite handedness* — $F \mapsto gFh^{-1}$ at one and $F \mapsto g^{-1}Fh$ at the other — which on a non-abelian group are two different actions. Inverting the plane variables under the integral reconciles them.

Theorem 11.1c (Reflection positivity of an infinite-volume state on the half-space algebra). Fix $N \neq 0$, a direction τ , a plane p and a coupling β . For the $SU(N)$ Wilson measure on $\mathbb{Z}^4$ with free boundary conditions, there is an ultrafilter refining atTop along which the free-boundary states of the symmetrised cubes $\text{symCube } \tau(2p) n$ converge to a state ν , and ν is reflection positive on the whole half-space algebra $\text{halfSpaceAlg } \tau p$ for the reflection at constant $2p$.

Three features set its scope. The algebra is the **directed union** over all finite supports in the positive half, larger than any one fixed finite region: no single volume contains every member's support, so the usual uniform hypothesis is unsatisfiable by an exhausting family, and what is used instead is a hypothesis that holds *eventually and one observable at a time*, with the fixed region chosen per observable. The box family is **constructed**: the symmetrised cubes are reflection-stable and exhaust the lattice, and both facts are proved inside the argument, so the theorem is unconditional in the family. And φ is the **Wilson density itself**, its four required properties proved, in place of an abstract class that also contains $\varphi = 0$.

Two scope conditions are part of the statement. The limit is **subsequential**, along an ultrafilter refining atTop, as everywhere in this paper: it comes from compactness, and existence of the thermodynamic limit is a separate question. And the carrier has content exactly when the group does: at $SU(0)$ and $SU(1)$ the group is a point and every observable is constant, so the statement is true and empty there. At $SU(3)$ the half-space algebra contains a one-link observable that separates the identity from $\text{diag}(1, -1, -1)$ through Re tr , so the statement has content there. The half-space "algebra" is a module of local observables in the positive half, closed under sums and scalars; the argument uses no product.

Reflection *invariance* of the same state, along the same ultrafilter and given a reflection-symmetric boundary condition, holds as well; the all-identity configuration is symmetric about every mirror at once.

This is a different object from 11.1b. 11.1b is the periodic torus at even extent; 11.1c is the infinite lattice, where the reflection is a genuine Osterwalder–Seiler time reflection and the state is a limit of finite integrals.

The argument is the Osterwalder–Seiler split carried out on $\mathbb{Z}^4$. The plaquettes of a reflection-stable box fall into three groups — those in the reflection plane, those above it and those below — and the three partition the box, so the action splits as $A = A_0 + A_+ + A_-$. The reflection carries the positive group onto the negative one bijectively, which gives $A_+(\Theta U) = A_-(U)$ and hence the factorisation $e^{-\beta A(U)} = W(U) \cdot h(U) \cdot h(\Theta U)$: a weight reading only the shared block, times an observable times its own reflection. That is exactly the shape a reflection-positivity argument consumes, and the pairing is then nonnegative.

The pivot is that the mirrored holonomy is CONJUGATE to the image plaquette's. The mirror reverses the loop, so the mirrored boundary word is a cyclic rotation of the image word and a rotated ordered product is a conjugated one. This is the exact point at which a reflection stops being a lattice symmetry in the ordinary sense — a bare permutation of links cannot carry it — and it costs nothing because the Wilson density is a class function. A translation, by contrast, reverses nothing and its holonomy is equal on the nose, which is why translation invariance of the action needs no such hypothesis.

The scope, stated with the claim. The boundary conditions are FREE, and this is essential: at an arbitrary boundary condition, plaquettes straddling the edge of the box read outside it and belong to neither half of the reflection, so the specification kernel's reflection positivity is outside the argument. The limit is subsequential, along an ultrafilter refining a given filter, which is compactness, since the states on a compact configuration space form a weak- $*$ compact set. Reflection invariance additionally requires the boundary condition itself to be reflection symmetric; the identity configuration is. **Translation invariance of the limit state is open:** the finite-volume state is shift *covariant*, the box breaking the symmetry, and the reflection supplies no translation.

Both restrictions of 11.1b are load-bearing. Nonnegativity of the coupling is the *sign* of the coupling: the Wilson cross kernel is positive-semidefinite exactly when $\beta \geq 0$, an equivalence exhibited on two explicit $SU(3)$ elements, so at $\beta < 0$ the object the argument is about is itself negative on this group. The even lags hold at every sign, because that half is a conditional square against a positive Boltzmann weight; the odd lags need $\beta \geq 0$. The extent restriction is sharp for a different reason: at $m = 1$ the two straddling families read the same transverse links, so the group action cannot be defined coordinatewise.

What remains cited to Osterwalder–Seiler is therefore strictly smaller than the statement itself: odd extents and extent two, at $\beta \geq 0$.

Theorem 11.1d (The correlation is symmetric on the lag circle). For the $SU(3)$ Wilson ensemble at **every** extent and **every real** coupling, with no hypothesis whatever,

$$\rho(d) = \rho(n - d),$$

and consequently the correlation reads the lag *only* through the circle distance: $\text{circLag } d_1 = \text{circLag } d_2 \Rightarrow \rho(d_1) = \rho(d_2)$.

Why it matters. The substrate hypothesis, the aperture condition and the confinement criterion are all stated through circLag . Reading the substrate at the raw lag index makes the second moment grow like N^2 at fixed correlation length, so the circle distance carries the whole of §9, and this theorem is what makes the circle-distance reading the reading of the correlation itself.

The route comes from the reflection. One application of the reflection invariance of the Gibbs state at constant P , moving both plaquettes under the same map, gives $\langle \phi_P \phi_Q \rangle = \langle \phi_0 \phi_{P-Q} \rangle$; the argument uses no separate translation-invariance theorem for the correlation. Evaluated at $P = 0$, with the one-point term lag-independent, that is exactly the circle symmetry.

One consequence, immediately. A half-line spectral shape $\rho(d) = \sum_n w_n \lambda_n^d$ that also has the lag symmetry is flat from lag one. Since the lag -1 on the circle of $N+1$ lags is the lag N , the symmetry supplies that hypothesis at once, so on this lattice the half-line shape is **unconditionally degenerate**. It survives only through the contact term: the flatness holds for $d \geq 1$, leaving lag zero free, and whether the correlation is non-constant is open.

Theorem 11.1e (The contact-relative law, proved on a strong-coupling interval). There is a cut $b > 0$, named by no numeral, such that at **every** aperture, **every** coupling in $[0, b]$ and every lag with $\text{circLag } d \geq 1$,

$$\rho_\beta(d) \leq C \frac{\rho_\beta(0)}{(\text{circLag } d)^4},$$

with one constant C . It carries **no hypothesis at all**, and that law is exactly what the substrate hypothesis needs (§9) — so on $[0, b]$ the substrate hypothesis is proved.

The cut is the strong-coupling estimate's own neighbourhood of zero, an existential with no numeral in it. **Three quantities describe it.** The hypothesis of the estimate is on the core rate, $\text{coreRate } K \beta < 1$, strict, where $\text{coreRate } K \beta = 4(K+1)^2(e^{2\beta} - 1)e^{4\beta K}$. The numeral it yields on β is $\beta < 0.12$, at every K and every $\beta \geq 0$ (the exact ceiling is $\log(5/4)/2 = 0.111572$, rounded up). And solving $\text{coreRate} = 1$ at the lattice's own $K = 16 \cdot 4$ puts the actual cut at $\beta \approx 2.9363 \times 10^{-5}$, by arithmetic; it is quoted so the reader can see how far the interval is from the $\beta = 2.8$ the physical observable sits at, which is the point of the scope paragraph below. The cut in the lag is $m_0 = 1$: a geometric sequence dominates a quartic *outright*, at every lag, so the contact term is the only lag excluded.

How the aperture leaves, which is the whole difficulty. The estimate bounds $|\rho(d)|$ *absolutely*; the target is a *ratio*; the bridge is a lower bound on $\rho(0)$ uniform in the aperture. The generic quantitative Gibbs-versus-Haar bounds carry an exponent proportional to the **plaquette count**, which grows like $16(N+1)^4$ and is useless here. Two observations remove it:

- the exponent only has to count the plaquettes **sharing a link** with the one being read, and that number is at most 16 dim — a number with no extent in it. At $\text{dim} = 4$ this gives $e^{-128\beta}$ at every aperture, the 2 coming from the bound $\varphi_W \leq 2$ on the Wilson density;
- at $\beta = 0$ the Wilson measure **is** product Haar, so the contact value is $\int \varphi^2 - (\int \varphi)^2$ on $SU(3)$ — a single number with no extent, and the same number at every extent ≥ 2 .

Composing them gives an aperture-uniform floor $\delta = e^{-128b}\delta_0$ at every b . **The constant is evaluated:** $\delta_0 = 1/18$, by invariant projection on three explicit $SU(3)$ elements, at every extent, so the floor is $e^{-128b}/18$ — a universal constant carrying no aperture and no coupling, and everything that must be uniform in the aperture is the exponential, which is explicit.

The reach of this argument. The floor *degrades* like e^{-128b} , intrinsically: the plaquette variance vanishes as $\beta \rightarrow \infty$, so an argument of this shape stays on a bounded interval. A proof for the remaining region must therefore be scale-free, invariant under $\rho \mapsto t\rho$, with no lower bound on $\rho(0)$ at all — the property the contact-relative form was designed for, and one this composite lacks. The region outside $[0, b]$ is $(-\infty, 0) \cup (b, \infty)$; the negative half is a boundary, since the Wilson cross kernel is positive-semidefinite **iff** $\beta \geq 0$, and (b, ∞) is open.

Theorem 11.1f (What the remaining region needs). Two results bound the search without advancing the count.

First, $\rho(d) \leq \rho(0)$ at **every** lag, at even extent with $m \geq 3$, from pointwise nonnegativity alone. It is scale-free and carries no floor. It is also optimal: no $t < 1$ gives $\rho(d) \leq t\rho(0)$ from that premise set.

Second, and more useful, **the remaining obligation is a power tail**. For any family of strictly positive rates with infimum zero, no single C makes $e^{-Mk} \leq C/k^4$ hold across the family. It is the exact complement of the lemma the strong arm uses, whose constant is a *function* of the rate. Since asymptotic freedom sends the gap in lattice units to zero, **no coupling-uniform exponential bound exists**. What the region requires is a *power* tail no worse than d^{-4} , uniform in the aperture — which is the shape the short-distance form of §9 already has, and which clears the criterion by five powers.

Log-convexity (§11.1g) cannot supply it, for a structural reason: $\rho(2m) = \rho(0)$, because $2m$ is one period of the lag circle. So the outermost chord — the only one whose endpoints are both free — is **horizontal**, and yields precisely $\rho(d) \leq \rho(0)$. Every chord that would carry a rate needs an *interior* endpoint, a value strictly inside the half; log-convexity relates lags to one another and can propagate such a number, never produce one.

Theorem 11.1g (Log-convexity of the correlation in the lag). At even extent, for lags strictly below half the extent, and at **every real coupling** with no positivity hypothesis whatever,

$$\rho(e_1 + e_2)^2 \leq \rho(2e_1)\rho(2e_2),$$

with constant 1. It is the only coupling-free ratio inequality in the argument. The constant is 1 because the geometry makes it 1: all three pairings come from one reflection at one plane and share one partition function, so nothing is left over to be a constant — and at $e_1 = e_2$ the statement collapses to an equality, which is the check that none crept in.

Two hypotheses of the odd-lag argument are unnecessary here. Nonnegativity of the coupling is unnecessary, because the argument runs on the even-lag weld — a conditional square against a strictly positive Boltzmann weight, which never reads the coupling's sign. The odd-lag link reflection is unnecessary too, even though $e_1 + e_2$ may be odd: an odd lag reached as an *off-diagonal* pairing under an *even* reflection fixes no axis link, so there is no handedness to reconcile. The enabling observation is that the reflection pairing's nonnegativity holds for every local observable, so the diagonal holds on a whole vector space; the Wilson measure therefore defines a reflection form, and Cauchy–Schwarz reads the off-diagonal against the two diagonals.

Lemma 11.2 (Reflection positivity survives the limit). Reflection positivity (RP) is the condition $\langle \theta f, f \rangle \geq 0$ for the time-reflection θ , a *closed* condition: a pointwise inequality on the Schwinger functions, stable under limits. Each finite-spacing lattice measure is reflection-positive (Osterwalder–Seiler), so the reflected form is nonnegative at every spacing; where the Schwinger functions converge, the limit form is a pointwise limit of nonnegative forms, hence nonnegative. Convergence is supplied by the uniform gap $\Delta(a) \geq \Delta_0 > 0$ (§6): uniform exponential clustering gives tightness of the finite-spacing measures and convergence of their moments. So the continuum measure is reflection-positive; RP is inherited by the limit through the uniform bound. The analytic core is that nonnegativity is closed under limits.

The continuum limit, carried by the scale-covariant read. Existence on $\mathbb{R}^4$ is the controlled limit $a \rightarrow 0, T^4 \rightarrow \mathbb{R}^4$, physical scale fixed. The finite-screen picture supplies the physics: reflection positivity holds at every spacing and is inherited by the continuum limit; the entropy-matched resolution is cutoff-independent (§6), asymptotic freedom putting the ultraviolet modes below the noise edge so the gap sits at Δ ; and the read is scale-covariant (§8.5). The uniform a-priori estimate the limit needs is the uniform gap itself: the reach-freeze monotone above the floor gives uniform exponential clustering, hence tightness of the finite-spacing Schwinger functions. What that tightness delivers is a subsequential pointwise limit of the moment functionals satisfying OS0–OS3; a measure on $\mathbb{R}^4$ and the non-triviality of the limit are separate, open steps. Balaaban's renormalisation-group programme supplies uniform effective-action bounds for the continuum limit; the stochastic-quantisation constructions (Chandra–Chevyrev–Hairer–Shen) establish the measure in two and three dimensions; the constructive four-dimensional $SU(N)$ measure is the named construction into which the existence half reduces. The finite-spacing OS-data family this limit consumes is instantiated for $SU(N)$ (§12).

The tightness step, from the gap. The entropy-matched read is a coarse-graining to the *resolved* content: it keeps the K_{signal} correlation modes above the reference null and treats the ultraviolet as sub-edge noise (asymptotic freedom). So the object is the measure on the finite signal content, of dimension K_{signal} , independent of the lattice mode count. The gap supplies the uniform bound: in a gapped theory at fixed physical volume the coherent states below the read scale are a finite family whose size K is fixed by the gap and the volume, **independent of the spacing**, so $K_{\text{signal}}(a) \leq K$ uniformly, the reflected forms are uniformly bounded, and a bounded reflection-positive family in a compact interval has a convergent subsequence inheriting RP: tightness with no ultraviolet renormalisation, only the gap. And K is derived from the gap as a momentum-space count: on a physical torus of size L the infrared momentum spacing is $2\pi/L$, fixed by the physical volume and independent of a . The gap keeps the supra-edge modes below a finite physical cutoff k_* , so the resolved dimension is $\leq \lceil k_* L / 2\pi \rceil$, independent of the lattice mode count, hence uniform in a . The one physical input is that k_* is finite, which is the finite correlation length, the gap. The tightness extends from a single reflected form to the **full Schwinger-function vector**: a countable family of reflected forms, each uniformly bounded by the gap's infrared count, has a *single* subsequence along which every form converges to a reflection-positive limit: one application of sequential compactness of the countable product $[-C, C]^J$. **Every Osterwalder–Schrader condition, being closed, survives it:** OS0 (regularity) is the uniform bound $|q_j| \leq C$; OS2 (RP) survives as above; OS4 (clustering) is the gap; OS1 (Euclidean invariance) and OS3 (permutation symmetry) follow because a symmetry of the finite-spacing forms passes to the limit by uniqueness of limits along the common subsequence. So from the gap (infrared cutoff) and the finite-spacing OS structure there exist a subsequence and a limit q with, for every test configuration, joint convergence, the temperedness bound (OS0), reflection positivity (OS2), Euclidean invariance (OS1), and permutation symmetry (OS3). Composed with the mass gap, the reduction data yields existence and the gap at once.

The reconstruction's operator core. Reconstruction is a chain: GNS separation (reflection positivity $\Rightarrow$ a physical inner product) $\rightarrow$ completion $\rightarrow$ the Euclidean-time transfer operator giving $H \geq 0 \rightarrow$ the vacuum (unique by clustering) $\rightarrow$ the Wightman functions by analytic continuation. The operator-theoretic segments are proved (§13). GNS separation, the step that consumes OS2, realises the reflection-positive form as a symmetric positive-semidefinite bilinear form B : Cauchy–Schwarz $(Bxy)^2 \leq (Bxx)(Byy)$ holds and its null cone $\{v : Bvv = 0\}$ equals the radical $\ker B$, so B descends to a positive-definite inner product on $V / \ker B$,

reflection positivity becoming the definite inner product of the quantum theory. The positive gapped Hamiltonian is the second: for a self-adjoint transfer operator T the Hamiltonian $H := -\log T$ (continuous functional calculus) is self-adjoint and, on a gapped positive contraction $\text{spec } T \subseteq [\varepsilon, 1]$, $H \geq 0$; when $\text{spec } T \subseteq [\varepsilon, e^{-\Delta}]$ the spectrum of H lies in $[\Delta, \infty)$. With the vacuum eigenvalue $1 \in \text{spec } T$ the ground-state energy 0 is attained, so $\text{spec } H \subseteq \{0\} \cup [\Delta, \infty)$: a vacuum at 0 and no spectrum in the open gap $(0, \Delta)$. Specialised to the finite-aperture margin $\text{spec } T \subseteq \{1\} \cup [\varepsilon, 3^{-1/4}]$ (the excited spectrum below $3^{-1/4} = e^{-\kappa_0}$), the reconstructed $H = -\log T$ has ground-state energy 0 and mass gap κ_0 : $\text{spec } H \subseteq \{0\} \cup [\kappa_0, \infty)$. The reconstruction assembles this as a **constructed gapped quantum theory**: the Hamiltonian on the physical Hilbert space, its vacuum, and its positive gap, produced as data from the transfer operator, and exhibited by the concrete finite-aperture operator $\text{diag}(1, 3^{-1/4})$ whose spectrum — and hence the gap κ_0 — is computed. Any transfer operator meeting the finite-aperture margin $\text{spec } T \subseteq \{1\} \cup [\varepsilon, 3^{-1/4}]$, the $SU(N)$ read $m_{\text{hi}} \leq 3^{-1/4}$, reconstructs to such a theory with gap κ_0 , carrying the finite-aperture read end-to-end.

The spectral gap from the decay. The spectral inclusion $\text{spec } H \subseteq \{0\} \cup [\Delta, \infty)$ is derived from the decay. Reflection positivity makes the Euclidean transfer operator T self-adjoint with a positive spectral measure, $C(\tau) = \int \lambda^\tau d\mu$ with $\mu \geq 0$ — equivalently a positive-semidefinite moment matrix, $C(\tau) = \sum_k w_k \lambda_k^\tau$ with $w_k \geq 0$. The **moment-support lemma** turns the decay $\|C_{\text{conn}}(\tau)\| \leq M e^{-\Delta\tau}$ into the spectral-support bound $\text{spec } T \subseteq \{1\} \cup [\varepsilon, e^{-\Delta}]$: a positive weight that survived a decaying total cannot sit above the decay rate. The continuous functional calculus then carries this to $\text{spec } H \subseteq \{0\} \cup [\Delta, \infty)$. Threaded through confinement, this reconstructs the $SU(N)$ Hamiltonian with spectral gap $\Delta_{YM} = \kappa_0 - \mu$. The $SU(N)$ witness places one excited mode at the certified margin $e^{-\Delta}$; the lemma takes a positive-weight spectrum of any finite width, so a multi-mode instantiation supplies the excited transfer eigenvalues with their reflection-positive weights and decay — together with the vacuum $\lambda = 1$ eigenspace and the cyclic-vector overlap, the companion inputs of the lemma. The lemma bounds every mode seen by the cyclic vector; lifting "seen by v " to the full spectral support needs Reeh–Schlieder totality and vacuum simplicity, cited companion facts to be proved.

Which operators that margin admits. The requirement $\varepsilon > 0$ is invertibility of T : keeping 0 out of the spectrum is exactly that, and a compact operator on an infinite-dimensional space is never invertible. The Euclidean transfer operator of a lattice gauge theory with a continuous compact structure group acts on an infinite-dimensional slice space through a continuous kernel on a compact manifold, hence is trace class and a fortiori compact — Lüscher (CMP **54** (1977) 283), Osterwalder–Seiler (Ann. Phys. **110** (1978) 440) — so it meets this margin at no $\varepsilon > 0$. That identification is cited; the implication it feeds is proved. Accordingly the witness exhibited above, $\text{diag}(1, 3^{-1/4})$, is finite-dimensional, and the reconstruction in this form is a statement about operators with a largest energy $-\log \varepsilon$. The contraction form, $\|T^n x\| \leq r^n \|x\|$ on the vacuum's complement, carries no such restriction: it needs no spectrum, no logarithm of an operator and no invertibility, and from it $\text{spec}_{\mathbb{R}}(T) \subseteq \{1\} \cup [-r, r]$ with 0 PERMITTED, so the invertibility requirement belongs to the $H = -\log T$ route and the gap conclusion holds without it. Two conditions travel with that form: the containment is informative only when $r < 1$, and the contraction itself is an input everywhere it appears — the same open intensive input §12 names. The Minkowski continuation of the Schwinger functions to the Wightman functions (tube domains / Bargmann–Hall–Wightman) is the classical Osterwalder–Schrader theorem, cited.

12. Existence and the mass gap

Two things characterise the quantum theory: its *existence* — a Hilbert space, a positive Hamiltonian, a unique vacuum, and Euclidean correlation functions satisfying Osterwalder–Schrader — and a positive *mass gap*.

The gap, stated precisely. Pure $SU(N)$ has a positive gap $\Delta > 0$, its scale the flux-tube bore $\Delta \propto \sqrt{\sigma}$ that confinement forces (§6), with $\Delta/a \rightarrow \Lambda > 0$ under refinement; equivalently the Euclidean transfer operator has an **isolated vacuum** with its remaining spectrum below $e^{-\Delta}$ ($\text{spec } T \subseteq \{1\} \cup [\varepsilon, e^{-\Delta}]$), the moment-support form of the decay, §11; equivalently the flow has **no interior fixed point** ($\beta \neq 0$ for every $g^2 > 0$); equivalently the centre-vortex tension stays below the floor $\mu(\beta) < \kappa_0$ (the **no-bump**); equivalently $a_{\text{IR}} = 0$. These are one statement (§6), and the lattice read makes $\beta \neq 0$ the directly computed quantity.

requirement	status here
the Clay statement for $SU(N)$, $N \geq 2$	reduced to named inputs; the statement itself is open. On one reconstructed Hilbert space, at every dyadic step m , a transfer gap at rate $e^{-(c/L)a_m}$ and a non-zero vector orthogonal to the vacuum follow from: the box gap at one coupling β_0 ; the UV step, below the box's rate; E, the uniform bound on separated renormalised Schwinger functions; the kernel convergence; and Y, the asymptotic-freedom short-distance law. The box is proved for $SU(2)$ and $SU(3)$ at $\beta_0 = 17N^2/(88\pi^2)$, so for those groups the statement rests on E, the kernel convergence, Y and the UV step alone. E is read only on reflected pairs, where it reduces to the reflection-diagonal bound; the kernel convergence with Y gives a pair lower bound, which a second form of the reduction takes in place of both, at the field-independent factor a^{-4} . Open: E, the kernel convergence, Y where it is a hypothesis, the UV step, and the box for $SU(N \geq 4)$. Outside this statement: non-triviality in the Clay sense (a connected function of order three or more away from zero) and rotation invariance off the hypercubic group.
mass gap $\Delta > 0$	$C(\tau) \rightarrow 0$ at every coupling from A1 (the reduction, on the four cited inputs) — read the scope with it: that conclusion is carried by the scalar μ alone. The finite-aperture model's single mode is DEFINED as $e^{-(\kappa_0 - \mu)}$, so $C(\tau) \rightarrow 0$ restates $\mu < \kappa_0$; and the whole conjunction — gap, $\mu - \kappa < 0$, $SO(4)$ and the OS0–OS3 continuum existential — holds for a model whose tension is the constant 0, with no read, no correlation, no lattice and no gauge group, the $SU(3)$ parameter record intact. Zero tension is the NON-confining case. What ties the conclusion to $SU(3)$ Yang–Mills is the spectral representability of the Wilson correlation (§12.6), reduced to two polynomial inequalities on (ρ_0, ρ_1, ρ_2) and proved at extent four for every $\beta \geq 0$; it carries no mode count and no rate. The HYPOTHESIS side concerns the genuine $SU(3)$ Haar Gibbs expectation on the 4-D periodic lattice; the quantitative rate $\Delta(\beta) \geq \kappa_0 - \mu(\beta) > 0$ with $ C(\tau) \leq (\sum P_k) e^{-\Delta\tau}$ is certified for the finite-aperture witness; reflection positivity gives $\rho'(n) \leq \rho'(1)^n$, and the single-plaquette gap $\geq \kappa_0$ ($\rho'(1) = m_{\text{cell}} \leq 3^{-1/4}$) with the read $m_{\text{hi}}(L) = \rho'(1)(L) \approx 0.33$ carries it to the physical modes; the decay reconstructs to $\text{spec } T \subseteq \{0\} \cup [\Delta, \infty)$ through the moment-support bridge. The margin $\mu < \kappa_0$ holds at both coupling ends (character bound below $\beta_* \approx 0.75$; asymptotic freedom above) and across the interior reduces to finite-volume analyticity (differentiability and a derivative bound) and a grid cover for the $SU(3)$ moment at extent 16, all three open (§9); the margin is intensive (it carries no lattice scale), the property the continuum limit uses (§11).
gap uniform in volume	two directions, two inputs. Reflection positivity makes the transfer operator self-adjoint, giving $\rho'(n) \leq \rho'(1)^n$ — a statement about n TIME-CUTS, with no volume index. A single cut $\rho'(1) < 1$ therefore gives the gap at every SEPARATION. Carrying it across VOLUMES is the separate intensive premise $\forall F, m_{\text{hi}}(F) \leq r$, an input of the volume-uniform statement: proved for the DECOUPLED product transfer and measured for the interacting one. Reflection positivity supplies the time direction; the spatial direction needs the intensive premise. With that

requirement	status here
	premise: the single-plaquette gap $\geq \kappa_0$ ($\rho'(1) = m_{\text{cell}} \leq 3^{-1/4}$) holds at $V=1$; the forward read $m_{\text{hi}}(L) = \rho'(1)(L)$ is 0.31–0.37 across the scaling window $L = 12\text{--}28$ (mean ≈ 0.33), the $L = 8$ and $L = 32$ endpoints outside that window
OS0–OS3 survive a tight subsequent limit over EXTENTS; existence on $\mathbb{R}^4$ open	the tight limit over extents (§11): from the finite-spacing Osterwalder–Schrader data (reflection positivity, the infrared mode-count bound, Euclidean and permutation invariance), the reflected forms are uniformly bounded and the full Schwinger vector is jointly tight, and every closed OS condition (OS0 bound, OS1 Euclidean, OS2 RP, OS3 symmetry) survives the limit. Cited inputs: the §2–§3 modelling identification (the $SU(N)$ Wilson ensemble is such a family, instantiated on a constructed realisation), Bałaban’s uniform effective-action bounds for the continuum limit, and the OS–Wightman reconstruction, which consumes Osterwalder–Schrader data whose translations act by $\mathbb{R}^4$. The tight limit supplies no such data: $\mathbb{R}^4$ is divisible, and at finite periodic extent the translations have finite order, which forces the identity, while on $\mathbb{Z}^4$ they are torsion-free and $\mathbb{Z}$ is not divisible, which forces zero. With the identity translation, OS1 holds of an ARBITRARY bilinear form, so such data carry no invariance content; the Osterwalder–Schrader data constructed here have the identity translation.
local fields, short-distance = asymptotic freedom, operator product expansion	open. The stress tensor and the operator product expansion are outside this paper, and the measure they would be built on is open (§11). The running side holds: $b_0, b_1 > 0$ and no interior fixed point, and asymptotic scaling is a genuine constraint — satisfiable AND refutable — whose realisation on the Wilson theory is open. These are standard <i>given</i> a constructed continuum measure.
any compact simple G	open beyond $SU(N)$. The continuum statement is for $SU(N)$, $N \geq 2$, and its box input is proved for $SU(2)$ and $SU(3)$; for $SU(N \geq 4)$ the box is open, and no other compact simple group is treated. Parts of the finite-spacing construction are general — the compact-gauge construction runs over an arbitrary compact group, the non-triviality of the Wilson plaquette variance is stated for an arbitrary compact G , and the floor, reflection positivity, asymptotic freedom and the band-limit lemma are N -general with centre twist Z_N — and none of these reaches the continuum statement.
clustering	follows from $\Delta > 0$

The load-bearing read. The tension bound $\mu < \kappa_0$ is *soft*: on the action-density read it holds with $\mu = 0$ in **both** phases (the F^2 : contrast ≈ 0.38 sits $\approx 3.5\times$ below the $3^{1/4}$ threshold, §8.6). The load-bearing input a lattice probe locates is the **finite aperture** (that the $SU(N)$ ensemble is the finite-mode model) read by the margin certificate. $SU(N)$ and confined $U(1)$ ($\beta < \beta_c$) read a *finite* aperture -- a decaying mode, correlation length $\xi = 1/\Delta$ finite = massive -- every row below the ceiling. Coulomb $U(1)$ ($\beta > \beta_c$) reads a finite aperture **too**: on the released $L = 8$ $U(1)$ ensembles, $m_{\text{hi}} = 0.172$ at $\beta = 1.70$ and 0.647 at $\beta = 2.50$, below the $3^{-1/4} = 0.7598$ ceiling and inside the unit circle (seven ensembles). The margin does move across $U(1)$ ’s transition — 0.166 confined against $0.172\text{--}0.647$ Coulomb — and that spread is read at an unmatched sampling ratio: the two Coulomb rows sit on opposite sides of the DMD truncation boundary (ratios 0.70 and 1.88), so the difference between them is an estimator artefact. At a matched ratio they read $0.166, 0.172$ and 0.197 — the first two are the $\beta = 0.9$ and $\beta = 1.7$ rows of the released margin table, while the third is the $\beta = 2.50$ ensemble re-measured at the matched $n = 48$, where the released table carries $n = 128$ (§14). It stays in the finite regime, and across groups it does not even order the phases: confined $SU(2)$ at $L = 24$ reads 0.342 , which sits *between* the two Coulomb rows. So on this read the finite aperture leaves the phase undecided — the same limitation as the tension bound above it, which is $\mu = 0$ in both phases — and $U(1)$ ’s deconfinement transition registers in the peak of the tension (§8.2) and in the disorder response (§8.4). What the reduction consumes is the weaker statement that survives: the $SU(N)$ read satisfies the aperture condition $m_{\text{hi}} \leq 3^{-1/4}$, verified on the ensemble the theorem concerns. The reading-fidelity identification (that the abstract read μ_{YM} is reading-A of the physical $SU(N)$ Wilson measure) is a theorem: reading-A is a concrete function and both reads are that function applied to the same ensemble correlation, so the identification holds by construction; its one structural property is the reflection positivity of the Wilson ensemble (Osterwalder–Seiler). Conditioning directly on the confinement read isolates the input into one hypothesis: the full bar (gap, non-triviality, $SO(4)$) follows from $\forall \beta \geq 0, \mu_{YM}\beta < \kappa_0$ alone, together with reflection positivity, which is proved at even extent at least four and nonnegative coupling (§11.1b) and cited outside that domain.

There are two routes to existence: an exact closed form, or a convergent sequence with uniform estimates. The argument here is **geometry and math on the screen**: the band-limit lemma (§4), the entropy floor (§7), and the reach-freeze monotone (§6), with the self-sourcing tiling driving the descent to $a_{\text{IR}} = 0$ and the band-limit turning that into $\Delta > 0$. The **direct read** takes φ off configurations with no sweep, fit, renormalisation, or predetermined scale, and confirms the descent across the crossover. Every input around the interior (the analytic step, the $1/L^2$ aperture scaling, and the finite-aperture condition) is a theorem; the uniform-in-volume gap follows from the L -stable intensive margin through a lemma; the crossover interior reduces to finite-volume analyticity (a bounded connected-correlator modulus) and a finite grid cover, both open (§9). The **intensive margin** is read directly: reflection positivity gives $\rho'(n) \leq \rho'(1)^n$, and $m_{\text{hi}}(L) = \rho'(1)(L)$ is 0.31–0.37 across the scaling window $L = 12\text{--}28$, below the floor ceiling $3^{-1/4}$; the finite crossover correlation length $\langle d^2 \rangle(\beta) \leq 1$ is its fixed-volume form, read on the $SU(N)$ ensemble and verified with margin.

The direction of the construction. The construction runs one way, from the finite aperture outward — the band-limit (§4), the entropy floor (§7), the reach-freeze descent (§6), the reflected forms and their continuum limit (§11) — each strand joining a standing classical result (reflection positivity, the character bound, asymptotic freedom, the Osterwalder–Schrader reconstruction) at a cited junction, with the gap read on the forward side. This is renormalisation’s own direction: the Connes–Kreimer Birkhoff factorisation of the Feynman character is a determined forward map whose inverse is a torsor under the renormalisation group (§1). Forward and inverse are different passes — the screen resolves a finite band of the continuum’s unbounded short-distance detail (§2–§3), so the read is a lossy projection whose inverse, reconstructing the continuum measure on $\mathbb{R}^4$ with its full renormalisation (Bałaban’s uniform RG bounds, §11), is underdetermined. The gap is the forward pass; the aperture supplies the gap.

The architecture is a reduction to an external law, and the gap is an entropy surplus. The argument is a single reduction: the mass gap, non-triviality, and $SO(4)$ follow from A1 ($\mu < \kappa_0$) and A2, and A1 rests on asymptotic freedom, $b_0 = 11N/3 > 0$, through the self-sourcing contraction against the entropy floor $\kappa_0 = \frac{1}{4} \ln 3$ (§5–§6). The base is a theorem, asymptotic freedom. The inequality that carries it is entropic on both sides: confinement is the disorder entropy density κ_0 exceeding the tension μ , and the gap is bounded below by the margin, $\Delta \geq \kappa_0 - \mu > 0$, the entropy surplus of disorder over tension.

Two statements follow for pure $SU(N)$: the lattice family has a limit over EXTENTS carrying the Osterwalder–Schrader structure — a tight subsequent limit of the moment functionals at a FIXED coupling, with reflection positivity and the axis-permutation invariance surviving it (§11; the index is the extent, the coupling held fixed) — and its Hamiltonian has a mass gap $\Delta \geq \kappa_0 - \mu > 0$ (§6). **And the plaquette variance is positive, with no hypothesis.** $0 < \rho(0)$ at **every** real coupling and **every** periodic extent, $\rho(0)$ being the connected correlation of the four-dimensional $SU(3)$ plaquette-energy observable with ITSELF, on the genuine Wilson ensemble; the proof uses a separating witness (two configurations whose holonomy differs by a commutator of non-commuting $SU(3)$ elements), and at $SU(1)$ the

same quantity is exactly 0. A non-vanishing connected correlator at lag ZERO is a variance, and a Gaussian measure has one too. Non-Gaussianity — a fourth cumulant, a connected four-point function, any comparison against a Wick factorisation — is outside this paper.

What is proved is that limit; a constructed measure on $\mathbb{R}^4$ is open, and the gap side is scoped to $\beta \geq 0$, where reflection positivity is proved. Each rests on the named inputs, listed in §13.

12.1 The gapless weight is bounded by the tension. The decay can be derived, in place of defined, from the aperture condition alone, with a RATE. A gapless theory carries a CONSTANT component in its correlation -- the transfer form $\rho(d) = \sum_n w_n e^{-E_n d}$ contributes $w_0 \cdot 1^d$ exactly when $E_0 = 0$ -- and on the lag circle a constant contributes NOTHING to the first moment, $\sum_{d < N+1} \cos(2\pi d/(N+1)) = 0$, while still adding $c(N+1)$ to the normalisation. It can only dilute. So clearing the floor bounds the gapless weight against the rest, $c(N+1) < (3^{1/4} - 1) \sum_d g(d)$, with a right-hand side carrying no aperture: if $\sum g$ stays bounded as the window widens, c is squeezed to zero. With no weight left at $\lambda = 1$ the maximum over a FINITE mode family is attained and is itself inside the unit circle, so the correlation obeys $\sum_k w_k \lambda_k^d \leq (\sum_k w_k) \rho^d$ with $\rho < 1$ -- decay at an explicit positive rate -- $\log \rho$, which is what a gap asserts and what $C(\tau) \rightarrow 0$ alone leaves open (a power law satisfies the latter). This holds for the CONSTRUCTED $SU(N)$ Wilson correlation -- the special unitary group, the real Wilson action, product Haar over links, ordered-loop holonomy -- on a geometry where no plaquette owns a private link. A plaquette with a private link integrates out to a free one, which is what a two-plaquette cell and a periodic ladder both turn out to be; in every $d \geq 3$ no plaquette of the hypercubic lattice owns a private link. The 3 is derived: the second plaquette must be non-degenerate, so its witness direction has to avoid both spanning directions, and a two-element set of directions has a complement exactly when there are more than two. $d \leq 2$ fails, which is why two-dimensional lattice gauge theory is exactly solvable.

The boundary is the finiteness. It is what carries the rate: for a COUNTABLE family the modes may accumulate at $\lambda = 1$, and that accumulation IS gaplessness. A finite lattice supplies finiteness through its finite-dimensional transfer matrix, so this is a gap at finite volume and finite spacing -- and it is exactly where the thermodynamic limit takes hold, since the spectrum fills in as $L \rightarrow \infty$ and the chain leaves that filling uncontrolled.

12.2 The criterion caps its own rate at $O(1/N)$. This can be made quantitative -- exactly, not asymptotically -- which settles what the aperture condition can and cannot deliver. For a single mode $\rho(d) = \lambda^{\text{circLag } d}$ BOTH sums in the cosine average are geometric, so with $n = N+1$, $m = n/2$, $k = 2\pi/n$ and $(\lambda e^{ik})^m = -\lambda^m$ the average is closed:

$$\langle \cos \rangle = \frac{\frac{2(1+\lambda^m)(1-\lambda \cos k)}{1-2\lambda \cos k + \lambda^2} - 1 - \lambda^m}{\frac{2(1-\lambda^m)}{1-\lambda} - 1 + \lambda^m}.$$

It is increasing in $1 - \lambda$, so demanding it clear $3^{-1/4}$ fixes a unique critical λ , and

$$(N+1)(1-\lambda) \gtrsim C = 2\pi a_*, \quad a_*^2 \coth(a_* \pi/2) = 3^{-1/4}(a_*^2 + 1),$$

giving $a_* = 1.7489138$ and $C = 10.9887497$. Both descend from κ_0 and the circle geometry; nothing is fitted. The finite-aperture sequence $(N+1)(1-\lambda_{\text{crit}})$ rises 5.689, 7.797, 9.245, 10.081, 10.526, 10.755, 10.930, 10.981 at $N+1 = 8 \dots 8192$ and is resolved to eleven digits against C by $N+1 = 2^{40}$. So $\mu < \kappa_0$ at aperture N certifies $\Delta \gtrsim 11/(N+1)$, a rate that **vanishes as the aperture widens**.

The structure of the closed form rests on four identities: the real cosine sum is the real part of a complex geometric series, hence $(z^m - 1)/(z - 1)$ off $z = 1$; a summand invariant under $d \mapsto 2m - d$ is fixed by its half-range; and that fold applied to $\rho(d) = \lambda^{\text{circLag } d}$ itself gives

$$\sum_{d < 2m} \lambda^{\text{circLag } d} \cos \frac{2\pi d}{2m} = 1 - \lambda^m + 2 \sum_{1 \leq d < m} \lambda^d \cos \frac{2\pi d}{2m},$$

and the $-\lambda^m$ is the ANTIPODAL term -- the contribution the unfolded shape λ^d has no counterpart for, and precisely the reason the folded ceiling is 10.9887 where the unfolded one is 11.1760. So the step that separates the two constants is proved. The remaining arithmetic -- assembling numerator and denominator into $\langle \cos \rangle$, its monotonicity in $1 - \lambda$, and the solve for the critical λ -- is computed, and the whole average is reproduced by direct summation over every lag (§14).

And a_* has a direct reading: the cap has content. A free MASSLESS field in the same periodic box is gapped at finite volume -- its lowest nonzero mode is $2\pi/(N+1)$, which is $a = 1$ in the scaling variable. Since $a_* = 1.7489 > 1$, that mode reads $\langle \cos \rangle = 0.5452$ against a floor of 0.7598 and **fails** the criterion, at every aperture. So where $\mu < \kappa_0$ holds for a spectral correlation it certifies a decay rate a_* times the gap a massless field would show in the same box; a_* is exactly that factor, and the criterion is strictly stronger than masslessness-in-a-box.

At the aperture actually simulated, the separation is wide. The cap C is the $n \rightarrow \infty$ limit; at finite n the requirement is weaker (7.797 at $n = 16$, against 10.989), so what the criterion decides on the released lattices is read from the closed form. It matters: at $\beta = 2.50$ the scaling cap would reject a physical glueball ($8.59 < 10.99$) where the exact form accepts it ($8.59 > 7.797$). Put a 0^{++} glueball of 1.7 GeV into lattice units through the measured spacing, at $n = 16$:

β	a [fm]	ma	λ	$\langle \cos \rangle$	vs floor 0.7598
2.30	0.1585	1.365	0.2553	0.9345	clears
2.40	0.1181	1.017	0.3616	0.8815	clears
2.50	0.0893	0.769	0.4634	0.8067	clears
massless	--	0	0.6752	0.5522	fails

So at the very aperture the ensembles carry, the criterion admits a physical glueball at all three couplings and rejects a massless field: at $n = 16$ it decides.

This is consistent with §9's compact $U(1)$ control, which found the MEASURED read unable to separate a confining theory from a massless one. That control concerns contact-scale correlations at the apertures actually simulated, where the criterion is satisfied for resolution reasons and the lag distribution is far from a transfer mode. The statement here is about the spectral object. Both hold, and the distance between them is the distance between what the criterion means and what the present measurement can deliver.

A neighbouring constant belongs to a different shape: $2\pi\sqrt{3^{-1/4}/(1-3^{-1/4})} = 11.1759763$ is the cap for the cosh correlator $\rho(d) = \lambda^d + \lambda^{n-d}$. The same number is often attributed to the **unfolded** shape $\rho(d) = \lambda^d$, which shares its limit — but that shape is asymmetric, and $\rho(d) = \rho(n-d)$ holds for the Wilson correlation (Theorem 11.1d), so the unfolded shape is inadmissible for this object. The value belongs to the symmetric shape. What separates the two constants is the minimum-image FOLD in ρ , which puts $\lambda^{n-d} > \lambda^d$ at large lag and contributes the $\coth(a\pi/2) > 1$ factor that lowers the requirement. The gap between them, 0.187, is smaller than the distance from the $N+1 = 256$ term of the CRITICAL-RATE sequence $n(1 - \lambda_{\text{crit}})$ to its own limit — 0.234 — so no aperture in the simulated range separates the two constants by eye. (Named because it matters which sequence: the companion mass sequence converges an order of magnitude faster, reaching 0.0012 of the limit at the same aperture, and the statement is false of it.) This criterion reads the folded shape and takes the smaller, conservative constant.

That is the quantitative form of "the tension bound is *soft*" (§12): the criterion is a RESOLUTION condition, and the size of the gap it certifies is set by the window.

The cap is an APERTURE limit, and the aperture is distinct from the box. This distinction decides which limits the bound survives. Writing $L_{\text{ap}} = (N+1)a$ for the screen's physical extent,

$$\Delta_{\text{phys}} = \frac{\Delta_{\text{lat}}}{a} \geq \frac{C}{(N+1)a} = \frac{C}{L_{\text{ap}}},$$

and the spacing has cancelled. The right-hand side names the SCREEN and nothing else -- neither the lattice spacing nor **the size of the system the screen looks into**. The three limits are therefore independent:

limit	what moves	effect on $\Delta \geq C/L_{\text{ap}}$
continuum	$a \rightarrow 0$	none -- a cancels
thermodynamic	$\text{box} \rightarrow \infty$	none -- the box does not appear
aperture	$L_{\text{ap}} \rightarrow \infty$	destroys it

The problem's $\mathbb{R}^4$ takes the second. The third is a limit no finite observer performs: the aperture's size is the observer's capacity, and an observer inside the substrate it reads has a window strictly smaller than that substrate, however large the substrate becomes. The substrate may be taken to infinity; the window cannot follow it.

That the bound is independent of the box is a theorem: the box is carried as a parameter and never used. What remains is empirical -- whether the tension still clears the floor when a FIXED window looks into an ever-larger system.

And the criterion polices the other direction, so a small window cannot manufacture a large gap. At aperture n the test passes only when $n\Delta > C$, so a window too narrow to resolve the decay returns no verdict. The sharpest bound comes from the narrowest window that still passes, and there it equals the gap itself. With $\sqrt{\sigma} = 0.44$ GeV (cited) the statement is

$$\Delta \gtrsim \frac{\kappa_n \hbar c}{L_{\text{phys}}}, \quad \kappa_n = n \Delta_{\text{crit}}(n) \uparrow C = 10.9887497.$$

A distinction matters here and it is worth 3%. The criterion constrains $1 - \lambda$; a MASS is $\Delta = -\log \lambda$. Both $n(1 - \lambda)$ and $n\Delta$ rise to the same C , but $n\Delta$ rises much faster and stays below it -- 10.689 at $n = 16$ against $C = 10.989$. So C/n is a CEILING on what the criterion can certify at any aperture -- a no-go for widening the APERTURE without limit, and for nothing else: the box does not enter it and the spacing cancels out of it. The bound certified at aperture n is κ_n/n , slightly smaller. Quoting C at a finite volume claims more than the criterion supports, so the numbers below use $\kappa_{16} = 10.689$:

$$\Delta \gtrsim \frac{2.109 \text{ GeV} \cdot \text{fm}}{L_{\text{phys}}} \quad (n = 16).$$

The same curve at every coupling: $\beta = 2.30, 2.40, 2.50$ at $L = 16$ give $L_{\text{phys}} = 2.536, 1.889, 1.429$ fm from the measured $a\sqrt{\sigma}$, and certified gaps 0.832, 1.116, 1.476 GeV, each reproducing $2.109/L_{\text{phys}}$ exactly. Across the physical volumes simulated here (1.4–5 fm) the criterion therefore certifies 0.42–1.48 GeV against a 0^{++} glueball of ≈ 1.7 GeV -- a valid bound, within a factor 1.15 of the physical value at the smallest volume and loosening as $1/L_{\text{phys}}$. Closing that limit needs an input bounding λ away from 1 UNIFORMLY in the volume -- which is the intensive premise $\forall F, m_{\text{hi}}(F) \leq r$, supplied by neither the geometric law in the separation nor the tension.

12.3 Where that leaves the definitional witness, quantitatively. The finite-aperture model sets its mode to $e^{-(\kappa_0 - \mu)}$, so it asserts $\Delta := \kappa_0 - \mu$, which tends to $\kappa_0 = 0.2747$ as $\mu \rightarrow 0$ with volume. Against the cap above, the two can be compared directly, and the comparison falls in two halves. At every volume this paper measures the criterion certifies MORE than the witness asserts -- $\Delta \geq 1.242, 0.873, 0.668, 0.452, 0.389, 0.341$ at $L = 8, 12, 16, 24, 28, 32$ against an asserted 0.226, 0.252, 0.261, 0.269, 0.270, 0.271 -- so on the data reported the witness is the conservative of the two. The curves cross near $L \approx 40$. Past it the assertion exceeds what the criterion supports, by a factor growing linearly with the volume: $3.2\times$ at $L = 128, 12.8\times$ at $L = 512, 51\times$ at $L = 2048$. So the witness's rate is supported by its own hypothesis only at small volume, and the thermodynamic limit is precisely where that support runs out -- the same boundary the finiteness of the mode family marks, reached from the other side. The measured input is also satisfied for a RESOLUTION reason (§9), so meeting it is by itself no evidence of confinement; what the theorem gives is a gap conditional on it. Evaluated on the ensemble, the bound of §12.1 is quantitative: at $\beta = 2.30, L = 32$ it puts the gapless weight under 0.0114, against a directly measured c of -2×10^{-5} , with $\sum g$ flat to 0.49% across a fourfold volume range.

12.4 Continuum scaling at MATCHED PHYSICAL VOLUME. The $1/L_{\text{phys}}$ cap above is a finite-volume limit and carries no spacing, so the continuum direction is the one it leaves open -- and it is testable on the released links, provided the box is held fixed while the spacing moves. Otherwise the comparison mixes a change of spacing with a change of volume. Two matched pairs exist: $(L=12, \beta=2.30)$ against $(L=16, \beta=2.40)$, with $La\sqrt{\sigma} = 4.241$ against 4.213 -- matched to 0.7% while the spacing changes by $1.342\times$ -- and $(L=8, \beta=2.30)$ against $(L=16, \beta=2.50)$, matched to 12.7% at a $1.77\times$ lever arm.

At fixed box a PHYSICAL mass has am falling with the spacing (ratio a'/a); a cutoff artifact holds am fixed (ratio 1). Read at FIXED (operator, smearing, τ) -- no minimum is taken, so no admissibility tolerance enters and this test stands outside the three-way disagreement of §8.7b. The separation is complete and it is by OPERATOR:

A series is RESOLVED when its own jackknife error is smaller than its own value -- the same criterion the m_{hi} scan already applies, carried here unchanged, so nothing is selected by hand. Three of the twenty-five series fail it and are quoted separately below; the remaining twenty-two are:

channel	pair 1 (predicts 0.745)	pair 2 (predicts 0.563)	prefers
plaquette, smeared ($n_s = 8, 24$)	0.644(27), 0.765(25), 0.657, 0.729	0.470(32), 0.736(28), 0.497	physical
plaquette, unsmeared ($n_s = 0$)	1.010(73), 0.898	1.195(94)	cutoff
action density (every n_s)	1.019–1.401	0.914–1.856	cutoff

Among resolved series the split is exceptionless and it is by OPERATOR: all seven smeared-plaquette series prefer the physical reading, all fifteen action-density and unsmeared-plaquette series prefer the cutoff one, on both pairs independently. The tightest points exclude the cutoff value at 13.0σ and 16.4σ . The three unresolved series decide nothing either way and are counted on neither side: action density $n_s=0, \tau=2$ on pair 1 reads 0.51 ± 0.65 (nominally physical), unsmeared plaquette $\tau=1$ on pair 2 reads 1.87 ± 2.50 and smeared plaquette $n_s=8, \tau=2$ on pair 2 reads 1.51 ± 4.68 (both nominally cutoff) -- one on each side, so setting them aside leaves the comparison balanced. This is §8.7b's operator dependence seen a second way, at fixed physical volume in place of across couplings, and without the variational minimum.

One systematic is outside those errors: the two tight plaquette points disagree with each OTHER -- $0.765(25)$ at $n_s=8$ against $0.644(27)$ at $n_s=24$, a 3.3σ gap -- so the smearing level shifts the ratio by more than the quoted precision. The VERDICT is robust across smearing (all four land below 0.78, against 1.000); the exact ratio is smearing-dependent.

12.5 The identification, measured. The reduction reaches its conclusion through a model whose mass is DEFINED as $e^{-(\kappa_0-\mu)}$, so its correlator decays exactly when $\mu < \kappa_0$: that link restates its own hypothesis and carries no physics on its own.

A second form of the reduction removes the definitional witness, quantifying over an ARBITRARY mode family and asking instead for two junction inequalities, $\kappa_0 - \mu \leq c$ and $c \leq \Delta$. What those jointly imply is a single statement in measured quantities — the margin lower-bounds the transfer gap:

$$\Delta \geq \kappa_0 - \mu.$$

Both sides are measured here: Δ is the variational transfer gap of §8.7b, read from the gap correlator with no model of the potential, and μ is the tension the reduction consumes. Neither measurement was produced with this comparison in mind. Held against each other at every coupling where both resolve, **on both gauge groups: all eight resolved points satisfy $\Delta - 2\sigma \geq \kappa_0 - \mu$ — six $SU(2)$ and two $SU(3)$ — the tightest by a factor 2.7 at the measurement's own lower error bar.** Five further points are excluded because their gaps do not resolve, by the resolution rule §8.7b already states; the worst of them reads $\Delta = 0.171 \pm 0.851$, an error five times its value.

That the composite holds on $SU(3)$ as well as $SU(2)$ is what makes it evidence about the argument and about more than one theory: a one-group result cannot separate the two.

The two junction inequalities themselves stay open, and the contraction constant c between them is unmeasured — only the composite they imply is measured. The step from $\mu < \kappa_0$ to a gap in the ACTUAL modes is therefore tested: where it can be checked, it holds with room.

Two routes reach this conclusion, and they ask for different things. The one above is the route whose residuals are measured; the other discharges the first junction inequality by proof, from the entropy floor's own count (§7): a physical vortex weight dominating the directed-cube count 3^k at area $4k + 6$ has free-energy density $\geq \kappa_0 - \mu$, so $\kappa_0 - \mu \leq c$ follows from the counted floor with no constant approximation, given that the directed surfaces are among the physical vortex family and the cited condensation–contraction scale duality. The count route is the stronger theorem: fewer inputs, more general, and one of the two junction inequalities replaced by the entropy floor's count. The measured route is the one whose two sides have been held against each other in measurement. Both are terminal, so this paper quotes the measured route for the empirical claim and names the count route as the stronger statement (§13 compares the two).

12.6 Spectral representability of the Wilson correlation. The link between the model's modes and the $SU(3)$ Wilson correlation is that the measured correlation admits a periodic transfer decomposition $\rho(d) = \sum_k w_k (\lambda_k^d + \lambda_k^{n-d})$ with $w_k \geq 0$ and $0 \leq \lambda_k \leq 1$. This holds at extent four for every $\beta \geq 0$, and the spectral sum then equals the ensemble's own correlation at every lag the period resolves — which is what turns a statement about a model's modes into a statement about the $SU(3)$ Wilson correlation.

The route is two reflection-positivity instances and no operator. Representability at extent four is characterised exactly: the triple is representable iff $\rho(3) = \rho(1)$, $0 \leq \rho(0)$, $0 \leq \rho(2)$, $\rho(2) \leq \rho(1)$ and $2\rho(1)^2 \leq \rho(2)^2 + \rho(0)\rho(2)$ — the convex cone on the curve $\lambda \mapsto (1 + \lambda^4, \lambda + \lambda^3, 2\lambda^2)$, whose hull is that arc plus the chord. Two of the four conjuncts are proved for the Wilson triple directly. The order relation $\rho(2) \leq \rho(1)$ is a LINK-reflection positivity instance read at the two-term half-space observable $G = F_1 - F_2$, for which $\langle \theta G \cdot G \rangle = 2(\rho(1) - \rho(2))$ — an identity, so the inequality IS that positivity. Its content is $\lambda_i \lambda_j \geq 0$ on every weighted pair, via $\lambda_i \lambda_j^3 + \lambda_i^3 \lambda_j - 2\lambda_i^2 \lambda_j^2 = \lambda_i \lambda_j (\lambda_i - \lambda_j)^2$: positivity of the SPECTRUM, which link reflection supplies and site reflection does not. The quadratic is Cauchy–Schwarz on the slab reflection form at the site reflection $s \mapsto -s$, and holds at every real β .

Its scope, stated with it. Representability carries no mode count and no rate: the constant triple is representable, with $\rho(2)/\rho(0) = 1$. It closes a named residue and supplies the family the resolved-mode count needs as input; the ratio obligation of §9 is separate from it.

12.7 The aperture is a variable of the theory. The window of $N + 1$ lags contributes the factor $(2\pi/(N+1))^2$ and the substrate contributes the second moment of the lag distance measured ON THE CIRCLE, while the floor $\kappa_0 = \frac{1}{4} \log 3$ is a per-area density with no window in it. Using the raw lag index instead makes that moment grow like N^2 even at a fixed correlation length, because a periodic correlation obeys $\rho(N) = \rho(1)$ — so a hypothesis bounding it is satisfiable by no physical correlation at all, and a theorem taking it is vacuous. The lag symmetry of the correlation itself (Theorem 11.1d) forces the circle distance, so reading the substrate there is a consequence of a proof.

Confinement follows from a single statement about the substrate: a bound $\exists B, \forall N \beta$ on the circle moment gives $\mu < \kappa_0$ at every large enough aperture and every coupling, with no number in its statement or its proof, and in fact $\mu \rightarrow 0$, so the margin $\Delta = \kappa_0 - \mu \rightarrow \kappa_0$ approaches the whole floor. The proof never uses boundedness: the aperture condition $(2\pi/(N+1))^2 \langle d^2 \rangle / 2 < 1 - 3^{-1/4}$ has a constant right-hand side while the left carries $(N+1)^{-2}$ — so the CEILING on $\langle d^2 \rangle$ grows like $(N+1)^2$ and the moment may grow with the aperture too, provided its coefficient stays under the ceiling's. The growth condition is the input, and the bounded case is its corollary. The same conclusion follows from geometric decay of the read's correlation in the lag, $\rho(d) \leq C r^d$ with $r < 1$ uniform in N and β , through the aperture-uniform moment bound and the equivalence $0 < \Delta_{YM}(\beta) \iff \mu_{YM}(\beta) < \kappa_0$; under reflection positivity that hypothesis is the transfer gap itself, with $r = e^{-\Delta}$, and the reduction takes it as an input.

That growth form matters. Measured in lattice units a correlation length diverges as the spacing goes to zero, so a hypothesis demanding a strictly bounded lattice moment would ask for more than a continuum limit supplies. Which correlation the read is taken on is part of supplying the hypothesis, and the reduction applies to whichever correlation is supplied.

It is open and tested. §9 reads the same moment at $L = 6 \dots 32$ over 15 couplings: twelve of the thirteen couplings testable across apertures are consistent with a constant B at $\chi^2/\text{dof} \leq 1.96$ against their own reproducibility, including 0.49 over seven apertures at $\beta = 2.30$; the raw-index moment grows $19.2\times$ on those same ensembles. A measurement reaches finitely many apertures and cannot discharge a hypothesis quantified over all of them — but the hypothesis has a measured value and a falsified alternative, which is the difference between an assumption and an open question.

13. The formal verification

The argument of §1–§12 is a derivation in mathematics and physics; it stands on the reasoning in the text and is self-contained, and a reader follows the text itself as the proof. This section records the formal verification that accompanies it. The reduction of the continuum statement for $SU(N)$ to named inputs — some classical results, some open (§12) — is machine-checked in Lean 4 / Mathlib, [sorry-free](#), its exact axiom footprint printed below. The machine check is a fact-check on the reduction, read with the same care as the rest. This section maps each formal fact the text states to the declaration that proves it; §14 locates the development in the repository. The read layer and its certification are the companion paper [E].

13.1 What is machine-checked. The finite-aperture mechanism — the band-limit lemma, the entropy floor and its counting bound, vortex condensation from that count, the single-cell spectral gap, the reflection-positivity carry $\rho'(n) \leq \rho'(1)^n$, and the reconstruction — is machine-checked on the three foundational axioms; the gauge-physics content enters as §1 states.

The following are theorems in Lean 4 / Mathlib on the three foundational axioms ([propext](#), [Classical.choice](#), [Quot.sound](#)), [sorry-free](#):

- the entropy floor $\kappa_0 = \frac{1}{4} \ln 3$ and its directed-cube counting bound $\kappa \geq \kappa_0$ ([Floor](#));
- vortex condensation from that count:** for $\mu < \kappa_0$ the directed-cube vortex sum diverges, with per-step counted log-rate exactly $4(\kappa_0 - \mu)$ ([Condensation](#));
- the band-limit step $\langle \cos \theta \rangle \geq 1 - \theta^2/2 \Rightarrow \mu < \kappa_0$ and the $1/L^2$ aperture scaling $\langle \theta^2 \rangle = (2\pi/L)^2 \langle d^2 \rangle$ ([Moment](#)), with the finite-aperture condition $(2\pi/(N+1))^2 \langle d^2 \rangle / 2 < 1 - 3^{-1/4}$ ([Read.tension_lt_floor_of_circ_moment](#));
- the **single-cell spectral gap** $\Delta_{\text{cell}} \geq \kappa_0$ at every coupling — hence $m_{\text{cell}} = e^{-\Delta_{\text{cell}}} \leq 3^{-1/4}$ — through a min-max eigenvalue-count lemma Mathlib lacks ([CellSpectrum](#), [CellEnclosure](#));
- reflection positivity making the Euclidean-time transfer operator self-adjoint, so $\rho'(n) \leq \rho'(1)^n$ on the vacuum-orthogonal subspace ([VolumeRate.rp_sub_geometric](#), fed to [Mixing.gap_of_maximal_correlation](#) by [VolumeRate.rp_gap_of_one_cut](#); the inequality is what is proved and what every consumer uses), and the moment-support bridge $\|C(\tau)\| \leq M e^{-\Delta \tau} \Rightarrow \text{spec } T \subseteq \{1\} \cup [\varepsilon, e^{-\Delta}]$ ([MomentSupport](#));
- the reconstruction $H = -\log T$ self-adjoint, $H \geq 0$, vacuum at 0, $\text{spec } H \subseteq \{0\} \cup [\kappa_0, \infty)$ ([Reconstruction](#)).
- reflection positivity of the whitened $SU(3)$ Wilson correlation itself**, at even extent ≥ 4 and nonnegative coupling — nonnegative at every lag with positive total mass, with no hypothesis about the measure ([OddLagSplit.corrClay_reflection_positive](#), recorded in [Complete](#) as [wilson_reflection_positive_at_even](#)). This is the axiom's own body, proved on that domain (§11.1b).

13.2 The inputs of the gap flagship. On these, the gap flagship reduces to four inputs: reflection positivity — proved outright for the whitened correlation at even extent ≥ 4 and $\beta \geq 0$ (§11.1b), and cited to Osterwalder–Seiler outside that domain (odd extents and extent two, at $\beta \geq 0$) — the strong-coupling character bound (Osterwalder–Seiler), the asymptotic-freedom plateau $\mu_\infty < \kappa_0$ (Gross–Wilczek–Politzer, the below-floor value proved by Wick), and the finite interior correlation length $\langle d^2 \rangle \leq 1$ (a finite-sample statistical bound at nominal 99.9% per coupling, largest upper 0.867; at nominal 99.9999% the largest upper is 1.372, which clears the aperture ceiling $B_{16} = 3.2480$ and exceeds 1; the bound reads $SU(2)$ ensembles and is evidence for the Lean hypothesis, which is stated on [d2At 15](#) and is open, §9). The single-plaquette aperture margin $\Delta \geq \kappa_0$ is a deterministic certificate; its spatial-volume carry to $V \rightarrow \infty$ is the open intensive bound, for which the forward read of §8.5 is evidence (§8.7b sets out what that read reports).

13.3 The gap side, with no named axiom at all. Reflection positivity is the one named axiom the gap half can carry, and it enters at exactly one place: the read's *certificate*. [wilsonCorrAt](#), the correlation itself, is clean; [readYMat](#) certifies it with the axiom. Substituting the proved certificate of §11.1b gives an aperture-restricted read whose entire chain is foundational:

```
'EvenAperture.existence_and_gap_of_substrate_even' depends on axioms: [propext, Classical.choice, Quot.sound]
'EvenAperture.mass_gap_rate_and_continuum_even'    depends on axioms: [propext, Classical.choice, Quot.sound]
```

So the mass gap, non-triviality, the axis-permutation invariance **and** the OS0–OS3 limit over extents follow from a single hypothesis with **no named axiom on either side** — the measure half carries none. What “carries none” means for the measure half: [FlagshipScope.flagship_measure_half_needs_no_hypothesis](#) proves the measure half takes **no hypothesis at all**, its proof term being the unconditional [ym_continuum_tension](#), with [hc](#) present in the type and unused. And [so4_clause_unconditional](#) proves the $SO(4)$ clause the same way, at every aperture and with no hypothesis; [R_is_the_same_term_at_every_aperture](#) does the same for the aperture-independence clause. So of the four clauses the single hypothesis reaches one ($\mu - \kappa < 0$) as itself, reaches $C(\tau) \rightarrow 0$ only through the defined mode, and reaches the other two not at all — [flagship_for_bogus](#) delivers the whole conjunction for a model of constant tension 0, the non-confining case. And the hypothesis is *weaker* than the one the axiom-bearing flagship takes: [substrate_even_of_substrate](#) proves the flagship's input implies the even-aperture input. That bridge is itself stated in terms of [d2At](#) and so reports the axiom in its own footprint; the chain it feeds does not. A reader who obtains the hypothesis *through* the bridge therefore holds an axiom-carrying composite, while the axiom-free statement is the one proved from the even-aperture hypothesis directly.

What the restriction costs. The proof needs $\beta \geq 0$ while the model's coupling field is total, so the read is built at $\max(\beta, 0)$. On the nonnegative half-line the two reads are *the same object*, nothing approximated ([readEven_eq_readYMat](#)); below it the model's column is the $\beta = 0$ column relabelled, with no claim about a negative-coupling ensemble ([readEven_eq_at_zero](#)). That is exactly the domain on which the Wilson cross kernel is *not* positive-semidefinite, so no claim is available there — and both flagships read the correlation at $\max(\beta, 0)$, so below zero each states its $\beta = 0$ content; they differ only in the aperture. The aperture is any even extent ≥ 4 , drawn from the same eventual set the original draws from, which is cofinal either way.

13.4 The scope of the flagship conclusion. The finite-aperture model of the text is [ymModel](#) ([ymModelAt N](#) at aperture N). Its single mode is DEFINED as $e^{-(\kappa_0 - \mu)}$, so its conclusion is carried by the scalar μ alone, and $C(\tau) \rightarrow 0$ restates $\mu < \kappa_0$. [FlagshipScope.flagship_for_bogus](#) machine-checks that the whole

conjunction — gap, $\mu - \kappa < 0$, $SO(4)$ and the OS0–OS3 continuum existential — holds for a model whose tension is the constant 0, with no read, no correlation, no lattice and no gauge group, the $SU(3)$ parameter record intact. Zero tension is the NON-confining case. The empty named-axiom footprint of `EvenAperture.existence_and_gap_of_substrate_even` is therefore an axiom-footprint claim; the strength of the conclusion is set by this scope, which `FlagshipScope` machine-checks. What ties the conclusion to $SU(3)$ Yang–Mills is `WilsonSpectral` (§12.6), which `SpectralFour` reduces to two polynomial inequalities on (ρ_0, ρ_1, ρ_2) and `SlabQuadratic.wilsonSpectral` proves at $N = 3$ for every $\beta \geq 0$. The HYPOTHESIS side is unaffected by the scope: `ConfinementAtAnAperture` unfolds to `corrClay`, a genuine $SU(3)$ Haar Gibbs expectation on the 4-D periodic lattice. Across the interior the margin reduces, in Lean, to finite-volume analyticity (`hdiff`, `hbn`) and a grid cover (`hcover`) on `d2At` 15, none of which the measured grid supplies (§9).

13.5 The named axioms. The mass gap and the OS continuum measure of the constructed realisation carry **one** named axiom, reflection positivity (`ym_mass_gap_of_junction`), and `existence_and_gap_of_model` and `mass_gap_rate_and_continuum` carry none. **The tree declares two named axioms in total.** The second is `os_reconstruction_wightman`, in `WightmanData.lean`: it states the Osterwalder–Schrader reconstruction against a bilinear Euclidean form with a Hilbert-space-valued conclusion. It is applied — `reconstructed_vacuum_energy_zero` reads a zero-energy vacuum out of its conclusion and `os_reconstruction_wightman_is_not_vacuous` proves both of its ends non-degenerate. The Wilson data it is applied to are `WilsonOS.wilsonOSData`, an `OSData` whose Schwinger form is the Gibbs reflected pairing of the constructed $SU(N)$ Wilson measure on the slab algebra of an even-extent lattice, its reflection positivity holding at every real coupling and its footprint the three foundational axioms alone; `WilsonOS.wilson_reconstructed_nontrivial` carries it through the reconstruction adding only `os_reconstruction_wightman`. Its translation field is the identity — the value `LatticeTranslNoGo.transl_eq_id_of_finite_order` forces at a finite periodic extent, where the translations have finite order — so Euclidean invariance is carried by the limit. The axiom is composed with the Wilson data and stays uncomposed with the gap side's lattice construction, because producing an `OSData` from that lattice needs a time translation that is an endomorphism of the slab algebra, and the shift carries a transverse link out of the module. The four-dimensional $SU(3)$ OS measure (derived Euclidean and permutation invariances) and its derived infrared datum both carry the three foundational axioms and nothing else. The exact-rational single-cell enclosure of §9 has a Lean port covering the truncated cells at every coupling and the strong-coupling window in general (the cell row of the ledger below).

The axiom footprint. Beyond the three foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), the development names ONE input that carries physics: reflection positivity of the Wilson ensemble at every aperture (`wilson_reflection_positive_at`, Osterwalder–Seiler). Reconstruction rests on the one cited Osterwalder–Schrader → Wightman axiom (`os_reconstruction_wightman`), applied to the Wilson data as above. Across every declaration that prints a footprint, no other axiom appears.

13.6 The theorem. The top-level theorem is the reduction with A1 and A2 discharged to the named inputs below: for the lattice $SU(N)$ witness at every physical coupling $\beta \geq 0$ it establishes clustering, $\|\sum_k P_k m_k^\tau\| \rightarrow 0$ (the gap in the form $C(\tau) \rightarrow 0$); non-triviality, $\mu - \kappa < 0$ (the area law); and $R(d) = R(d')$ (Euclidean $SO(4)$), carrying no A1/A2 hypothesis. **Both conclusions — the finite-spacing Osterwalder–Schrader data and the gap in the form $C(\tau) \rightarrow 0$ — hold for a constructed $SU(N)$ object;** the gap conclusion is about `ymModel`, carried by μ alone (§13.4), and existence on $\mathbb{R}^4$ is outside them (§12 table). A constructed $SU(N)$ realisation (`ym_wilson_of`) instantiates the finite-spacing Osterwalder–Schrader data family, its reflected forms built from the same `wilsonCorr` as the gap side (so `os_rp` is the reflection-positivity axiom and the two are one physical model), with its Euclidean and permutation clauses holding STRUCTURALLY rather than by an invariance of the measure: `QYM` reads `j` only through the label `j.2.2`, while `actEYM` and `actPYM` move `j.1` and `j.2.1`, so `os_euc` and `os_perm` close by `rfl` over actions on components the form never reads. The invariance that is DERIVED is `ymFamilyGauge`'s, where `WilsonGauge.QG_eq` obtains it from `Symmetry.expect_invariant` on an $SU(3)$ Haar expectation. `ym_existence_and_gap_of_junction` then delivers the mass gap AND the OS0–OS3 limit over extents (§12 table), on that constructed realisation. `ym_existence_and_gap_of_junction` (gap and OS0–OS3 measure together) carries the one named axiom (`wilson_reflection_positive_at`). No new axiom, no `sorry`. The quantitative strengthening $\|C(\tau)\| \leq M e^{-\Delta\tau}$ with $\Delta = \kappa_0 - \mu > 0$ is a separate certified theorem carrying the same footprint. The spectral form `ym_mass_gap_spectral` builds the correlator from the witness transfer modes m_k , each carrying the certified bound, and gives $\|\sum_k P_k m_k^\tau\| \rightarrow 0$ from the finite-aperture margin $\|m_k\| \leq 3^{-1/4} = e^{-\kappa_0}$, taken as an EXPLICIT hypothesis over an arbitrary mode family rather than discharged by a pinned witness. Satisfiability is exhibited separately and without naming a value: `spectral_bar_nonvacuous` takes any family size and any magnitude strictly positive and at or below the derived ceiling $3^{-1/4} = e^{-\kappa_0}$. The ceiling is grounded in the exact-rational single-plaquette enclosure of §9 (a certificate outside the Lean footprint, §14), which brackets the single-plaquette gap $\Delta \geq \kappa_0$ at every coupling ($\Delta \geq 1.633$ at mid-crossover $\lambda = 1$, giving $m_{hi} < \frac{1}{5}$ there) so $m_{hi} = e^{-\Delta} \leq 3^{-1/4}$. The measured $SU(N)$ read sits below the margin ($m_{hi}(L) = \rho'(1)(L)$ is 0.31–0.37 across the scaling window $L = 12$ –28, below $3^{-1/4}$), which is the statement the Lean step consumes. The $U(1)$ –Coulomb read sits below it as well ($m_{hi} = 0.172$ –0.647, §12), so the certificate bounds the $SU(N)$ aperture and leaves the phases to the geometric discriminator of §5.

The reduction. A1 (strong end: the character bound below the derived threshold $\beta^2 < \frac{1}{2} \ln 3$; weak end: $a^2 \mu \rightarrow 0$; interior: the analytic step and the $1/L^2$ aperture scaling, with $\mu(\beta) < \kappa_0$ a theorem `confinement_of_growth_ratio` at every large enough aperture) and A2 (discrete, spatial, and axis-role isotropy, the last via an orthogonal congruence of the correlation Gram) are certified lemmas. The reads' identification with the physical Wilson measure (`readYM_is_wilson`), the coupling-scale positivity (`rArgYM_pos`), the character ratio $I_2/I_1 > 0$, the Nyquist $O(4)$ transport (`Otr_iso`), and the finite-aperture condition (`Read.tension_lt_floor_of_circ_moment`) are theorems.

Existence. Reflection positivity survives the continuum limit as a closed condition; the finite-spacing infrared mode-count bound (an Osterwalder–Schrader datum) makes the full Schwinger vector jointly tight; OS0–OS3 survive the joint limit; GNS separation and the reconstructed QM core produce the gapped Hamiltonian $H = -\log T$ ($H \geq 0$, vacuum at 0).

13.7 The formal lattice. *The gauge-measure fidelity.* The Osterwalder–Schrader invariances of the constructed measure are machine-checked as *derived* from the gauge group. A finite lattice gauge measure over a compact group — the product Haar measure on the link variables — has Gibbs correlations invariant under any lattice symmetry that permutes plaquettes, because Haar-invariance of the product measure composes with the plaquette-permutation invariance of the Wilson action (`LatticeGauge.Symmetry.expect_invariant`). A compact Hausdorff group carries a canonical left-invariant probability measure (`CompactGauge`, normalized Haar), and the compactness, topological-group, and Borel structure of $SU(N)$ (`specialUnitaryGroup`) are supplied (`SUN`: $SU(N)$ is closed in the matrix algebra and entrywise bounded, hence compact by a product-of-closed-balls Tychonoff argument), so the derivation lands on the actual $SU(N)$ Haar measure. The plaquette holonomy is the genuine ordered link-product Wilson loop (`WilsonLattice`, each plaquette a boundary word of oriented links), the action density the Wilson density $1 - \frac{1}{N} \text{Re tr}(\text{WilsonAction})$, with conjugation invariance $\varphi_W(hgh^{-1}) = \varphi_W(g)$ the gauge-invariance seed, and a concrete two-plaquette $SU(2)$ system with a plaquette-swap symmetry exhibits the invariance non-vacuously (`WilsonReal`). The OS0–OS3 continuum measure on this genuine gauge measure (`WilsonGauge.ym_continuum_gauge`) is taken **at the Clay problem's own group and dimension**: its reflected form is a Gibbs expectation of the Wilson plaquette energy over `WilsonHypercubic.sysWilson` 3 4 n , four-dimensional periodic $SU(3)$, against the canonical probability Haar measure, and its Euclidean/permutation invariance is the lattice's own axis symmetry (`axisSymmetry`) composed with Haar-invariance. It carries the three foundational axioms only; the existence-and-gap companion on it (`WilsonGauge.ym_existence_and_gap_gauge`) carries the same one named axiom as the flagship (`wilson_reflection_positive_at`), on the gap side — the measure side contributes none. The defining symmetry of the gauge theory is machine-checked on this

measure: the Wilson correlation is invariant under a gauge transformation $U \mapsto gUg^{-1}$ (`WilsonReal.sysReal_gauge_invariant`), because the conjugation preserves the Haar measure — the compact group is unimodular, its probability Haar left- and right-invariant (`isMulRightInvariant_probHaar`, via the modular character) — and preserves the Wilson action, whose plaquette holonomy conjugates and whose density $1 - \frac{1}{N} \text{Re tr}$ is a class function (`WilsonAction_gauge_invariant`). On this measure the Wilson correlation is a genuine *state*: the partition function is strictly positive ($Z > 0$, the Boltzmann weight bounded and integrable on the probability Haar measure), so $\langle \cdot \rangle = \int (\cdot) e^{-\beta S} / Z$ is well-defined, normalized ($\langle 1 \rangle = 1$), positive ($\langle O \rangle \geq 0$ for $O \geq 0$), linear, monotone, and bounded ($|\langle O \rangle| \leq \sup |O|$) — a positive normalized linear functional on the observables (`WilsonReal.sysReal_expect_{one,nonneg,mono,smul,add,abs_le}`), with the integrability side-conditions discharged for every bounded measurable observable (`sysReal_mul_boltz_integrable`). Applied to the physical plaquette-energy order parameter $\varphi_W(\text{hol}_p)$ this gives a well-defined, bounded, gauge-invariant mean plaquette energy $0 \leq \langle \varphi_W(\text{hol}_p) \rangle \leq 2$ (`expect_plaqObs_mem`). All of this holds at general N and any finite lattice (`wilsonSystem_partition_pos`, `wilsonSystem_expect_{one,nonneg,smul,add,mono,abs_le}`), the $SU(2)$ two-plaquette system being the concrete instance. The clustering that underlies the gap appears already at zero coupling: the $\beta=0$ Wilson measure is the bare product Haar measure (`sysReal_expect_at_zero`), under which the two disjoint plaquettes' link blocks are independent (`coords_indep`, from `iIndepFun_pi` grouped over the disjoint blocks), so the connected correlation of the two plaquette energies vanishes exactly — $\langle \varphi_0 \varphi_1 \rangle = \langle \varphi_0 \rangle \langle \varphi_1 \rangle$ (`expect_plaqObs_factor_at_zero`), the product-measure seed of clustering. In fact for this two-plaquette system the Boltzmann weight itself factorizes over the disjoint blocks ($e^{-\beta S} = e^{-\beta \varphi_0} e^{-\beta \varphi_1}$, `sysReal_boltz_factor`), so the factorization holds at *every* coupling ($\langle \varphi_0 \varphi_1 \rangle = \langle \varphi_0 \rangle \langle \varphi_1 \rangle$ for all β , `expect_plaqObs_factor`) — the plaquettes are non-interacting, and this is exactly why the clustering is trivial. Nontrivial exponential-in-separation clustering — the quantitative mass-gap statement — requires plaquettes that *share* links; that interacting object is itself a genuine gauge theory (`sysInt`, two plaquettes sharing a link, with `sysInt_partition_pos` and `sysInt_gauge_invariant`), and its link-blocks overlap (`blockInt_not_disjoint`), so the independence kernel above is inapplicable — the machine-checked construction marks precisely the boundary at which the quantitative decay must take over. The strong-coupling input to that decay is itself machine-checked: the exact Haar moments of the $SU(2)$ fundamental — $\int U_{ij} dU = 0$ and the second moment $\int U_{ij} \bar{U}_{kl} dU = \frac{1}{2} \delta_{ik} \delta_{jl}$ (the Schur orthogonality relation, `MassGap.HaarMoments`) — are derived by an invariant-projection argument (Peter–Weyl being absent from the formalization library) on two explicit group elements (`diag(i, -i)` and the swap $[[0, 1], [-1, 0]]$), yielding the character norm $\int |\text{tr } U|^2 dU = 1$ (`haar_su2_char_norm`) and the two-point contraction $\int \text{tr}(UA) \text{tr}(U^*B) dU = \frac{1}{2} \text{tr}(AB)$ (`haar_su2_two_point`); this last, applied to the environments of two link-sharing plaquettes, is the leading non-factorizing correlation. The volume-uniform continuation of that decay is the open intensive input: reflection positivity carries the separation direction only, and the measured single-plaquette gap is evidence for the volume direction, which stays an open input. The centre-vortex count-injection is likewise machine-checked at the entropy floor's exact constant: a physical vortex weight dominating the directed-cube count 3^k (`directed_surface_count`) at area $4k + 6$ has free-energy density $\geq \kappa_0 - \mu$, so the self-sourcing junction $\kappa_0 - \mu \leq c$ is sourced from the counted floor with no constant approximation (`VortexCount`, `selfSourcingJunction_of_physical_count`), the two inputs reduced to the sub-family cardinality (directed surfaces embed into the vortex ensemble) and the cited condensation–contraction scale-duality.

The lattice is genuinely d -dimensional, and d is an argument. The systems above are small by design -- two plaquettes, sharing a link or not -- because their job is to exhibit a mechanism. The mechanism having been exhibited, the same construction is instantiated on a periodic hypercubic lattice of arbitrary dimension and extent (`WilsonHypercubic`). A site is a coordinate per direction (`Site = Fin d → Fin n`), a link a (direction, site) pair, and a plaquette an ORDERED PAIR of directions together with a site -- the pair rather than a normal, because a plane in d dimensions needs two directions and only in $d = 3$ does one suffice. Degenerate pairs $\mu = \nu$ are carried in the type rather than excluded, and cost nothing: the boundary word retraces itself, so the holonomy is the identity and the Wilson density vanishes (`bd_diag_hol_one`). Carrying the pair ordered means both (μ, ν) and (ν, μ) are in the type and both are summed, so this action counts every plane twice — the two orderings have inverse holonomies and hence the same $\text{Re tr}(\text{PlaqCount.plaq_ordered_double_counts})$ — and the coupling carried here is accordingly half the standard Wilson one (`§8.1`, `PlaqCount.boltz_eq_std`). The boundary word is the elementary Wilson loop $U_\mu(x) U_\nu(x+\hat{\mu}) U_\mu(x+\hat{\nu})^{-1} U_\nu(x)^{-1}$, and it is the only place the geometry enters.

What this buys is a cardinality. A system whose link type is `Fin d` has d links and is not a lattice in any dimension; this one has $d \cdot n^d$ links and $d^2 n^d$ plaquettes, machine-checked as such (`card_link`, `card_plaq`), so at $d = 4, n = 8$ that is 16,384 links and 65,536 plaquettes, growing with the volume as a four-dimensional gauge theory must. The action density is the genuine Wilson density rather than the zero function (`sysWilson_phi`), which is what makes β move the Gibbs measure at all; the holonomy is the ordered loop product (`sysWilson_hol`). Neither d nor n nor N is a literal anywhere in the construction: $d = 4, N = 3$ is an instantiation of it, and $d = 3$ is the same construction at a different dimension.

The hypercubic lattice's discrete Euclidean symmetry is then *derived*. An axis permutation $e \in \mathfrak{S}_d$ relabels directions and carries site coordinates with it, and the entire content is that this commutes with the unit shift (`shift_axis`) -- from which the boundary word transports compatibly (`bd_axis`) and `WilsonLattice.wilsonSymmetry` returns a genuine *Symmetry* of the gauge system (`axisSymmetry`). That is precisely the datum an `os_euc/os_perm` field of a `LatticeYMFfamily` requires, so on this system those OS invariances are consequences of the lattice's own geometry composed with Haar invariance, with no axiom added (`#print axioms shift_axis, bd_axis`: foundational only). What remains on this route is the arithmetic of transporting the finite-aperture read onto a $d \cdot n^d$ -link system -- the mechanism, the symmetry, and the dimension-generalty are proved; the read's instantiation at $d = 4, N = 3$ is the open step, and it is the step the lattice campaign of §8 is generating data for.

The infrared datum is SOURCED on that measure. A `LatticeYMFfamily` carries four Osterwalder--Schrader data, and on the gauge measure three of them are theorems about $SU(3)$ Haar-invariance. The fourth, `os_gap`, asks that every mode standing above the noise edge sit below a spacing-independent index cutoff -- and supplying that directly is asserting where the resolved modes are. `Measure.familyOfSortedCount` refuses to take it: it asks instead for an ORDERED spectrum and a bound on HOW MANY of its modes clear the edge, and derives `os_gap` from the pair. The count is exactly what `ZeroMode.resolved_count_le_of_subset` obtains from the measured tension. `WilsonGauge.ym_continuum_gauge_counted` is the composition -- the OS0--OS3 continuum limit on the four-dimensional $SU(3)$ Wilson measure, with every OS datum either a theorem about that measure or a quantity a read reports, and none of them an assertion about the infrared. Foundational axioms only.

Reflection positivity: the mechanism is proved, and the cited content is one step. RP enters this development as a single named axiom, `wilson_reflection_positive_at`, credited to Osterwalder and Seiler. The parts of the cited theorem differ in difficulty, and this records which part is load-bearing.

Mechanically, RP says an observable paired with its own reflection has nonnegative expectation, and the reason is algebraic: such a pairing is a SQUARE. Split the links into a positive half S and a negative half T , disjoint and exchanged by a reflection θ . Then for any function h supported on the positive half,

$$\int h(U|_S) h((U \circ \theta)|_S) d\mu = \left(\int h(U|_S) d\mu \right)^2 \geq 0,$$

because the two halves read disjoint coordinates so the integral factorises, and the measure does not distinguish the halves so the two factors are equal (`ReflectionPositivity.pairing_with_reflection_nonneg`, from `WilsonReal.block_integral_factor` and `relabel_measurePreserving`). At $\beta = 0$ there is

no cross term and RP follows outright, with nothing cited (`reflection_positive_at_zero`).

The splitting picture is proved too, and its content is LOCALITY. A plaquette's holonomy reads only the links its own boundary word names (`hol_congr_on_support`); the action divides by any plaquette partition (`action_split`); and lifting locality to a set gives " S_+ depends only on $U|_+$ " as a theorem (`action_on_congr_of_support`). So

$$e^{-\beta S} = g(U|_+) g'(U|_-) e^{-\beta S_{\text{cross}}},$$

and the first two factors are already of the paired shape the mechanism consumes.

The one step this module leaves open is expanding $e^{-\beta S_{\text{cross}}}$ over the plaquettes straddling the reflection plane into a convergent sum of paired products, and concluding by summing nonnegatives. Everything around it -- locality, the split, the pairing-is-a-square mechanism, and the free case -- is machine-checked here with no axiom, eight theorems carrying the three foundational ones only.

\$11.1b proves that step, on even lattice extent ≥ 4 at $\beta \geq 0$, by a different route than the character expansion: an action split into reflected halves, a three-block Haar factorisation holding the shared block outside the square, mirror transport, and the handedness reconciliation the link reflection forces, with no representation theory (`OddLagSplit.corrClay_reflection_positive`, surfaced as `wilson_reflection_positive_at_even`). The Osterwalder--Seiler theorem is what the development cites **outside** that domain -- odd extents and extent two, at $\beta \geq 0$. At negative coupling `CharacterExpansion.su3_kernel_nonneg_iff` shows the Wilson cross kernel fails to be positive-semidefinite, and `readYMat` reads the correlation at $\max(\beta, 0)$.

The two halves are easy to conflate: in a different setting an antiunitary pairing can hold while positivity fails -- on an odd cycle the identity can be restored and the positivity cannot (\$14 names the study). Invariance of the measure under the reflection is the easy half; reflection positivity is the nonnegativity of the pairing.

And the count is a theorem from the tension, with nothing left in between. The composition above takes the resolved-mode count as a hypothesis, and the tension supplies it. `ZeroMode.resolved_count_le_of_subset` produces that count as a PRODUCT inequality -- the count multiplied by everything it is bounded against -- which is the form the proof yields; a continuum limit consumes the quotient. `ScreenedGap.resolved_count_ceiling` performs the division once, giving

$$\text{card}\{\text{resolved}\} \leq \frac{12 W M}{\varepsilon \lambda_0^{k+1} (2(k+1))^2 S},$$

and `resolvedDim_ceiling` states the same on `resolvedDim` itself, the quantity `Measure.familyOfSortedCount` reads. In that expression M/S is the weighted mean of clag^2 over the correlation -- which IS the substrate, and the substrate is exactly what the tension bounds. Composing them (`ScreenedGap.resolvedDim_le_of_tension`):

$$\mu < \kappa_0 \implies \text{card}\{\text{resolved}\} \leq \frac{12 (1 - 3^{-1/4}) / 8 \cdot W}{\varepsilon \lambda_0^{k+1}} = \frac{0.360246 W}{\varepsilon \lambda_0^{k+1}}.$$

Every symbol on the right is the read's own -- W its total weight, ε its noise floor, λ_0 the lower edge of the band the resolved modes occupy. The 12 is the sum-of-squares denominator and $(1 - 3^{-1/4})/8$ is the entropy floor composed with $\cos x \geq 1 - x^2/2$; no third constant enters, and no fit. Foundational axioms only.

That is the count hypothesis `Measure.familyOfSortedCount` asks for. The chain from a measured tension to an OS0--OS3 limit over EXTENTS on the four-dimensional $SU(3)$ Wilson lattice is therefore closed as a chain of theorems: $\mu < \kappa_0 \rightarrow$ substrate cap $\rightarrow$ resolved-count ceiling $\rightarrow$ `os_gap` $\rightarrow$ `continuum_of_family`. What is left is instantiation -- a read at each EXTENT whose tension clears the floor; no further inequality is needed. Read the endpoint precisely: `OSFamily.osFamily_Q_eq` evaluates the family at index a as `corrClay` (`extent a`) with `extent a = 2a+2` and the coupling a fixed PARAMETER, so the index sets the periodic extent and no lattice spacing occurs in `LatticeYMFfamily` anywhere. This chain reaches the infinite-VOLUME limit; the continuum direction is separate from it. And note where ε sits: in the DENOMINATOR, which is why the residual-edge theorems carry weight. An inflated floor makes this ceiling smaller, and a smaller ceiling is a stronger claim than the data supports.

One constructor, and the scope of the constructed realisation. The OS-data interface has one route: `ymFamilyGauge`, the unconditional family, IS `ymFamilyGaugeCounted` at a spectrum with a single supra-edge mode, with the three hypotheses discharged. That spectrum is therefore load-bearing: it is what shows the conditional theorem is about something.

The realisation is read at one rank, and at $SU(3)$ that rank is the Clay problem's. The parameter record takes its rank as an argument (`ymParams N hN`), so a realisation reads the gap side and its own physical parameters at the SAME N . `ym_wilson_gauge_su3` is that realisation at $N = 3$, where the measure's rank joins them: $\mu_{\text{YM}}(3, \beta)$, `ymParams 3`, and the four-dimensional periodic $SU(3)$ Wilson lattice `sysWilson 3 4 n`. The rank appears once as a symbol and every use is that symbol, so no two numerals can disagree; `ym_wilson_gauge_su3_rank` states the identity as a theorem.

The general- N form makes the coincidence visible where it holds: away from $N = 3$ the gap side is a statement about a correlation's decay and the measure side one about $SU(3)$ Haar-invariance, and `existence_and_gap_of_model` conjoins them with neither referring to the other. Tying the ranks changes what the flagship is ABOUT and nothing else: the two junction residuals `hfe` and `hgap` are open at every rank, and the measure side proves a subsequential pointwise limit, with a measure on $\mathbb{R}^4$ open. Neither depends on the rank.

And the noise edge itself has a direction of error, proved. The count bound divides by `edge`, so the edge carries weight: a floor taken as a share of the TOTAL spectral weight inherits a share of the near-unit vacuum component, while the modes a gap argument concerns live in the residual. The two floors differ by exactly the vacuum's share (`ScreenedGap.edge_raw_sub_residual`, an identity rather than an estimate), the raw floor is therefore the larger (`edge_residual_le_raw`), and a larger floor resolves no more modes (`resolvedDim_antitone_edge`). Composing: reading the edge off the raw spectrum UNDER-reports the resolved count (`resolved_count_under_reported_of_raw_edge`), which makes the bound look tighter than the modes present justify -- the failure mode it guards against. The inequality has content: `under_report_is_strict` exhibits the condition under which a mode is counted by the residual floor and missed by the raw one. This is an effect reported from measurement elsewhere (\$14); here it is a theorem, with no constant introduced.

The gap side and the OS measure are ONE lattice. The correlation read here (`Complete.wilsonCorrAt`) is `WilsonBridge.corrClay`, the connected correlation on `WilsonHypercubic.sysWilson 3 4 n`: four-dimensional, $SU(3)$, plaquettes carrying the two spanning directions -- the lattice the Osterwalder--Schrader measure is built on. So the reflection-positivity citation and the measure concern one construction at the Clay problem's own dimension and group. A plaquette indexed by its NORMAL, the geometry of `WilsonBridge.corr3`, determines a plane only in three dimensions, which is why the read uses the plane-pair lattice. The plane and the lag direction are PARAMETERS of the general form (`corrHyper`), because `Fin d` for a variable d carries no numerals -- the type system declining to let $d \geq 3$ hide

inside three literals. Footprint: the three foundational axioms, with reflection positivity the one cited input, asserted ABOUT a constructed object. Both hypotheses are shown satisfiable (`Witness.chain_hypotheses_satisfiable`), so the implication has content.

The closed form of §12.2. Its structure is machine-checked -- four theorems, each on the three foundational axioms:

theorem	statement
<code>sum_geom_cos_eq_re</code>	the real cosine sum IS the real part of a complex geometric series
<code>sum_geom_cos_closed</code>	hence $(z^m - 1)/(z - 1)$ off $z = 1$, via Mathlib's <code>geom_sum_eq</code>
<code>sum_range_antipodal_fold</code>	a summand invariant under $d \mapsto 2m - d$ is fixed by its half-range
<code>circLag_cos_sum_fold</code>	that fold applied to $\rho(d) = \lambda^{\text{circLag } d}$ itself

The last is the fold identity of §12.2, the step that separates the folded constant 10.9887 from the unfolded-shape 11.1760. The assembly of numerator and denominator into $\langle \cos \rangle$, its monotonicity in $1 - \lambda$, and the solve for the critical λ are outside the formalisation and are computed (§14).

The two routes of §12.5, formally. The measured route is the first; the count route discharges one of its two inputs by proof:

	<code>Complete.ym_mass_gap_of_junction</code>	<code>VortexCount.existence_and_gap_of_floor_count</code>
assumes	<code>hfe</code> and <code>hgap</code>	<code>hM + hdual + hgap</code> — it proves <code>hfe</code>
axioms	4 (carries <code>wilson_reflection_positive_at</code>)	3 — foundational only
generality	the YM model, mode index <code>Unit</code>	abstract in the mode index and direction
the evidence here	§12.5's measured residuals, on both groups	the count is machine-checked; <code>hdual</code> is cited

The second is the stronger theorem: fewer axioms, more general, and one of the two inputs the table lists for `ym_mass_gap_of_junction`, `hfe`, replaced by the entropy floor's own count (§7, constant-free in both factors). The first is the one whose two sides have been held against each other in measurement. Neither is load-bearing for anything else in the development — both are terminal — so this paper quotes the measured route for the empirical claim and names the count-injection route as the stronger formal statement.

The substrate chain. Confinement follows from a single statement about the substrate: `confinement_of_bounded_substrate` takes $\exists B, \forall N \beta$ with the circle moment below B , and returns $\mu < \kappa_0$ at every large enough aperture and every coupling. No number appears in its statement or its proof. `tension_tendsto_zero_of_bounded_circ_moment` strengthens the conclusion from "eventually below the floor" to $\mu \rightarrow 0$, so `margin_tendsto_floor` gives $\Delta = \kappa_0 - \mu \rightarrow \kappa_0$: the margin approaches the whole floor. `ym_mass_gap_of_substrate` carries that to the mass gap, non-triviality and $SO(4)$, with `#print axioms` returning the three foundational axioms **plus** `wilson_reflection_positive_at` **alone**. The development compiles with **no sorry**. That is an axiom-footprint statement and the scope above applies to it unchanged: the conclusion is about `ymModel`'s defined mode, so it converts $\mu < \kappa_0$ rather than deriving decay of the Wilson correlation.

Complete.ym_mass_gap_of_lag_decay has one open input. Its conclusion is about `ymModelAt N` (the scope stated above), so this input list is the flagship's; the Clay statement's is §13.10. `Complete.ym_mass_gap_of_lag_decay` delivers the whole conjunction — $C(\tau) \rightarrow 0, \mu - \kappa < 0$ and $SO(4)$ — at every large enough aperture and every coupling, from the single hypothesis that the read's correlation decays geometrically in the lag, $(\text{readYMAAt } N \beta).p d \leq C r^d$ with $r < 1$ uniform in N and β , on the three foundational axioms plus `wilson_reflection_positive_at` and nothing else. Everything between that hypothesis and the conjunction is proved: the aperture-uniform moment bound (`Moment.circ_moment_le_of_geometric`), the entropy-floor comparison (`Complete.gap_pos_iff_confinement: 0 < \Delta_{YM}(\beta) \iff \mu_{YM}(\beta) < \kappa_0`, so the gap IS the tension below the floor), and the model assembly (`Model.mass_gap_of_model`). No aperture, no threshold and no measured constant appears in the statement or the proof. Under reflection positivity that hypothesis is the transfer gap itself, with $r = e^{-\Delta}$. The reduction takes it as an input.

One thing is open **on this route**, a statement about the substrate weaker than an aperture-independent bound (§12.7). `ClayAssembly` states the development's open obligations in one place, against the genuine objects, and names more than this one. What `Read.tension_lt_floor_of_circ_moment` asks for is the aperture condition $(2\pi/(N+1))^2 \langle d^2 \rangle / 2 < 1 - 3^{-1/4}$; `confinement_of_growth_bound` takes the growth form, and the bounded case is its corollary.

13.8 The declarations behind §7–§12. Each formal fact the text states, by section.

§7, the entropy floor. Every step of Theorem 7.1 is machine-checked. Steps (3) and (4) are `Floor.directed_paths_card`, `Floor.cubeConfig_injective` and `Floor.directed_surface_count` ($= 3^k$); injectivity at the level of the SURFACES is `CubeArea.boundaryFaces_cubeConfig_injective`, and `CubeArea.directed_surfaces_count_and_area` states the theorem whole: 3^k distinct surfaces, each of area exactly $4k + 6$. Closedness, step (1), is `CubeClosed.edge_parity_all`, its per-cube count `CubeClosed.cube_edge_even` and its per-face count `CubeArea.owners_one_or_two`. The fixed face is `CubeClosed.origin_face_mem_boundary`, through `CubeClosed.mem_boundaryFaces_iff_floor`. Connectedness is `CubeConnected.boundary_connected`. The family is `VortexFamily.vortexFamily`, and `VortexFamily.three_pow_le_vortexCount` is $3^k \leq N$ against it with no hypothesis. The cardinality over `Plaq 4 n` inside a box closes with `three_pow_le_card_of_plaq_subfamily`, with no containment hypothesis left. Step (2), the area, is `CubeArea.boundary_card_eq`, from `card_faces`, `card_cubeConfig`, `card_cubes_with_face_le_two`, `step_rel_of_shares`, `sum_dist_one_of_shares`, `shared_face_eq` and `card_sharedFaces`. Foundational axioms only throughout.

§8.1, the coupling convention of the formal lattice. The Lean development indexes a plaquette by an **ordered** pair of directions (§13.7), so its sum runs over (μ, ν) and (ν, μ) alike and every plane is counted twice (`PlaqCount.plaq_ordered_double_counts`). Its coupling is consequently $\beta_{\text{Lean}} = \beta/2$: the Lean Boltzmann weight at β_{Lean} is the standard Wilson weight at $2\beta_{\text{Lean}}$ (`PlaqCount.boltz_eq_std`). The two-loop spacing is read at the same coupling: `AsymptoticScaling.aRun N \beta` is the standard two-loop spacing at $2\beta_{\text{Lean}}$ (`AsymptoticScaling.aRunStd`), so every statement pairing the spacing with the Lean Wilson family measures lengths in that family's own lattice units.

§8.2. The termwise bound $I_2/I_1 \leq \beta/4$ is `Bessel.ratio_le_quarter`.

§8.6. The formal chain consumes the tension of the limitation paragraph exactly: `confinement_at_of_substrate_sharp` returns $\mu < \kappa_0$ from the correlation and is indifferent to which theory produced it. The $U(1)$ measurement uses the Lean's $-\log(\cos \theta)$.

§8.7c. The limit $11.176/L$ is `Complete.continuum_mass_bound_tendsto`, the unfolded-shape constant; the lag symmetry is `MomentShape.wilsonCorrAt_neg`; the fold is `ZeroMode.circLag_cos_sum_fold`.

§9, *A1 and A2*. The read-level form of Euclidean $SO(4)$ is the 24-element hypercubic axis group `DYM = Equiv.Perm (Fin 4) (Complete.card_DYM)`; the transport is that permutation's own matrix and `permMatrix_orthogonal` proves it orthogonal, so it lands in $O(4)$. Non-triviality is proved in the form $\mu - \kappa < 0$; no fourth cumulant, connected four-point function or Wick comparison appears in the tree. The *A1* input is the hypothesis `d2At N $\beta \leq B$ of Complete.confinement_at_of_d2_sharp, with B under the sharp ceiling; Interior.ym_crossover_confinement_of_grid_sharp states it on a finite grid, and Interior.ym_mass_gap_grid_certified_sharp assembles from that grid and the two ends, taken as the bare hypotheses hstrong and hweak, the gap for ymModelAt 15, the model on the periodic extent 16, its axiom footprint the three foundational axioms + reflection positivity. d2At 15 is the circle-distance moment of the Lean correlation WilsonBridge.corrClay. The grid capstone needs hcover, hdiff and hbnd; no theorem in the development supplies hdiff or hbnd for d2At 15. The grid lemma is Interior.d2_le_of_analytic_grid.Moment.Read N indexes lags by Fin(N+1). The cosine route is confinement_at_of_cosAvg and confinement_at_iff_cosAvg, stated on cosAvgYMat 15, the $SU(3)$ correlation; the continuum statement is the substrate hypothesis of confinement_of_bounded_substrate. The layer identity is ContactDominance.circ_moment_eq_layer. The contact-relative law gives the conjunct through ShareEnvelope.substrate_of_contact_relative_decay and the full statement through yang_mills_of_contact_relative_decay; exponent 3 fails by ShareEnvelope.cubic_contact_relative_gives_no_bound, and no statement is formalised strictly between the two exponents. The coupling-direction family is ContactDominance.aperture_uniformity_does_not_give_coupling_uniformity.Moment.Reads satisfying every clause wilson_reflection_positive_at asserts, in which wilsonCorrAt does not appear. The closed alternative routes are Substrate.substrate_bound_needs_more_than_positivity, CellCouple.chain_budget_fails and CellCouple.form_perturbation_reaches_finitely_many; the Knabe control chain is LocalGap.laplace_window_gap_le and laplace_no_uniform_gap; the frustration-free assessment is carried in the source as prose; the volume cancellation is StrongCoupling.nonbridging_sum_eq_zero and wilsonCorrConn_eq_bridging_sum, and the local rate facts are core_rate_lt_one_of_small (an unnamed neighbourhood $\exists b > 0$) and coreRate_mono. The unconditional quarter is Substrate.substrateRatio_le_quarter, from d2At_le_quarter_sq, attained by antipodeRead_moment_eq_quarter_sq; the growth criterion is confinement_of_growth_ratio, its sharp threshold substrateThreshold, and Substrate.quarter_ratio_fails_aperture_criterion proves the quarter lands on the wrong side. The cut split is ContactDominance.circ_moment_le_cut. The vanishing tension under a bounded circle moment is tension_tendsto_zero_of_bounded_circ_moment, and confinement_of_substrate_bound exploits the L^2 ceiling to reach the limit. The pointwise criterion is confinement_at_of_substrate_sharp, which takes substrateRatio N $\beta < \text{substrateThreshold}$ at ONE aperture and ONE coupling and returns $\mu < \kappa_0$ there; substrateRatio N β is a moment of WilsonBridge.corrClay; confinement_at_coupling_of_ratio is the same statement at a fixed coupling over apertures. The flagship ym_mass_gap_of_ratio gives the gap, non-triviality and $SO(4)$ from a bound on the ratio, and ym_mass_gap_of_substrate is its corollary; both carry the three foundational axioms + wilson_reflection_positive_at and nothing else. The divergence of the grid constant is CompactBeta.clay_covariance_constant_not_aperture_uniform. The two-state cell truncation of the enclosure is machine-checked on a foundational-only footprint (CellSpectrum, ledger below).`

§11. Theorem 11.1b is `Complete.wilson_reflection_positive_at_even` (through `OddLagSplit.corrClay_reflection_positive`), whose axiom footprint is the three foundational axioms and nothing else. Theorem 11.1c is `ReflectionHalfSpace.wilson_reflPositive_limit_exists`; the eventual, pointwise hypothesis is `InfiniteReflection.reflPositive_of_eventually_pointwise`; `symCube_refl_stable` and `symCube_exhausts` enter `wilson_reflPositive_limit_exists` as proof terms; the Wilson density's four properties are supplied by `WilsonAction`; the non-constant observable is `ReflectionHalfSpace.halfSpaceAlg_has_nonconstant`; reflection invariance is `reflection_facts_on_halfSpaceAlg`; the translation holonomy is `InfiniteShift.wilsonHol_ishiftPlaQ.halfSpaceAlg` is a `Submodule` of the continuous functions, not a multiplicatively closed algebra; the name is inherited. Theorem 11.1d is `MomentShape.corrHyper_neg`, its corollaries `corrClay_neg` and `wilsonCorrAt_neg`, and `MomentShape.wilsonCorrAt_circLag_congr`, all on the three foundational axioms alone; the reflection step is `LogConvex.EW_plaqE_pair_shift`, and the tree carries no separate translation-invariance theorem for the correlation. The half-line degeneracy is `Spectral.flat_of_aperiodic`. Theorem 11.1e is `ContactFloor.contact_relative_unconditional`, feeding `ShareEnvelope.substrate_of_contact_relative_decay`; its cut is `StrongCoupling.core_rate_lt_one_of_small_hypercubic`. The machine-checked hypothesis is on the core rate, not on β : `coreRate K $\beta < 1$; the only machine-checked numeral it yields on β is $\beta < 0.12$ (ClayAssembly.coreRate_lt_one_forces_small_beta, whose own derivation note records the true ceiling 0.111572 and rounds up deliberately); no declaration in the tree carries the numeral 2.9363×10^{-5} . The lag cut is StrongArm.exists_geom_quartic_bound; the touch degree is StrongCoupling.touchDeg_bd_le; the 2 is WilsonAction.wilsonDensity_le_two, through ContactFloor.wilsonCorrConn_self_ge_haar; the contact value is ContactFloor.corrClay_zero_at_zero_eq and corrClay_zero_at_zero_const; the floor is contactFloor_holds; $\delta_0 = 1/18$ is ContactValue.corrClay_four_zero_zero, carried to every extent by corrClay_zero_at_zero_value. The composition is open: contactFloor_holds routes through the non-constructive exists_haar_floor, and composing it with the evaluated value is the unwired step. The scale-free obstruction is ShareEnvelope.mass_floor_is_not_scale_free; the sign equivalence is CharacterExpansion.su3_kernel_nonneg_iff. Theorem 11.1f is WeakArm.corrClay_le_at_zero, WeakArm.no_strict_lag_bound_from_shape, WeakArm.no_uniform_quartic_constant_of_vanishing_rate and WeakArm.corrClay_at_period. Theorem 11.1g is LogConvex.corrClay_log_convex, the tree's only coupling-free ratio inequality; ActionSplit.pairing_nonneg_of_local is quantified over observables, and LogConvex.wilsonReflForm is the first Transfer.ReflForm constructed from the Wilson measure. The tightness assembly carries the three foundational axioms only; the operator-theoretic segments of the reconstruction are formalised on foundational axioms only. Invertibility is TransferInvertibility.isUnit_of_spectral_hypothesis and not_isUnit_of_isCompactOperator, composed in spectral_hypothesis_fails_for_compact; the trace-class identification is cited and not formalised. The contraction form is TransferGap.GapAt, and GapToOperator.spectrum_opt_subset gives the real spectrum inside $\{1\} \cup [-r, r]$; the gap conclusion from it is proved in seven places, and no declaration in the tree produces a GapAt — it is a hypothesis everywhere it appears. The Osterwalder–Schrader continuation is entered as the named axiom os_reconstruction_wightman. The moment-support lemma is MomentSupport.le_of_positive_weight_decay; the functional-calculus step is GapOfDecay.gapped_of_positive_decay (foundational axioms only); threaded through confinement, ym_spectral_gap reconstructs the $SU(N)$ Hamiltonian with spectral gap $\Delta_{YM} = \kappa_0 - \mu$, and #print axioms returns exactly the reduction's own footprint — the moment-support bridge adds no axiom. gapped_of_positive_decay takes a positive-weight spectrum of any finite width; the companion inputs are named in MomentSupport.`

§12, *the requirements table*. The volume row: $\rho'(n) \leq \rho'(1)^n$ lives in `Mixing.lean`, which carries no volume index; `Certify.gap_uniform_in_volume_of_intensive` TAKES the intensive premise as a hypothesis, and nothing in the tree supplies it unconditionally — it is machine-checked for the DECOUPLED product transfer; the single-plaquette gap is machine-checked at $V=1$. The OS row: the compactness assembly carries the three foundational axioms only; the OS→Wightman reconstruction is stated as a named axiom and consumes an `OSData` whose `transl` acts by $\mathbb{R}^4$; `LatticeTranslNoGo` closes both horns by proof — `transl_eq_id_of_finite_order` at finite periodic extent and `addHom_to_int_lattice_eq_zero` on $\mathbb{Z}^4$; once `transl` is the identity,

OS1 holds of an ARBITRARY bilinear form (`os1_holds_of_everything_when_transl_trivial`), and the only `OSData` in the tree is the trivial one. The local-fields row: no stress tensor and no operator product expansion appears anywhere in the development; `Running.lean` proves $b_0, b_1 > 0$ and no interior fixed point, and `AsymptoticScaling.lean` makes asymptotic scaling a real constraint — `AsymptoticScalingAt` is proved satisfiable AND refutable — with no producer for it on the Wilson theory. The any- G row: the capstone `ClayRoutes.clay_continuum_of_boxPatchGap` is stated for $SU(N)$, $N \geq 2$, with its box input `PairCorrSharp.boxPatchGap_floor_su2` and `boxPatchGap_floor_su3`; `BoxPatch.BoxPatchGap` is open for $SU(N \geq 4)$; `CompactGauge` runs over an arbitrary compact group and `clay_nontriviality_of_wilson_variance` is stated for an arbitrary compact G . **Lie-algebra simplicity cannot be stated against Mathlib v4.31** — `SimpleGroup.lean` settles only the abstract-group reading, where $SU(N)$ is not simple because it has a centre. The load-bearing read: the formal development consumes the aperture condition $m_{hi} \leq 3^{-1/4}$, and the full bar from $\forall \beta \geq 0$, $\mu_{YM} \beta < \kappa_0$ carries the three foundational axioms plus reflection positivity. The limit over extents is `Measure.continuum_of_family`. The plaquette variance is `PlaquetVariance.corrClay_zero_pos` and `corrClay_zero_eq`, with the negative control `wilsonCorrConn_self_eq_zero_of_trivial` at $SU(1)$; non-Gaussianity is absent from the development.

§12.1–§12.7. The gapless-weight bound is `ZeroMode.lean`, which derives the decay in place of defining it: `zero_mode_lt_of_tension` ($c(N+1) < (3^{1/4} - 1) \sum_d g(d)$) and `no_zero_mode_of_tension_lt_floor` (c squeezed to zero); `wilson_correlation_gap` states the rate on the CONSTRUCTED $SU(N)$ Wilson correlation of `WilsonBridge.lean` — Mathlib's `specialUnitaryGroup`, the real Wilson action, product Haar over links, ordered-loop holonomy. `WilsonHypercubic.link_not_private` rules out private links in every $d \geq 3$, and `bd3_link_not_private` is its $d = 3$ case. The box independence of the cap is `ScreenedGap.gap_bound_box_independent` and `gap_survives_thermodynamic_limit`. The definitional witness of §12.3 is `ymModelAt`. The flagship of §12.5 is `ym_mass_gap_of_ratio`; the junction form is `ym_mass_gap_of_junction`, with residuals `hfe` ($\kappa_0 - \mu \leq c$) and `hgap` ($c \leq \Delta$), neither of which is proved. The representability of §12.6 is `Complete.WilsonSpectral N \beta`, which is `Nonempty (Spectral.PeriodicSpectralForm (N+1) \ (\\mathhtt{wilsonCorrAt}) \ N \ \beta)`; `SlabQuadratic.wilsonSpectral` proves it at $N = 3$ for every $\beta \geq 0$, foundational axioms only, and `Complete.wilson_sum_eq_corr` makes the spectral sum equal the ensemble's correlation. The characterisation is `SpectralFour.four_iff`, the order relation `LagOneDominates.pairRef1Positive_iff`, and `SpectralFour.fourRepresentable_const` proves the constant triple representable. No theorem in the tree takes `WilsonSpectral` as a hypothesis.

13.9 The status ledger. Status labels are by provenance: *proved* (analytic or Lean), *certified* (a deterministic or exact-rational computation), *certified statistical* (a finite-sample empirical-Bernstein bound that assumes independent configurations and takes the sample range as the support width, so its confidence level is nominal), *verified* (measured, reproducible). The rows produced by code are in §14.

claim (section)	Lean	status
entropy floor: directed-path count 3^k , density $\kappa_0 = \frac{1}{4} \ln 3 > 0$ (§7)	<code>Floor.lean</code> , <code>CubeArea.lean</code>	proved (foundational), both factors: the count by <code>cubeConfig_injective</code> and <code>directed_surface_count</code> ($N(A) \geq 3^k$), the area by <code>boundary_card_eq</code> ($ \partial C = 4k + 6$, from <code>card_faces</code> , <code>card_cubeConfig</code> , <code>card_cubes_with_face_le_two</code> and <code>card_sharedFaces</code> through <code>incidence_double_count</code>). So $\frac{1}{4}$ and $\ln 3$ are each a theorem; κ_0 takes no input. The surfaces are also CLOSED (<code>CubeClosed.edge_parity_all</code> , $\partial\partial = 0$ over $\mathbb{Z}_2$ at EVERY edge, by a double count over cubes), CONNECTED (<code>CubeConnected.boundary_connected</code>), and all pass through one fixed face (<code>origin_face_mem_boundary</code>), so each carries the four properties Theorem 7.1's family asks of it. The packaging is done too: <code>VortexFamily.vortexFamily</code> DEFINES the family by those four conditions on the physical lattice's own plaquette type, and <code>VortexFamily.three_pow_le_vortexCount</code> is $3^k \leq N$ against it with no hypothesis. <code>VortexCount.kappa0_is_the_surface_entropy_density</code> assembles the two factors into the density itself: $\log(\text{card surfaces})/\text{area} \rightarrow \frac{1}{4} \ln 3$
gap reduction $c > 0 \Rightarrow a_{IR} = 0 \Rightarrow \Delta \geq c > 0$ (§6)	<code>ReachFreeze.lean</code>	proved
asymptotic-freedom sign $\beta < 0$, no interior fixed point (§5–6)	<code>Running.lean</code> (<code>b0_pos</code> , <code>b1_pos</code> , <code>beta_neg</code> , <code>no_interior_fixed_point</code> , <code>sun_beta_neg</code> , <code>sun_no_interior_fixed_point</code>)	proved, as arithmetic: each quantifies over bare reals and a bare N , with no gauge group, action, measure or correlation function in the statement. That b_0, b_1 are the physical beta-function coefficients is the cited perturbative computation, and only the root aggregate imports the module
forgetting property: margin $\Rightarrow$ decay $\wedge$ summable $\wedge \Delta_{\text{Ces}}$ (§4, 6)	<code>Forgetting.lean</code>	proved
$A1 \wedge A2 \Rightarrow$ gap + non-triviality + $SO(4)$; $A1$ ends; $A2$ isotropy (§9)	<code>Apriori.lean</code>	proved
the reduction: $A1, A2$ as theorems $\Rightarrow$ gap with no $A1/A2$ hypothesis (§9)	<code>Complete.lean</code>	proved (footprint: three foundational + <code>wilson_reflection_positive_at</code>); A1 is replaced by a substrate or grid input rather than removed, and the conclusion is the three clauses about <code>ymModel</code>, whose single mode is defined as $e^{-(\kappa_0 - \mu)}$ — scope in §13.4
$A1$ interior: analytic step $\cos x \geq 1 - x^2/2 \Rightarrow \mu < \kappa_0$, the $1/L^2$ aperture scaling, and the grid route (§9)	<code>Moment.lean</code> , <code>Complete.lean</code> , <code>Interior.lean</code>	proved; the named input is a bound on the aperture-free ratio $\langle d^2 \rangle / (N+1)^2$ (<code>substrateRatio</code> , via <code>confinement_of_growth_ratio</code>), or, dropping it, finite-volume analyticity (a bounded β -derivative) plus a finite grid under the sharp ceiling (<code>Interior.ym_crossover_confinement_of_grid_sharp</code> , <code>d2_le_of_analytic_grid</code> ; the gap for <code>ymModelAt</code> 15 from it and the two ends is <code>Interior.ym_mass_gap_grid_certified_sharp</code>); the measured grids they are read against are evidence for those hypotheses, which stay open (they read

claim (section)	Lean	status
		$SU(2)$ ensembles, distinct from <code>d2At 15</code> , at nominal confidence, and leave <code>hcover</code> , <code>hdiff</code> and <code>hbnd</code> open; §9 states where the grid ends, §14 the scripts)
modified Bessel $I_n(x) > 0 \Rightarrow I_2/I_1 > 0$ (§8.2)	<code>Bessel.lean</code>	proved
$\Delta = \kappa_0 - \mu \Rightarrow C(\tau) \leq Me^{-\Delta\tau}$ (§6, §12)	<code>GapRate.lean</code>	proved
RP survives the continuum limit (§11)	<code>Existence.lean</code>	proved
continuum measure tight from the gap; OS0–OS3 survive the joint limit (§11)	<code>Existence.lean</code> , <code>Measure.lean</code>	proved; the OS-data family (<code>LatticeYMFFamily: os_rp, os_gap, os_euc, os_perm</code>) is instantiated for $SU(N)$ (<code>ymFamily</code>), its reflected forms built from the same <code>wilsonCorr</code> as the gap side (<code>os_rp</code> = the RP axiom), with <code>os_euc/os_perm</code> STRUCTURAL for <code>ymFamily</code> — <code>QYM</code> reads only <code>j.2.2</code> while the two actions move <code>j.1</code> and <code>j.2.1</code> , so both close by <code>rfl</code> — and DERIVED for <code>ymFamilyGauge</code> via <code>QG_eq</code> from <code>Symmetry.expect_invariant</code>
existence and the gap for one model: gap AND OS continuum measure (§11–12)	<code>ym_existence_and_gap_of_junction</code>	proved for a constructed $SU(N)$ instance (<code>ym_wilson_of</code>): <code>ym_existence_and_gap_of_junction</code> delivers both conditionally on its two residuals, which are open : <code>hfe</code> ($\kappa_0 - \mu \leq c$) and <code>hgap</code> ($c \leq \Delta$), named as open in §12.5 and open everywhere in the tree (footprint = three foundational + the one named axiom <code>wilson_reflection_positive_at</code>), no new axiom, no <code>sorry</code> ; both built on one <code>wilsonCorr</code> model, the §2–§3 physical identification cited
the constructed gauge measure: OS Euclidean/permutation invariances derived from $SU(N)$ Haar-invariance , on a genuine ordered-loop Wilson holonomy and the Wilson action density (§11–12)	<code>LatticeGauge.lean</code> , <code>CompactGauge.lean</code> , <code>SUN.lean</code> , <code>WilsonGauge.lean</code> , <code>WilsonAction.lean</code> , <code>WilsonLattice.lean</code> , <code>WilsonReal.lean</code>	proved (foundational): <code>Symmetry.expect_invariant</code> (Haar-invariance $\circ$ plaquette-permutation); the $SU(N)$ compact/topological-group/Borel instances (<code>SUN</code> , closed + bounded $\Rightarrow$ compact); <code>WilsonGauge.ym_continuum_gauge</code> (OS0–OS3 continuum measure on genuine $SU(3)$ Haar over the four-dimensional periodic Wilson lattice <code>WilsonHypercubic.sysWilson 3 4 n</code> — the Clay problem's group and dimension — with the Euclidean/permutation invariance supplied by the lattice's own axis symmetry <code>axisSymmetry</code> , derived from <code>shift_axis/bd_axis</code> ; three foundational axioms only); <code>WilsonGauge.ym_existence_and_gap_gauge</code> (gap + measure, the same one named axiom <code>wilson_reflection_positive_at</code> , on the gap side; measure side axiom-free); non-vacuous by <code>WilsonReal.sysReal_invariant</code> (concrete $SU(2)$ two-plaquette system, plaquette-swap symmetry)
the gauge system on a genuine d -dimensional periodic hypercubic lattice, with the axis symmetry derived (§11–12)	<code>WilsonHypercubic.lean</code>	proved (foundational): <code>sysWilson</code> is the ordered-loop Wilson system on <code>Site = Fin d $\rightarrow$ Fin n</code> , links (μ, x) and plaquettes $((\mu, \nu), x)$, with $d \cdot n^d$ links and $d^2 n^d$ plaquettes machine-checked (<code>card_link</code> , <code>card_plaq</code> : 16,384 and 65,536 at $d=4, n=8$) and the action density the genuine Wilson density, not the zero function (<code>sysWilson_phi</code>), so β moves the measure; degenerate planes $\mu=\nu$ contribute the identity holonomy and need no exclusion (<code>bd_diag_hol_one</code>); the discrete Euclidean invariance is DERIVED from the geometry --- an axis permutation commutes with the unit shift (<code>shift_axis</code>), hence transports the boundary word (<code>bd_axis</code>), hence is a <code>Symmetry</code> of the system (<code>axisSymmetry</code>), which is exactly the <code>os_euc/os_perm</code> datum. d, n, N are arguments: $d=4, N=3$ is an instantiation of the one construction. What remains is transporting the finite-aperture read onto it
the OS-measure's infrared datum DERIVED from the tension, on the 4-D $SU(3)$ measure (§11–12)	<code>WilsonGauge.lean</code> , <code>Measure.lean</code> , <code>ZeroMode.lean</code>	proved (foundational): <code>WilsonGauge.ym_continuum_gauge_counted</code> builds the family through <code>Measure.familyOfSortedCount</code> , which takes an ORDERED spectrum and a bound on its resolved count in place of <code>os_gap</code> and derives <code>os_gap</code> from them (<code>os_gap_of_sorted_count</code>); the count is the quantity <code>ZeroMode.resolved_count_le_of_subset</code> obtains from the measured tension. The reflected form, and hence <code>os_rp/os_euc/os_perm</code> , is the same genuine $SU(3)$ Gibbs expectation as <code>ym_continuum_gauge</code> . Two hypotheses are named: at most c modes clear the edge (the tension's), and at least one does (arity -- the clamp $Q \leq 1$ needs something to be bounded by, and a theory with a vacuum resolves it)
no plaquette owns a private link, in ANY dimension $d \geq 3$ — the structural reason the theory is interacting (§11–12)	<code>WilsonHypercubic.lean</code>	proved (foundational): <code>link_not_private</code> . A private link can be integrated out first and left-translation by it carries its plaquette's holonomy through Haar, so that plaquette becomes independent, the connected correlation collapses to a contact term, and the aperture condition is satisfied trivially by a free theory. <code>WilsonBridge.bd3_link_not_private</code> said this for $d = 3$ on the normal-indexed geometry, which only exists in three dimensions; this says it for every $d \geq 3$ on the plane-pair lattice the OS measure uses. The 3 is derived: the witness plaquette must be non-degenerate, so its direction must avoid both

claim (section)	Lean	status
		spanning directions, and a two-element set of directions has a complement exactly when $d > 2$
reflection positivity: the MECHANISM proved, the cited content narrowed to one step (§11–12)	ReflectionPositivity.lean	proved (foundational, 8 theorems): <code>pairing_with_reflection_nonneg</code> — a positive-half function paired with its own reflection integrates to EXACTLY the square of its integral, from disjointness (<code>WilsonReal.block_integral_factor</code>) and the measure not distinguishing the halves (<code>relabel_measurePreserving</code>); <code>reflection_positive_at_zero</code> — unconditional at $\beta = 0$, nothing cited; <code>hol_congr_on_support</code> — a plaquette's holonomy reads only the links its boundary word names, the locality every splitting argument rests on; <code>action_split</code> and <code>action_on_congr_of_support</code> — the action divides by any plaquette partition and each piece is a function of its own half's links. So $e^{-\beta S} = g(U _+)g'(U _-)e^{-\beta S_{\text{cross}}}$ with the first two factors already paired, and <code>reflection_positive_of_expansion</code> sums the expansion the citation produces — nonnegative coefficients times paired products, each a nonnegative multiple of a square, so the summing is proved. What is cited is therefore narrower than "reflection positivity holds": it is that the Osterwalder–Seiler expansion of $e^{-\beta S_{\text{cross}}}$ EXISTS with nonnegative coefficients, a statement about characters of a compact group rather than about the theory. One scope limit: <code>reflection_positive_of_expansion</code> sums a FINITE index set. The cited expansion is an infinite convergent sum; a finite truncation is what finitely many straddling plaquettes produce at each order, and the passage to the limit is the analytic half that remains outside
the resolved-mode COUNT derived from the measured tension — the count hypothesis <code>Measure.familyOfSortedCount</code> asks for (§9, §11–12)	ScreenedGap.lean, ZeroMode.lean, Moment.lean	proved (foundational): <code>resolved_count_ceiling</code> divides <code>ZeroMode.resolved_count_le_of_subset</code> through into an explicit ceiling, <code>resolvedDim_ceiling</code> states it on <code>resolvedDim</code> (what <code>Measure.familyOfSortedCount</code> reads), and <code>resolvedDim_le_of_tension</code> composes it with <code>Moment.Read.substrate_lt_of_tension_lt_floor</code> — because the ceiling's M/S IS the substrate the tension bounds. Result: $\mu < \kappa_0 \Rightarrow \text{card}\{\text{resolved}\} \leq 0.360246 W / (\epsilon \lambda_0^{k+1})$, every symbol the read's own, the two constants being the sum-of-squares denominator and the entropy floor composed with $\cos x \geq 1 - x^2/2$
the noise edge belongs to the identity-removed RESIDUAL, and the raw edge under-reports (§9, §11)	ScreenedGap.lean	proved (foundational): <code>edge_raw_sub_residual</code> (the two floors differ by exactly the vacuum's share — an identity), <code>edge_residual_le_raw</code> (so the raw floor is the larger), <code>resolvedDim_antitone_edge</code> (a larger floor resolves no more modes), composing to <code>resolved_count_under_reported_of_raw_edge</code> ; non-vacuous by <code>under_report_is_strict</code> , which exhibits the mode counted by one floor and missed by the other. The direction matters — an under-reported count makes a bound that divides by <code>edge</code> look tighter than the data supports. Reported empirically elsewhere (§14); proved here, with no constant introduced
gauge invariance of the Wilson correlation, $\langle O \circ (U \mapsto gUg^{-1}) \rangle = \langle O \rangle$ — the defining symmetry, on the genuine ordered-loop Wilson action (§11)	WilsonAction.lean, WilsonLattice.lean, CompactGauge.lean, WilsonReal.lean	proved (foundational): <code>wilsonHol_conj</code> (the holonomy conjugates) + <code>wilsonDensity_conj</code> (the action density is a class function) $\Rightarrow$ the action is gauge-invariant; <code>isMulRightInvariant_probHaar</code> (compact-group unimodularity: probability Haar is left- and right-invariant) $\Rightarrow$ the conjugation <code>confConj_measurePreserving</code> preserves the measure; composed via the general <code>expect_invariant_of_mp</code> into <code>sysReal_gauge_invariant</code>
centre-vortex count-injection at the entropy-floor exact constant: $\kappa_0 - \mu \leq c$ from the machine-checked directed count (§7, §10)	VortexCount.lean	proved (foundational): <code>floor_count_injection</code> (the injection's counted weight is exactly <code>directed_surface_count</code> 's 3^k) and <code>floorTerm_exponent_is_the_area</code> (the exponent's $4k + 6$ is <code>CubeArea.boundary_card_eq</code> for the same configuration, so count and area speak about one object); <code>three_pow_le_card_of_embeds</code> and <code>three_pow_le_card_of_directed_surfaces</code> (the physical count dominates 3^k when the directed surfaces are among it — stated over sets of FACES, the type the vortex family has, and injective there by <code>CubeArea.boundaryFaces_cubeConfig_injective</code>), with <code>directed_surfaces_all_have_area</code> fixing all 3^k of them at the one area $4k + 6$. The 3-D $\rightarrow$ 4-D embedding IS built (<code>SurfaceEmbed.lean</code>): <code>faceToPlaq</code> sends a face to the plaquette spanned by the two axes its normal is not, at its corner, in the time-zero slice, and <code>directed_plaq_surfaces_count_and_area</code> states the count and the area over <code>WilsonHypcubic.Plaq 4 n</code> — the type the vortex family has. Its one hypothesis is that the path fits the box, $k + 1 < n$, which is why <code>three_pow_le_card_along_boxes</code> runs the injection along a GROWING sequence of boxes: <code>hM</code> is needed at every k and no fixed volume supplies it. Closedness is built too (<code>CubeClosed.edge_parity_all</code> : every edge lies in an even number of boundary faces, which is $\partial\partial = 0$ over Z_2), so what remains unbuilt is the containment alone — that these surfaces are among the

claim (section)	Lean	status
		physical vortex family, which is the physics; <code>junction_of_physical_count</code> and <code>selfSourcingJunction_of_physical_count</code> (discharging the flagship <code>Capacity.SelfSourcingJunction</code> $\kappa_0 - \mu \leq c$ with the count-injection proved and no constant fudge)
reconstruction produces a gapped quantum theory as data ($H \geq 0$, vacuum, gap), exhibited by the concrete operator <code>diag(1, 3^{-1/4})</code> with gap κ_0 computed, and any operator meeting the finite-aperture margin reconstructs to such a theory with gap κ_0 (§11)	<code>Reconstruction.lean</code> , <code>GappedTheory.lean</code> , <code>GappedExample.lean</code>	proved (foundational axioms only)
the contact-relative law PROVED on a strong-coupling interval, and the remaining region characterised (§11.1e–g, §13)	<code>ContactFloor.lean</code> , <code>StrongArm.lean</code> , <code>WeakArm.lean</code> , <code>LogConvex.lean</code>	proved (foundational): <code>ContactFloor.contact_relative_unconditional</code> gives $\rho(d) \leq C\rho(0)/\text{circLag}(d)^4$ at EVERY aperture and EVERY lag on a derived $[0, b]$, carrying nothing — so on that interval the substrate hypothesis is proved. Its one-time hypothesis, an aperture-uniform floor on $\rho(0)$, is discharged by <code>contactFloor_holds</code> : the exponent counts only plaquettes SHARING A LINK with the one read (<code>touchDeg_bd_le</code> , 16 dim, no extent), and at $\beta = 0$ the measure IS Haar so the contact value is one $SU(3)$ number at every extent ≥ 2 (<code>corrClay_zero_at_zero_eq</code>). SCOPE: the floor degrades like e^{-128b} and provably cannot reach the half-line, because the plaquette variance vanishes as $\beta \rightarrow \infty$; the open region is $(-\infty, 0) \cup (b, \infty)$. Two results bound the search there: <code>WeakArm.corrClay_le_at_zero</code> ($\rho(d) \leq \rho(0)$ at every lag, scale-free, and optimal by <code>no_strict_lag_bound_from_shape</code>), and <code>WeakArm.no_uniform_quartic_constant_of_vanishing_rate</code> , which proves NO coupling-uniform exponential bound exists to be found — the obligation is a POWER tail, not a rate. <code>LogConvex.corrClay_log_convex</code> is the tree's only coupling-free ratio inequality, and <code>WeakArm.corrClay_at_period</code> ($\rho(2m) = \rho(0)$) is why its chord cannot supply one
reflection positivity PROVED, and the gap side made axiom-free (§11.1b, §13)	<code>OddLagSplit.lean</code> , <code>CrossingIntegration.lean</code> , <code>EvenAperture.lean</code> , <code>Complete.lean</code> (<code>wilson_reflection_positive_at_even</code>)	proved (foundational): <code>OddLagSplit.corrClay_reflection_positive</code> proves the body of the <code>wilson_reflection_positive_at</code> axiom at even lattice extent ≥ 4 and $\beta \geq 0$ — action split, three-block Haar factorisation with the shared block outside the square, mirror transport, handedness reconciliation; no representation theory — the step Schur orthogonality performs in Osterwalder–Seiler is carried instead by <code>CrossingIntegration.wilson_crossing_pairing_nonneg</code> , applied at <code>OddLagSplit.lean:4303</code> , whose own no-go <code>straddling_word_not_paired_at_odd_lag</code> shows the straddling word is not of paired form and so closes the shortcut that would have avoided it. Substituting that certificate at the one definition that consumed the axiom gives <code>EvenAperture.existence_and_gap_of_substrate_even</code> and <code>mass_gap_rate_and_continuum_even</code> , both on the three foundational axioms ALONE. The bridge <code>substrate_even_of_substrate</code> (original hypothesis $\Rightarrow$ even-aperture hypothesis) is stated through <code>d2At</code> and so reports the axiom itself; the chain it feeds does not. Scope: the read is built at $\max(\beta, 0)$, identical to the unrestricted read on $\beta \geq 0$ (<code>readEven_eq_readYMA</code>) and the $\beta = 0$ column relabelled below it (<code>readEven_eq_at_zero</code>), which is exactly where <code>CharacterExpansion.su3_kernel_nonneg_iff</code> shows the cross kernel is not PSD
the moment-support step: a positive-weight exponential sum bounded by a decaying total forces every observed mode $\leq \rho$ — the "seen by v " half of the reconstruction hypothesis $\text{spec } T \subseteq \{1\} \cup [\varepsilon, 3^{-1/4}]$ (§11, §13)	<code>MomentSupport.lean</code> (<code>le_of_positive_weight_decay</code>)	proved (foundational, axiom-free). Lifting "seen by v " to the full spectral support $\text{spec } T$ needs Reeh–Schlieder totality and vacuum simplicity — cited companion facts, open, to be <i>proved</i> ; <code>ym_reconstructed_gap</code> stays a clean conditional theorem on the spectral input hsp
finite-aperture gap; certified $\mu < \kappa_0 \Rightarrow$ gap; contraction bound (§4, 6, 9)	<code>Aperture.lean</code> , <code>Certify.lean</code>	proved
single-cell ($V=1$) spectral gap $\geq \kappa_0$ of the $SU(2)$ Kogut–Susskind cell (§9)	<code>CellSpectrum.lean</code> (<code>atMostOne_eigenvalue_lt</code> ,	certified (exact-rational Sturm/Schur, all couplings; §14) and Lean-proved for the dominant two-state truncation at every coupling (<code>Hcell12_clears_floor</code> : $m_{\text{cell}} \leq 3^{-1/4}$) and the three-state cell on $\lambda^2 \leq (\frac{3}{4} - \kappa_0)(2 - \kappa_0)$; and —

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	<code>eigenvalues_gap_of_codim1_form, Hcell12_gap, Hcell13_gap</code>	extending these to <i>every</i> truncation — the general $j_{\max}$-truncated tridiagonal cell has $\text{gap} \geq \kappa_0$ at all $j_{\max}$ on the strong-coupling window $\lambda \leq (\frac{3}{4} - \kappa_0)/2$ (<code>cell_general_gap</code>, via <code>interacting_cell_gap</code>), the machine-checked form of the $j_{\max}=30$ truncation-completeness statement on that window. Beyond that window, to the full coupling range, the abstract Sturm/LDL⁺ inertia engine is machine-checked and foundational: a completing-the-square factorization $H - sI = L^\top \text{diag}(p)L$ with nonnegative pivots gives $E_0 \geq s$ (<code>posSemidef_of_ldl</code>, with the reconstruction <code>ldl_entry/tridiag_ldl_of_recurrence</code>), exactly one negative pivot gives $E_1 \geq s$ through a codimension-1 kernel (<code>gap_of_ldl_one_neg_pivot</code>, <code>ldl_quadform</code>, <code>ldl_form_ge_on_kernel</code>), and a tight variational E_0 upper bound (<code>exists_eigenvalue_le_of_form</code>, least eigenvalue $\leq$ Rayleigh quotient) combines with the one-negative-pivot bound for the <i>relative</i> gap in the large-coupling regime where both E_0, E_1 fall below κ_0 (<code>gap_of_ldl_one_neg_pivot_rel</code>); this reduces the physical $SU(2)$ cell gap to a rational pivot certificate (<code>Hcell1R_gap_of_certificate</code>), exhibited non-vacuous in both regimes — the strong-coupling absolute gap (<code>hcell1R_gap_demo</code>, $\lambda=1, E_1 \geq \kappa_0$) and the large-coupling relative gap (<code>hcell1R_rel_gap_demo</code>, $\lambda=3$, where <i>two</i> eigenvalues fall below κ_0 and the absolute bound provably fails, yet the gap $\geq \kappa_0$ holds). The per-λ certificate for the full crossover range (§14) certifies 61 of 61 couplings on $\lambda \in [0.16, 6.76]$ — 17 by the absolute route, 44 by the relative one — in exact rational arithmetic with no floating point in any bound. The recurrence is solvable in closed form backwards from the last index ($p_{n-1} = d_{n-1} - \mu, p_i = d_i - \mu - \lambda^2/p_{i+1}, e_{i+1} = -\lambda/p_{i+1}$), so the pivots are ratios of integers and the sign claim is checkable by inspection. The shift $\mu = 2747/10000$ is rational and strictly above κ_0 — no rational equals $\frac{1}{4} \ln 3$ — and it is checked against the certified enclosure of κ_0, so the conclusion is $\Delta \geq \mu > \kappa_0$. Above $\lambda \approx 2.1$ two pivots go negative, the absolute statement is false there, and the relative route carries it (<code>gap_of_ldl_one_neg_pivot_rel</code>). Every certified coupling is transcribed into Lean (<code>CellPivot.lean</code>, §14): 61 theorems, 17 by the absolute route and 44 by the relative one, at $j_{\max}=8$, one per λ, each foundation-only. The truncation is set by the TAIL, not by the gap value: $j_{\max}=8$ is the smallest at which the tail bound lifts every transcribed theorem from $H_{\text{cell}}^{(j_{\max})}$ to the cell it truncates (measured: 4 lifts 17 of 61, 6 lifts 70 of the cover's 75 anchors, 8 lifts all), and no shallower truncation is emitted. The relative half needed a reduction of its own (<code>Hcell1R_gap_of_certificate_rel</code>: a Sturm shift bounding E_1 from below, a trial vector's Rayleigh quotient bounding E_0 from above) and short rational witnesses for both — the theorem asks for A valid shift and ANY nonzero vector, so rounding the bisected shift and the trial direction gives a weaker certified gap that is still true and still clears the floor, at shifts of 0, $-1/2, -1, -3/2$ and pivots of five digits. The file names the couplings it declines and why, so the range it reaches is stated. And the certificate covers the coupling range continuously. The cell is LINEAR in λ — Casimir diagonal, $-\lambda$ times the path-graph adjacency off it — so Weyl's inequality moves each eigenvalue by at most $\Delta\lambda \ A\$ with $\ A\ \leq 2$, and the gap by at most $4 \Delta\lambda$. Bisection for the LARGEST shift each route supports rather than the default μ (which only just clears the floor) gives each certified coupling a neighbourhood of radius $(\mu_{\max} - \kappa_0)/4$ on which the gap cannot have fallen to κ_0. Walking right by each point's own radius, the absolute route covers $[0.16, 1.99166]$ in 68 points and stops exactly where $\mu_{\max}$ falls to κ_0 — its own boundary, set by the route itself — and the relative route covers $[1.99166, 7.88673]$ in 7, its margins being far larger. Together: $[0.16, 6.76]$ continuously. And the Lipschitz constant this rests on is machine-checked, without any Weyl inequality. Mathlib has no eigenvalue-perturbation Weyl, but the two spectral facts the certificate is built from take QUADRATIC FORMS and not eigenvalues (<code>exists_eigenvalue_le_of_form</code>, <code>atMostOne_eigenvalue_lt</code>), so transporting a certificate to a neighbouring coupling only has to move the form. <code>MassGap.CellPerturb</code> proves it does so by at most $2 \Delta\lambda$ per unit norm: the coupling enters H only through the 0/1 path-graph adjacency (<code>Hcell1R_sub_smul_adj</code>), whose form is bounded by $2\ v\ ^2$ by AM--GM against its row and column sums (<code>adj_form_bound</code>), giving <code>Hcell1R_form_lipschitz</code>; the two transports <code>cell_form_le_of_form_le</code> and <code>cell_form_ge_of_form_ge</code> are one per eigenvalue, which is where the 4 comes from. All nine are foundation-only. And the cover is a THEOREM. <code>MassGap.CellCover</code> (generated, §14): 19 anchors, each carrying its own pivot certificate and its own EARNED radius (certified $-\kappa_0$)/4, chained into <code>cell_gap_on_range</code> (<code>lam :</code>

claim (section)	Lean	status
		<code>R) (h1 : 4/25 <= lam) (h2 : lam <= 169/25) : CellGapAtLeastR 8 lam kappa_0</code> --- for EVERY REAL $\lambda \in [0.16, 6.76]$ the cell has a spectral gap at least κ_0's certified upper bound, foundation-only, and checked non-vacuous in the file at $\lambda = 1$. Nineteen anchors suffice, against seventy-five for the two separate walks, because each step takes whichever route earns the larger radius: the relative route dominates almost everywhere, including at weak coupling where the absolute one was assumed natural. Overlap of consecutive balls and coverage of the range are checked before the kernel sees them (§14) and again by the kernel. The coupling is real and not rational: the anchors stay rational because the certificate produces rational pivots, but each ball ranges over $\mathbb{R}$ (<code>cellGapAtLeastR_of_ball</code>, via <code>HcellR_eq_HcellRr</code> --- the equality between the two independently built cells). Tail liftability BETWEEN anchors is carried by the certificate; the truncation-independent bound <code>CellTail.pivotSeqFrom_ge_of_selfconsistent</code> makes it an argument, and closing it as one statement is open. Half of that is immediate --- with $\varepsilon = \lambda$ the tail-start condition reduces to $d_{m+1} \geq s + 2\lambda$, monotone in both λ and the shift, so a truncation deep enough at an interval's right-hand end is deep enough throughout it --- and what remains is the sign condition on the tail-inclusive pivots across an interval. And the truncation tail is bounded. Truncating the backward recurrence at n sets $p_{n-1} = d_{n-1} - \mu$, DROPPING the $-\lambda^2/p_n$ term, so a truncated pivot is larger than the true one and the error runs in the one direction that could report a positive pivot where the cell has a negative one. The recurrence is monotone in its seed (<code>CellTail.pivotSeqFrom_mono</code>, <code>pivotSeqFrom_mono_above</code>, <code>pos_above_of_tail</code>), so seeding it with a lower bound on the cell's pivot at n bounds every pivot below; that bound is self-consistent exactly when $\varepsilon(d_{m+1} - \mu - \varepsilon) \geq \lambda^2$, which closes at $m \leq 6$ across the whole crossover because $d_i = i(i+2)/4$ grows while λ is fixed. The question this settles is asked AT THE SHIFT EACH ROUTE CERTIFIES AT --- the absolute route's bisected $\mu_{\max}$, the relative route's Sturm shift, which is negative above $\lambda \approx 2.6$ --- and the test is that the tail-inclusive sequence carries at most one negative, not that its signs match the truncated one's. That is the form the theorem consumes: the tail-inclusive pivots are a pointwise LOWER BOUND on the cell's, so a nonnegative entry is a nonnegative pivot in the cell, and at most one negative gives at most one eigenvalue below the shift for the CELL. Measured this way at $j_{\max}=30$, all 75 cover anchors and all 61 certified couplings lift (§14). The $j_{\max} = 4, 8, 12, 16$ spread of 3.4×10^{-4} is corroboration; the claim rests on the tail bound. The Lean wiring from a bounded pivot SEQUENCE to an inertia count on the untruncated cell is open; foundation-only footprint
gap uniform in volume from the cell (§9, §12)	<pre>VolumeRate.lean (gap_rate_at_floor_uniform_in_volume, product_volume_gap_rate, product_volume_gap_rate_concrete); CellSpectrum.lean (Hcell12_clears_floor, product_subvacuum_le)</pre>	proved (foundation-only): reflection positivity gives $\rho'(n) \leq \rho'(1)^n$ at one volume, so the single cut $\rho'(1) < 1$ carries the gap uniform in SEPARATION; the volume direction is the separate intensive premise, and where that premise is supplied ONE rate $\kappa_0 = \frac{1}{4} \log 3$ bounds every volume's correlator — the radius being the machine-checked cell ceiling $3^{-1/4}$ (<code>Hcell12_clears_floor</code>). The <code>CellSpectrum</code> forms state no common rate: <code>gap_uniform_of_cell</code> is indexed by the separation and carries no volume index at all, while <code>gap_uniform_of_cell_intensive</code> and <code>product_volume_gap</code> conclude $\exists \kappa, \kappa > 0 \wedge \dots$ in which κ does not occur in the body it binds, a shape <code>VolumeRate.exists_pos_and_iff</code> machine-checks to be equivalent to dropping the existential. <code>cell_volume_bar_nonvacuous</code> records a constant quantified over F, which gives no volume uniformity; its own docstring states this, and that the physical statement $\forall F, m_{hi}(F) \leq 3^{-1/4}$ for the ensemble read is OPEN. The intensive input $m_{hi}(F) \leq 3^{-1/4}$ is machine-checked for the decoupled product transfer (<code>VolumeRate.product_volume_gap_rate/product_margin_le_cell_ceiling</code>: the volume-uniform gap runs on the genuine product-of-cells spectrum with its intensive margin derived, not a constant witness — an F-cell product's sub-vacuum magnitude stays at the single-cell ceiling uniformly in F, vacuum factors contributing 1 not growth; concretely <code>product_volume_gap_concrete</code> exhibits the F-cell product of $\text{diag}(1, 3^{-1/4})$) and is the measured continuum read for the interacting model (Figure 13); the interacting (coupled) transfer's volume gap reduces — through the dimension-general corank-1 form (<code>coupled_gap_of_coupling_bound</code>) — to two proved form bounds: the Casimir margin (<code>diag_form_margin</code>, excited energies $\geq \frac{3}{4} > \kappa_0$) and the coupling bound (<code>coupling_form_lower</code>, a symmetric coupling with row sums $\leq c$ has form $\geq -c$) — assembled as <code>interacting_cell_gap</code>, the interacting cell gap $\geq \kappa_0$ for coupling row-sum $c \leq \frac{3}{4} - \kappa_0$. The volume-uniform interacting gap follows from the Gosset-Mozgunov finite-size criterion (Knabe, <i>Commun. Math.</i>

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		<i>Phys.</i> 1988; Gosset–Mozgunov, <i>J. Math. Phys.</i> 2016), whose operator-algebra construction is machine-checked and axiom-free: the equivalence $H^2 \succeq \gamma H \Leftrightarrow$ (frustration-free gap $\geq \gamma$) (<code>gap_of_operator_sq_ge</code>, <code>operator_sq_ge_of_gap</code>); the deformed-window identities on the periodic chain in the pair-sum (T_δ) basis (<code>chain_sq_expand</code>, <code>weighted_window_sq_sum</code>, <code>weighted_window_sum</code>, <code>regroup_by_fiber</code>); the entire eq.-21 left side collapsed to a single net-coefficient sum $H^2 + \beta H - \alpha \sum_k B_{n,k}^2 = \sum_\delta (1 + \beta[\delta=0] - \alpha W_\delta) T_\delta$ (<code>eq21_lhs</code>); the eq.-23 operator inequality $H^2 + \beta H \succeq \alpha \sum_k B_{n,k}^2$ modulo per-term positivity (<code>eq23_of_terms</code>); the far-offset positivity $T_\delta \succeq 0$ for commuting projectors (<code>commuting_proj_mul_posSemidef</code>, <code>Tdelta_posSemidef</code>, <code>psd_nonneg_smul_sum</code>); the α, β cancellation at $\delta = 0, \pm 1$ (<code>gm_alpha_beta</code>); and the state-level gap assembly $\lambda \geq \alpha \gamma' c - \beta$ (<code>gm_gap_of_eq23</code>). the entire abstract Knabe/Gosset–Mozgunov finite-size criterion is machine-checked and axiom-free. The capstone <code>gm_gap_c1</code> proves: for a frustration-free nearest-neighbour projector chain with the plain Knabe windows $c_j=1$, given (i) the commuting structure $T_\delta \succeq 0$ for $\delta \notin \{0, \pm 1\}$, (ii) a low-energy eigenstate ψ ($H\psi = \lambda\psi$, $\lambda > 0$), and (iii) GM's Lemma 4, the eigenvalue obeys the Knabe bound $\lambda \geq (m-1)^{-1} \gamma' m - (m-1)^{-1}$. This assembles the abstract eq.-23 operator inequality <code>gm_eq23_c1</code> ($(\sum_b h_b)^2 + \alpha \sum_b h_b - \alpha \sum_k B_k^2 \succeq 0$, via the fiber sums <code>fiber_sum_pos/neg/empty</code>, the net-coefficient nonnegativity <code>hnn_c1</code> where $A_x = m - x$ trivialises the autocorrelation, and the $\delta=0, \pm 1$ cancellations), the operator-to-form conversion <code>eq23_form_of_psd</code>, and the state-level gap <code>gm_gap_of_eq23</code>. Every scalar, combinatorial, operator-algebraic, and gap-extraction step of the Knabe/GM argument is formalized, none as axioms (foundational footprint). The theorem is exhibited non-vacuous by <code>knabe_gap_demo</code> (a concrete commuting-projector chain satisfying all hypotheses and yielding a genuine gap). What remains is only the physical frustration-free binding — instantiating the abstract chain with the actual Kogut–Susskind/Wilson bond projectors, which supplies the three physical hypotheses ($T_\delta \succeq 0$ via <code>Tdelta_posSemidef</code>, the eigenstate, and Lemma 4); this presumes the physical transfer admits a frustration-free nearest-neighbour projector form
system identification: confinement $\iff$ contrast $< 3^{1/4} = e^{\kappa_0}$ (§8.6)	<code>Certify.lean</code> (<code>confinement_iff_contrast_lt_rpow</code>)	proved
weak-coupling limit: the free-field plateau below the floor, $\mu_\infty(8) = 0.0326 < \kappa_0$ (Wick, §8.6)	<code>FreeField.lean</code> (<code>muInf_lt_floor</code>)	proved
the read layer	[E]	companion paper, own Lean/Mathlib certification

13.10 What is proved of the Clay statement. The Clay problem asks, for every compact simple G , for a non-trivial Yang–Mills theory on $\mathbb{R}^4$ satisfying the Wightman (equivalently Osterwalder–Schrader) axioms, with a mass gap. For $G = SU(N)$, $N \geq 2$, the capstone `ClayRoutes.clay_continuum_of_boxPatchGap` proves the gapped part of that statement, `ClayRoutes.ClayContinuum` — on one reconstructed Hilbert space, at every dyadic step of the continuum tower, a transfer gap of physical rate at least c/L and a non-zero vector orthogonal to the vacuum — from named inputs, each a Lean proposition. For $SU(2)$ and $SU(3)$ the box input is a theorem at the floor coupling $\beta_0 = 17N^2/(88\pi^2)$ (`PairCorrSharp.boxPatchGap_floor_su2`, side 2606, gap 307/10000; `PairCorrSharp.boxPatchGap_floor_su3`, side 545, gap 367/2500), and `PairCorrSharp.clay_continuum_floor_su2` and `clay_continuum_floor_su3` carry the remaining inputs on the three foundational axioms. Those inputs are open: **E**, the uniform bound on separated renormalised Schwinger functions `ContinuumSep.UniformBoundSep`, which the proof reads only on reflected pairs, where it reduces to the reflection-diagonal bound `ContinuumSep.DiagBoundSep`; **the non-zero vector**, from the kernel convergence `ContinuumNontrivial.KernelConvergesSep`, which with **Y** gives `ClayReduce.PairLowerBound`, all the capstone reads of either; **Y**, the asymptotic-freedom short-distance law `ShortDistanceY.AFShortDistance`, a hypothesis of the floor capstones and absent from `ClayReduce.clay_continuum_floor_su2_rhoA` and `_su3_rhoA`, which, at the field-independent factor `ClayReduce.zA N = a-4`, take `PairLowerBound` in its place; and **the UV step** `ClayRoutes.UVBelowIR` below the box's rate $\min(\text{boxKnabe } n \gamma, 1)/(508 a_{\text{run}}(\beta_0))$, which with the box gives `WeakCouplingWindow.FixedWindowDecay`. `FixedWindowDecay` has two proved equivalent forms: the edge step (`EdgeStep.fixedWindowDecay_iff_edgeStep`), in which each block step lengthens the physical correlation length by a summable amount, and the exponential-moment tower (`MomentTower.fixedWindowDecay_iff_momentTower`), in which for some $M > 0$ and every large β the series $\sum_t C(t) e^{\sigma t}$ of every gauge-invariant observable's connected correlator converges for all $\sigma < M a_{\text{run}}(\beta)$. For $SU(N \geq 4)$ the box `BoxPatch.BoxPatchGap` is open as well, and no compact simple group other than $SU(N)$ is treated. Non-triviality in the Clay sense — a connected function of order three or more away from zero (`ThreePointN.WilsonThreePointSeparation`) — and rotation invariance beyond the hypercubic group (`ContinuumHypercubic.RotationOpen`) lie outside `ClayContinuum`. In the capstone's own terms, `ClayRoutes.ClayContinuum` is a transfer gap at rate $e^{-(c/L) a_m}$ at every dyadic step m (`ContinuumClay.GappedAt`) together with a non-zero vector orthogonal to the vacuum (`ContinuumClay.NontrivialVacuum`); the reduction of **E** to reflected pairs is `ClayReduce.clay_continuum_of_diagBoundSep`; and the box's rate is `HeatBathLocal.boxRate`. The development leaves the Clay statement open.

14. Data, code and reproducibility

The instrument. Every empirical quantity in this paper is a deterministic read of a raw lattice configuration by the open-source *Entroptics* instrument, v0.2.6 at commit cf9edee [S], github.com/Agience/entroptics; the resampling, the empirical-Bernstein intervals and the lag-ratio rates are the library's too, and only the elementary statistics (standard errors, descriptive fits) are local, as set out below. The empirical reads are deterministic: every number in §8 reproduces by rerunning the named script on the archived ensembles.

The repository. The development lives under [research/](#): [code/](#) (the reader and the generators, with [code/certify/](#) the certificates, deterministic and statistical, cited below as [certify/...](#)), [data/](#) (the numbered run scripts, each writing its own CSV and figure), and [lean/](#) (§13).

What runs through the library. Every read in §8–§9 goes through the Entroptics library [S], and only through the wrapper [research/code/entroptics_adapter.py](#); no other file imports the library ([test_provenance.py::test_nothing_reaches_the_entroptics_library_except_the_adapter](#)). That includes the resampling — [bootstrap](#) (e.g. [certify/ym_crossover_confinement_of_grid.py](#), [data/9_1_run_d2_bound.py](#)) and the delete-one-bin [jackknife](#) (e.g. [data/8_7_run_transfer_gap.py](#), [data/8_7_run_mhi_lscan.py](#)); the empirical-Bernstein interval, [empirical_bernstein](#) (called by [eb](#) in [certify/ym_crossover_confinement_of_grid.py](#), which [d2_upper](#) and [certify/ym_confinement_of_cos_average.py](#) use, and by [Reads.k_signal_certificate](#) in the wrapper; both pass the sample range as the support width, which is why those confidences are nominal); and the lag-local decay rate $\log(C(t)/C(t+1))$, [effective_rates](#) (in [data/8_7_run_gap_correlator.py](#), and averaged over lags by [lag_budget](#) in [code/aperture_reads.py](#)). The standard errors of ensemble means and the descriptive fits the scripts report are elementary statistics, computed where they are reported. Lattice physics — configuration generation, plaquettes and the action density, smearing, Wilson loops, and the matrices the theory defines — is computed in the repository, separately from the reads.

Data availability. The empirical reads run on a frozen release of Monte-Carlo ensembles of the gauge-invariant local action density $\phi(x) = \sum_{\mu < \nu} (1 - \frac{1}{N} \text{Re tr } U_{\mu\nu}(x))$ for compact $U(1)$, $SU(2)$, and $SU(3)$ in four Euclidean dimensions: **81 ensembles in (group, L, β) across 229 density shards (187 action-density fields, 42 operator-reduced histories), 20{,}668 configurations, 6.13 GiB** (stored records: a configuration kept at several smearing levels counts once per level), each a `float32` array of shape (n, L, L, L, T) with $T = 2L$, except where an experiment holds the time extent FIXED on purpose: the fixed-aperture box series holds $T = 32$ throughout — the action density at $L = 8, 12, 16, 20$, and the smeared-plaquette amplitudes at $L = 8\text{--}20$ (APE 16) and $L = 8\text{--}24$ (APE 32) — because moving the box while moving the aperture would not separate them, and its 12 operator-reduced shards carry shape (n, T) rather than the field (the zero-momentum projection is taken before the links are discarded, which is what makes a shard kilobytes instead of gigabytes). The §8.7 transfer-gap read uses a separate archived raw-link ensemble, [configs_links_su2](#), at $L = 8, 12, 16, 20 \text{ -- } 16^3 \times 28$ at $\beta = 2.30, 2.40, 2.50$, the other three at $\beta = 2.30$ with $T = 2L$. Coverage: $SU(2)$ at $L = 8\text{--}32$ (including the $\beta = 2.30$ volume tower $L = 8, 12, 16, 20, 24, 28, 32$), $SU(3)$ at $L = 6\text{--}12$ across $\beta = 5.0\text{--}7.0$, and $U(1)$ at $L = 8$ across its deconfinement transition, $\beta = 0.4\text{--}2.5$. Generation is deterministic in (group, L, T, β , seed, therm, method): every field regenerates from the seed manifest. The release ships a data card, the per-campaign generation record and seed map, a NumPy reader, a manifest generator, [manifest.csv](#) (per-shard SHA-256, carrying the channel and smearing level of each operator-reduced shard), and [SHA256SUMS](#) (all 265 shards verified: the 229 density shards and the 36 raw-link shards, 16.14 GiB, of [configs_links_su2](#)). Dataset: CC-BY-4.0, on Zenodo, DOI [10.5281/zenodo.22850110](https://doi.org/10.5281/zenodo.22850110) (v0.2.0, open access), derived from the Entroptics software [S]. Every figure and certificate in §8–§9 regenerates from the frozen ensembles by the named script, reading them with Entroptics v0.2.6, source at commit cf9edee [S].

Where each number is produced. Each empirical claim of §8–§12, with the script that produces it and the artifact it writes. Status labels are as in §13.9.

claim (section)	code / artifact	status
per-ensemble equilibrium check, 105 comparisons, bracketed and hot-start-only references (§8)	store_dat_thermalisation.csv	verified; one row per (group, L, T, β , collection, field); the reference column records, per row, whether it rests on a bracketed reference or a hot-start-only one, and the test column records whether the verdict is a level comparison or a drift test. The $U(1)$ Metropolis proposal width is fixed at 1.0 radian for every β by _ops("u1")
the coupling convention: two-loop $a\Lambda_{\text{lat}}$; the generator held to $\beta/2$, $\beta/4$, $\beta/18$ and the one-loop slope (§8.1)	lattice_scales.py ; tests/test_generator_physics.py	verified
compact- $U(1)$ tension on 20 couplings, no coupling above the floor (§8.2, §8.6, §9)	9_5_dat_u1_discriminator.csv	verified
K_{signal} of $SU(2)$ $\beta = 2.30$ on 8×8 and 16×16 planes, each against the null pinned at its own plane shape (0.0722, 0.0727), with the within-plane-shuffled control (§8.3)	8_3_run_ksignal_planescale.py → 8_3_dat_ksignal_planescale.csv	verified
the compact- $U(1)$ bands of K_{signal} : confined ceiling 0.0868 above Coulomb floor 0.0705, with the within-plane-shuffled control (§8.3)	8_3_run_confinement_order_parameter.py → 8_3_dat_confinement_order_parameter.csv	verified (pinned read)

claim (section)	code / artifact	status
disorder response with standard errors (§8.4)	8_4_run_disorder_response.py	verified
the gap correlator and the variational minimum (§8.7b)	8_7_run_gap_correlator.py	verified
the scaling test at fixed (operator, smearing, τ), taking no minimum and so needing no tolerance, a fourth reading outside the three-way disagreement (§8.7b, §12.4)	certify/fixed_selection_of_scaling.py → 9_8_dat_fixed_selection.csv , whose <code>matched_volume</code> rows carry §12.4	verified
the string tension at $L = 8$: an exactly-determined three-parameter fit, which the certification refuses to report (§8.8)	8_8_run_string_tension.py	verified
the smeared channel (§9)	9_7_dat_smeared_channel.csv	verified
the measured interior grids: $\langle d^2 \rangle$ and $\rho'(1)$ (§9)	certify/ym_crossover_confinement_of_grid.py ; certify/interior_mixing_of_analytic_grid.py → 9_4_dat_interior_mixing_grid.csv	certified <i>statistical</i> , nominal; the certificate refuses a partial grid
the single-plaquette enclosure, $\Delta \geq 1.633$ at $\lambda = 1$ (§9, §13.6)	certify/small_volume_enclosure.py	certified (exact rational), outside §13
the per-coupling pivot certificate: 61 of 61 couplings on $\lambda \in [0.16, 6.76]$, and the continuous cover (§9, §13.9)	certify/cell_pivot_certificate.py → 9_14_dat_cell_pivot_certificate.csv	certified (exact rational). The script CHECKS the rational shift $\mu = 2747/10000$ against the certified enclosure of κ_0 ; it writes each certified coupling as a theorem for §13 (<code>--emit-lean</code>) and writes the cover module (<code>--emit-lean-cover</code>); it refuses a truncation shallower than the one at which the tail bound lifts every transcribed theorem, and refuses to emit the cover unless consecutive balls overlap and their union contains the range. The <code>tail_lifts</code> and <code>tail_shift</code> columns record the tail-inclusive check at $j_{\max}=30$
the margin across $U(1)$'s transition (§12)	certify/gap_of_margin.py ; 9_2_dat_margin_aperture.csv	verified; the matched-ratio value 0.197 at $\beta = 2.50$, re-measured at $n = 48$, is recorded in certify/gap_of_margin.py and is outside the released artifact, which carries $n = 128$
the aperture cap: the closed-form average, its monotonicity and the solve for the critical λ , with the whole average reproduced by direct summation over every lag (§12.2)	aperture_cap_of_floor.py → 9_10_dat_aperture_cap.csv	certified; the script refuses if the closed form and the direct sum disagree. The <code>aperture_cap</code> rows carry the critical-rate sequence and the <code>aperture_cap_mass</code> rows the mass sequence; each of the two cap constants is bound to its own row
the gapless-weight bound on the ensemble (§12.3)	9_9_dat_zero_mode_bound.csv	certified; refuses if the bound and the measured weight ever contradict
the junction residuals, both groups (§12.5)	9_6_dat_junction_residuals.csv	verified
the box at fixed aperture (§9)	certify/gap_of_box_at_fixed_aperture.py → 9_12_dat_gap_of_box.csv ; smearing wired into the generator (<code>--smear</code>)	measured, no verdict
centre dominance (§9)	certify/string_tension_eq_centre.py → 9_7_dat_centre_dominance.csv	measured, partial
K_{signal} , its within-plane-shuffled read and their paired difference on	8_3_regen_from_store.py → 8_3_dat_nobump.csv	verified (pinned read)

claim (section)	code / artifact	status
$U(1)$ and $SU(2)$, 8×8 planes (§8.3)		
disorder response: peak separates $U(1)$ from $SU(2) \approx 20\times$ (§8.4)	8_4_regen_from_store.py	verified
free-scalar gap calibration (§8.5)	8_5_run_gap_calibration.py	verified (recovers E_0 at every mass to within a fraction -- 0.02 to 0.35 -- of the read's own four-block reseeding spread; the spread itself falls from 19% of E_0 at $m = 0.20$ to 4.5% at $m = 1.10$)
instrument vs established: three cited panels (§8.6)	8_6_run_benchmark.py	verified
system-ID: confinement $\iff$ contrast $< 3^{1/4} = e^{\kappa_0}$ (§8.6)	8_6_run_probe.py	certified (gap identity to 1.9×10^{-11})
weak-coupling limit: free-field plateau $\mu_\infty(8) = 0.0326 < \kappa_0$ (Wick, §8.6)	certify/free_field_muinf.py	certified (exact Wick circulant on the L^4 free propagator, no Monte Carlo: $\mu_\infty(L)L^2 = 2.09$ at $L = 8$ rising to 2.18 by $L = 32$, $M_2 = 0.111$, lattice axes identical to 4×10^{-17} , λ_2 a doublet to 3×10^{-16}); the below-floor value is also proved (§13.9)
interior read $\langle d^2 \rangle(\beta)$ across the crossover, peak 0.192 (§9)	9_1_run_d2_bound.py	verified (circle-distance read over the full periodic extent, $\pm 2\sigma$ bootstrap, $SU(2)$ $L=16, n \geq 96$)
interior read $\langle d^2 \rangle(\beta)$ for $SU(3)$ across $\beta = 5.0 - 7.0$, below the aperture ceiling (§9)	9_1_run_d2_su3.py	verified (same direct-lag read, $\pm 2\sigma$ bootstrap, $SU(3)$ $L=6, 8$; $\langle d^2 \rangle \leq 0.231$ vs ceiling 0.457/0.812)
$\langle d^2 \rangle$ under the derived sharp aperture ceiling $B_{16} = 3.2480$ at nominal 99.9999%, nothing pinned (§9)	9_1_run_d2_certify.py	certified <i>statistical</i> , nominal (finite-sample empirical-Bernstein; it assumes independent configurations and takes the sample range as the support width; largest nominal-99.9999% upper 1.372, clearing the ceiling by $2.3\times$; nominal joint $\approx 99.9987\%$; $SU(2)$ reads, evidence for the open input on the $SU(3)$ moment, §9). For the confinement claim the row below is the stronger certificate: it certifies $\mu < \kappa_0$ itself for the $SU(2)$ read
confinement itself, $\mu < \kappa_0$, from the read's own cosine average (§9)	certify/ym_confinement_of_cos_average.py $\rightarrow$ 9_2_dat_confinement_cos.csv	certified <i>statistical</i> , nominal (every coupling confined at nominal $\delta = 10^{-30}$; worst $\mu \leq 0.1767$ against $\kappa_0 = 0.2747$)
the substrate ratio across apertures $L = 6 \dots 32$ (§9)	certify/ym_substrate_bound_of_aperture.py $\rightarrow$ 9_3_dat_substrate_of_aperture.csv	verified (13 couplings at more than one aperture; one $c = 6.40 \times 10^{-3}$ covers all, $2.0\times$ under the ceiling)

Related studies. The antiunitary-pairing counterexample cited in §13.7 is the [entropitics-positivity](#) study; the raw-edge under-report proved in §13.7 is reported empirically by [entropitics-jlens](#).

Declaration of generative AI use

The author used Anthropic's Claude Opus (versions 4.8 and 5) in the preparation of this work. Its contribution was to write code, and to generate and validate content. The ideas, the construction and the claims are the author's. No other generative AI tool was used. The author reviewed and edited all output and takes full responsibility for the content of this publication.

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